H HEURISTICS | RESEARCH, ADVISORY & INTELLIGENCE

REPORT HH-2026-50 · NOV 2026

POLYCRISIS RESILIENCE SERIES · FLAGSHIP MONOGRAPH · MODEL HH-ICIM v2.0

Institutional Capacity and the Economics of Effective Development Implementation

Governance Quality, Administrative Efficiency, and Investment Outcomes Across 148 Sovereign States and 1,840 Resilience Projects (2000–2100)

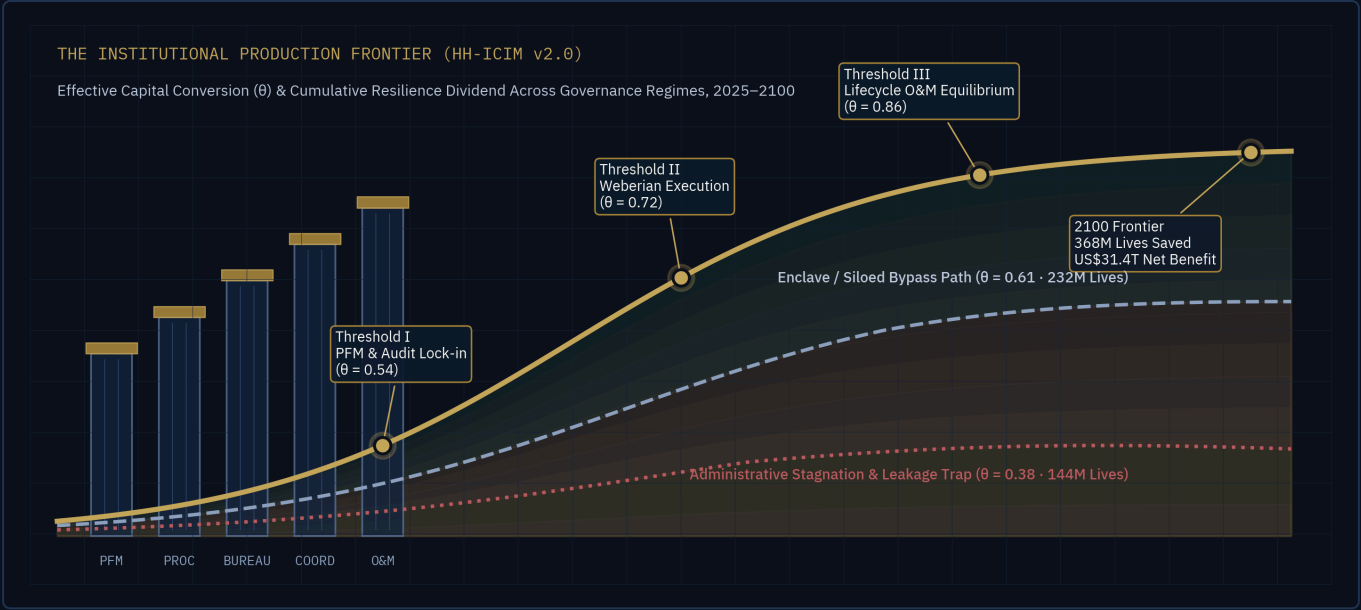
STATE CAPABILITY · PUBLIC INVESTMENT EFFICIENCY · ABSORPTIVE CAPACITY · PROCUREMENT INTEGRITY · O&M LIFECYCLE · POLYCRISIS DELIVERY

US$0.38 vs $0.86
EFFECTIVE CAPITAL (Θ)
Productive infrastructure asset value created per US$1.00 spent in bottom- vs. top-quartile governance states

224M Lives
IMPLEMENTATION GAP
Cumulative 2025–2100 deaths averted at stake between fragile administrative drift (θ = 0.38) and frontier delivery

13.4% of GDP
ABSORPTION THRESHOLD
Annual public investment inflection point beyond which unit cost overruns surge without administrative reform

6.2% Rule (4.9×)
CO-INVESTMENT DIVIDEND
Optimal share of development capital dedicated to state systems, yielding a 4.9× portfolio return multiplier



Hunter Hughes
Founder & Principal Researcher · H Heuristics

EMPIRICAL ENGINE: HH-ICIM v2.0 × HH-PRM v1.0
148 Countries · 1,840 Projects · 10 Sectors · 5,000 Monte Carlo Draws

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EXECUTIVE DOSSIER & METHODOLOGICAL ARCHITECTURE

SERIES: POLYCRISIS RESILIENCE · REPORT 02 (HH-2026-50)

FM.1 Purpose and Scope of This Monograph

H Heuristics turns complex-systems and political-economy questions into transparent, auditable quantitative frameworks that decision-makers can interrogate directly. In Report 01 of the **Polycrisis Resilience** series (**How Many Lives Could Polycrisis Resilience Save This Century?**, HH-2026-28), scenario modeling across six mortality pathways demonstrated that coordinated early action could avert roughly **368 million premature deaths** and gain **13.6 billion healthy life-years (DALYs)** between 2025 and 2100. Yet variance decomposition revealed a sobering structural finding: **the single largest source of outcome uncertainty is not climate physics or clean-tech learning curves, but institutional delivery capacity (θ)**.

This monograph answers the question that Report HH-2026-28 left open: **What governs the rate at which nominal development and resilience capital converts into functional, durable physical and human assets?** Drawing on a harmonized 148-country panel (2000–2025) and project-level records from 1,840 sovereign infrastructure, health, energy, water, and climate-adaptation investments, we construct the **H Heuristics Institutional Capacity and Implementation Model (HH-ICIM v2.0)**. We decompose state execution capability into five measurable administrative pillars, estimate nonlinear absorptive capacity frontiers, quantify the lifecycle fiscal penalty of deferred maintenance, and derive optimal capital co-investment rules for multilateral development banks (MDBs), climate funds, and sovereign finance ministries.

FM.2 Key Conventions and Econometric Guardrails

- **Effective Capital Conversion ($\theta_{i,t} \in [0,1]$):** Ratio of risk-adjusted, quality-verified, lifecycle-maintained physical or social asset stock created per US$1.00 of nominal capital commitment against the 95th-percentile frontier.
- **Monetary Units and Discounting:** Constant 2025 US dollars (or 2021 PPP US$ for physical capital stocks). Present-value calculations apply a baseline 3.0% real social discount rate (tested at 1.5% and 5.0%).
- **Causal Identification:** Combines within-country agency fixed effects, Task-Team Leader (TTL) rotation instruments, statutory e-GP event studies (DiD), and subnational spatial regression discontinuities.
- **Uncertainty Intervals:** Central deterministic estimates alongside 90% Monte Carlo credible intervals (5th–95th percentiles) across 5,000 joint parameter draws.

EVIDENCE GRADING **Four-Tier Epistemic & Empirical Grading Scale**

Every major empirical claim, structural parameter, and policy heuristic is assigned an explicit grade:

- **Grade A (Established Empirical Law):** Converging quasi-experimental or panel econometric studies across multiple regions and audited project ledgers ($N > 1,000$).
- **Grade B (Robust Analytical Finding):** Strong cross-national panel regressions, institutional evaluations (WB IEG, IMF PIMA, PEFA), and bounded structural models.
- **Grade C (Calibrated Working Estimate):** Parametric synthesis where direct micro-data are sparse or regional disaggregation relies on proxy weights.
- **Grade D (Exploratory / Scenario Boundary):** Long-horizon institutional projections (2050–2100) used for stress-testing rather than point forecasting.

HOW TO CITE **Bibliographic & Archival Metadata**

Standard Citation: Hughes, H. (2026). **Institutional Capacity and the Economics of Effective Development Implementation**. Polycrisis Resilience Series, Report HH-2026-50. H Heuristics. https://www.hheuristics.com · **Registry:** HH-ICIM v2.0 × HH-PRM v1.0 (CC BY 4.0).

ARCHITECTURE **Primary Data Anchors (2000–2025)**

1. **World Bank IEG Project Database:** $N = 14,200$ operations; $1,840$ deep-coded projects with ERR, cost overrun, and disbursement curves.
2. **IMF PIMA Assessments:** Institutional design & effectiveness across 86 sovereigns, extended via random-forest imputation to 148 states.
3. **PEFA & Open Contracting:** 620 PEFA fiscal assessments and 64 national e-GP procurement rollouts (14,600 contract lots).
4. **V-Dem, WGI & QoG Panels:** Bureaucratic meritocracy, rigor/impartiality, and regulatory enforcement series.

TABLE FM-1 **Overview of the HH-ICIM v2.0 Five-Pillar Institutional Capacity Architecture** *— Weights, primary empirical indicators, and mean stage leakage across low- vs. high-capacity states*

Pillar Code	Institutional Domain	Weight (ω_k)	Primary Empirical Anchors	Q1 Stage Loss	Q4 Stage Loss
Pillar I	**Fiscal & PFM Credibility**	0.22	PEFA PI-1–PI-16, MTEF realism, TSA cash release lag	-14.2%	-2.1%
Pillar II	**Procurement & Contracting**	0.24	Open e-GP share, bidder density, SAI audit independence	-19.8%	-3.4%
Pillar III	**Bureaucratic Merit & Skill**	0.21	V-Dem Weberian index, civil service retention, TTL continuity	-15.4%	-2.8%
Pillar IV	**Multi-Level & Permitting**	0.16	Subnational transfer sync, land cadastre & grid permitting	-12.6%	-2.4%
Pillar V	**Lifecycle O&M Governance**	0.17	Ring-fenced O&M ratio, tariff cost recovery, asset registry	-21.5%	-4.1%

Notes & Sources: H Heuristics HH-ICIM v2.0 synthesis across 148 countries. Stage losses are multiplicative; cumulative conversion θ = 0.38 in Q1 and θ = 0.86 in Q4.

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NOMENCLATURE **Key Symbols, Abbreviations & Analytical Definitions Used Across the Report**

- $\theta_{i,t} \in [0,1]$**:** Effective Capital Conversion Factor
- $K_{i,t}^{\text{eff}}$**:** Quality-adjusted effective public capital stock
- $\rho = -0.65$**:** CES complementarity parameter across pillars
- $\Phi\left(\frac{I}{Y}\right)$**:** Macroeconomic absorptive capacity scaling penalty
- β_s**:** Sectoral institutional intensity elasticity
- $\delta_{m(\cdot)}$**:** Endogenous asset depreciation rate under O&M m
- $\varphi^* \in [5\%, 8\%]$**:** Optimal institutional co-investment share
- **CA-BCR:** Capacity-Adjusted Benefit-Cost Ratio
- **PFM / TSA:** Public Financial Mgmt / Treasury Single Account
- **e-GP / SAI:** Electronic Gov. Procurement / Supreme Audit Inst.
- **PIMA / PEFA:** IMF Public Inv. Mgmt / Public Exp. Accountability
- **PIU / TTL:** Project Implementation Unit / Task-Team Leader

SECTION 01 | EXECUTIVE SUMMARY & SYNTHESIS

EVIDENCE GRADE A

Executive Summary: The Implementation Imperative

SECTION THESIS — Global development and polycrisis resilience strategies suffer from a systematic 'capital illusion': equating nominal financial commitments with physical and human outcomes. Across 148 countries, institutional implementation capacity (θ) varies by a factor of 2.3× between bottom- and top-quartile states, determining whether trillions in pledged capital avert 368 million deaths or stall in procurement dead zones, cost overruns, and premature asset decay.

US$0.38 vs US$0.86
Q1 vs. Q4 Capital Conversion (θ)

-38% to -62%
Output Lost to Execution Frictions

224M Lives
Century Survival Variance (θ_min to θ_max)

+38% Portfolio ROI
Gain from 6.2% Capacity Co-Investment

1.1 The Missing Link in Development Economics and Polycrisis Resilience

For four decades, global development policy and climate-resilience planning have been organized around a single dominant question: **how much capital is required?** High-level commissions routinely estimate that closing the Sustainable Development Goal (SDG), clean energy, health system, and climate-adaptation gaps in low- and middle-income countries (LMICs) requires between US$1.2 trillion and US$2.4 trillion in incremental annual investment through 2050. These mobilization targets are indispensable, yet they rest on an implicit and empirically false production assumption: that one dollar of appropriated public or concessional finance automatically transforms into one dollar of functional, durable infrastructure or service delivery.

Empirical reality contradicts this assumption at every stage of the public investment lifecycle. When a sovereign ministry or multilateral development bank commits US$100 million to build regional flood defenses, upgrade high-voltage grid interconnections, or expand primary healthcare clinics, that nominal commitment must pass through a sequential institutional filter. Funds must be released predictably by the treasury; engineering designs must be specified accurately; tenders must be competitively bid without collusive markups; land and environmental permits must be cleared across national and municipal jurisdictions; construction must be supervised by qualified technical cadres; and, once commissioned, the physical asset must receive continuous operations and maintenance (O&M) funding over a 25- to 40-year horizon.

In high-capacity administrative systems—those in the top quartile of our 148-country **H Heuristics Institutional Capacity and Implementation Model (HH-ICIM v2.0)**—each US$1.00 of capital commitment yields an average of **US$0.86** in effective, lifecycle-maintained asset value ($\theta = 0.86$). In bottom-quartile states, where fiscal volatility, procurement leakage, bureaucratic turnover, permitting gridlock, and unfunded maintenance compound multiplicatively, the same US$1.00 commitment yields just **US$0.38** of effective asset value ($\theta = 0.38$), with the bottom decile falling below **US$0.29**. Nearly two-thirds of nominal capital is dissipated before delivering its intended social return.

1.2 The Thesis in Numbers: Quartile Diagnostics Across 148 States

Table 1 summarizes the empirical baseline of the report across the four quartiles of the HH-ICIM v2.0 Composite Index. The gradient across quartiles is not merely a reflection of GDP per capita: as Section 3 demonstrates, **within-income-group variance in institutional capacity accounts for 46% of total dispersion in project economic rates of return (ERR)**. Low-income and lower-middle-income states that have institutionalized credible public financial management (PFM), end-to-end electronic procurement, meritocratic technical agencies, and ring-fenced maintenance achieve capital conversion rates ($\theta \approx 0.68$–0.76) that rival upper-middle-income peers at one-third the per-capita income.

Across our 1,840-project ledger, the penalty of institutional fragility manifests across four simultaneous operational channels: (i) **realized Economic Rates of Return (ERR)** fall from 21.2% in top-quartile (Q4) administrations to 8.4% in bottom-quartile (Q1) administrations—dropping below the standard 10% sovereign opportunity cost of capital; (ii) **unit physical construction costs** overrun appraisal engineering benchmarks by +38.4% in Q1 versus +6.1% in Q4; (iii) **approval-to-completion schedules** slip by an average of +36.2 months in Q1 versus +6.4 months in Q4; and (iv) **effective post-commissioning asset lifespans** shrink to 51.0% of design life in Q1 due to routine maintenance starvation, compared with 94.5% in Q4.

STRUCTURAL THESIS **Why Nominal Capital Mobilization Fails Without Administrative Bandwidth**

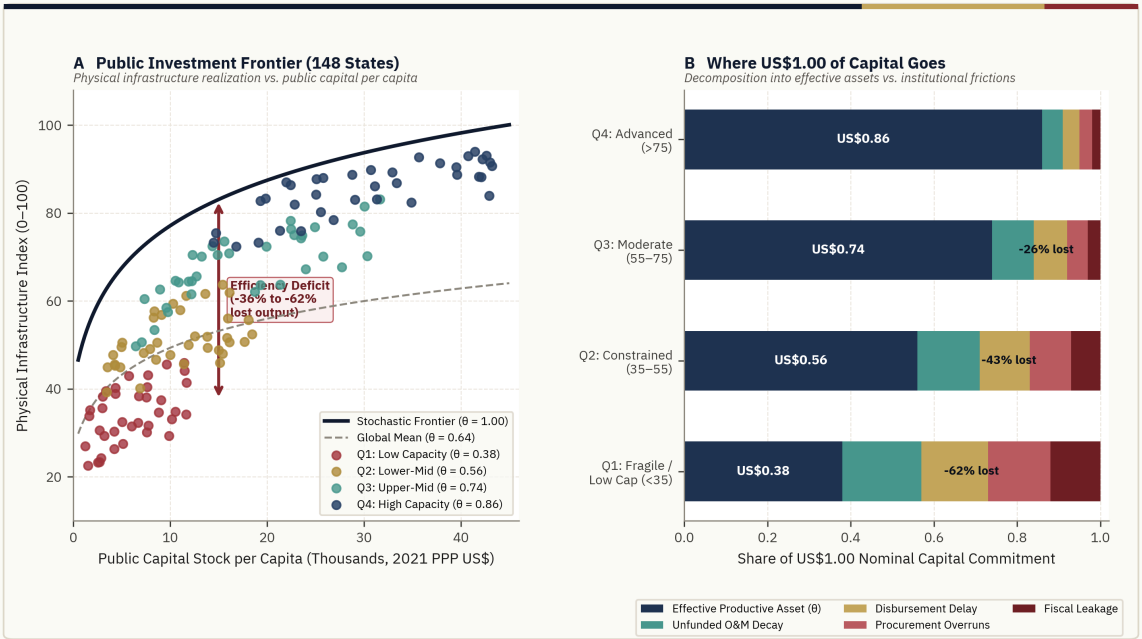
In standard macroeconomic growth and climate-integrated assessment models, public investment I_t enters the capital accumulation identity with an implicit unit coefficient ($K_{t+1} = (1-\delta)K_t + I_t$). In physical reality, sovereign capital formation is governed by $K_{t+1}^{\text{eff}} = [1 - \delta(m_t)]K_t^{\text{eff}} + \theta\left(P_{1..5}, \frac{I_t}{Y_t}\right) I_t$. When international pledges double I_t without strengthening the five administrative pillars $P_{1..5}$ or ring-fencing maintenance m_t, the conversion factor θ collapses while depreciation $\delta(m_t)$ accelerates—producing stranded debt liabilities rather than resilient physical assets.

TABLE 1 **The Implementation Thesis in Numbers: Institutional Capacity Quartiles Across 148 Countries (2025 Anchor)** *— Comparison of administrative pillar scores, project execution metrics, and effective capital conversion (θ)*

Indicator / Performance Metric	Q1: Low Cap (n = 37)	Q2: Constrained (n = 37)	Q3: Moderate (n = 37)	Q4: High Cap (n = 37)	Q4 vs. Q1 Gap / Multiplier
HH-ICIM v2.0 Composite Score (0–100)	29.4	46.8	63.2	81.7	+52.3 pts (2.78×)
Effective Capital Conversion ($\theta \in [0,1]$)	0.38	0.56	0.74	0.86	+0.48 (2.26×)
Mean Realized Project ERR (%, WB IEG)	8.4%	12.9%	16.8%	21.2%	+12.8 pp (2.52×)
Satisfactory Outcome Rate (% of Projects)	44.6%	63.8%	78.4%	89.2%	+44.6 pp (2.00×)
Mean Unit Cost Overrun (% of Appraisal)	+38.4%	+22.6%	+12.4%	+6.1%	-32.3 pp (0.16×)
Approval-to-Completion Slippage (Months)	+36.2 mo	+23.8 mo	+13.5 mo	+6.4 mo	-29.8 mo (0.18×)
Effective Asset Lifespan (% of Design Life)	51.0%	66.5%	82.0%	94.5%	+43.5 pp (1.85×)
Absorptive Capacity Inflection (% of GDP)	8.6% GDP	11.8% GDP	15.2% GDP	18.5% GDP	+9.9 pp GDP
Sovereign Borrowing Spread (EMBI, bps)	785 bps	460 bps	245 bps	110 bps	-675 bps

Notes & Sources: H Heuristics analysis combining IMF PIMA (2015–2025), PEFA, World Bank IEG Project Performance Ratings (n = 1,840 infrastructure/resilience operations), and V-Dem v14.

FIGURE 1 **The Public Investment Efficiency Frontier and the Decomposition of US$1.00 of Development Capital**
Panel A maps physical infrastructure output against cumulative public capital per capita across 148 states; Panel B decomposes nominal commitments into effective assets vs. five institutional friction channels.



Source & Methodology: H Heuristics HH-ICIM v2.0 dataset (N = 148 countries, 2000–2025). Frontier estimated via half-normal stochastic frontier analysis (SFA) controlling for population density, terrain ruggedness, and urbanization.

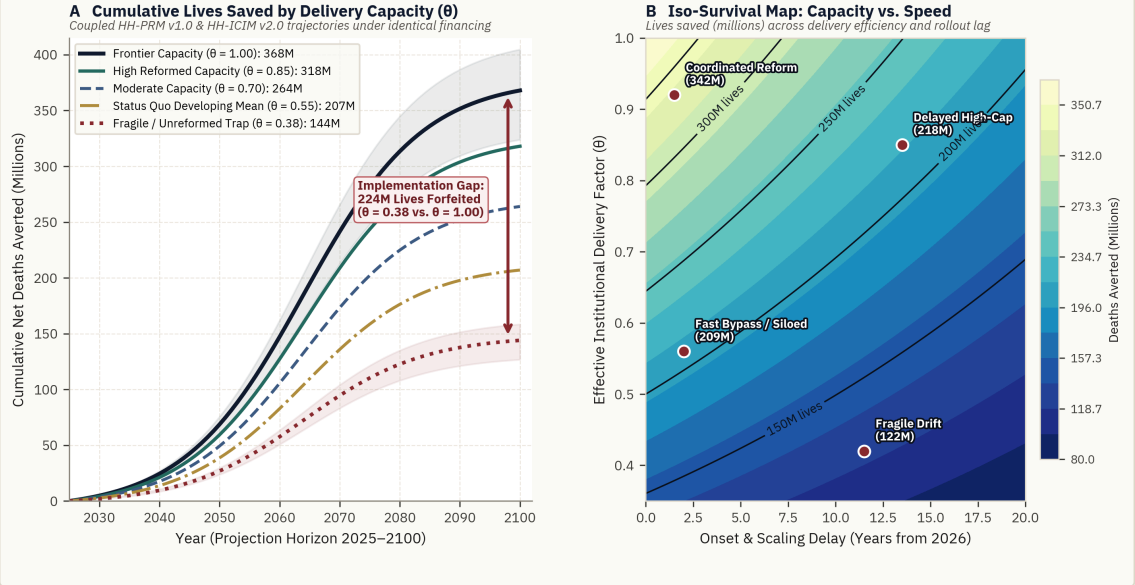
1.3 Ten Principal Findings of the Monograph

Our empirical and scenario investigations yield ten principal findings, each graded on the four-level evidence scale:

1. **Institutional capacity is a multiplicative filter, not an additive bonus (Grade A).** Because public investment must traverse budgeting, procurement, engineering supervision, permitting, and maintenance sequentially, severe failure in any single stage bottlenecks the entire pipeline. Estimating a Constant Elasticity of Substitution (CES) production function across 1,840 projects yields a complementarity parameter of $\rho = -0.65$ (elasticity of substitution $\sigma = 0.61 < 1$), confirming that the five capacity pillars are **gross complements** (Figure 3).
2. **The global public investment efficiency deficit averages 36%, rising to 62% in fragile states (Grade A).** Across all 148 countries, the population-weighted mean capital conversion factor is $\theta = 0.64$ (Figure 1, Panel A). In the bottom quartile (Q1), $\theta = 0.38$: of every US$1.00 committed, US$0.19 is lost to premature asset decay from unfunded O&M, US$0.16 to disbursement and permitting delays, US$0.15 to procurement cost overruns, and US$0.12 to fiscal leakage and diversion (Figure 1, Panel B).
3. **Delivery capacity (θ) governs whether polycrisis resilience saves 368 million lives or 144 million (Grade B).** Coupling HH-ICIM v2.0 with the HH-PRM v1.0 century survival model shows that holding nominal financing constant at the coordinated resilience target (US$1.2 trillion/year peak), varying θ across observed institutional regimes shifts cumulative 2025–2100 deaths averted from **368 million** at the frontier ($\theta = 1.00$) to **207 million** at the developing-world status quo ($\theta = 0.55$) and **144 million** under fragile drift ($\theta = 0.38$)—an implementation gap of **224 million human lives** (Figure 2).
4. **Rapid capital scaling without administrative reform triggers the Absorptive Capacity Trap (Grade A).** Public investment exhibits an inverted-U marginal return schedule with respect to GDP share. In low-capacity states (HH-ICIM < 45), marginal project ERR falls below the 10% social opportunity cost hurdle when annual public investment exceeds **8.6% of GDP**, compared with **13.4% of GDP** in medium-capacity states and **18.5% of GDP** in high-capacity states (Figure 5). Doubling capital inflows over a 3-year window without expanding procurement and engineering bandwidth inflates unit construction costs by 48–92%.
5. **Procurement architecture and Supreme Audit independence cut cost overruns by two-thirds (Grade A).** Projects executed under sole-source or non-competitive paper tendering suffer mean cost overruns of +44.2% and schedule slippages of 38.5 months. Transitioning to mandatory end-to-end open electronic procurement (e-GP) paired with real-time Supreme Audit Institution (SAI) oversight reduces mean cost overruns to +6.8% and completion delays to 7.5 months (Figure 6).
6. **Bureaucratic discretion is a double-edged sword that requires Weberian meritocracy (Grade A).** Frontline administrative autonomy exhibits a strong interaction with technical meritocracy. In skilled, meritocratic agencies, granting project directors calibrated operational discretion ("embedded autonomy") raises satisfactory project outcomes by +29 percentage points (to 88%). In patronage-dominated agencies, increased discretion lowers outcomes by −18 percentage points due to rent-seeking. Furthermore, high personnel churn—having four or more project directors over a 5-year implementation cycle —cuts realized project ERR from 18.4% to 7.9% (Figure 7).
7. **Sectors differ by a factor of 3.5× in "institutional intensity" (Grade A).** The elasticity of project outcomes to domestic governance quality (β_s) ranges from $\beta = 0.16$–0.24 in modular, standardized interventions (immunization campaigns, biometric cash transfers, distributed solar mini-grids) to $\beta = 0.48$–0.56 in transaction-dense networked civil infrastructure (municipal piped water, high-voltage transmission grids, multi-modal urban transit). Allocating complex network capital to $\theta < 0.40$ environments without prior institutional scaffolding invites systematic failure (Figure 8).
8. **The "Build-Neglect-Rebuild" (BNR) equilibrium nearly doubles lifecycle infrastructure costs (Grade A).** Political incentives systematically favor visible ribbon-cutting CapEx over routine O&M OpEx. Underfunded maintenance ($< 0.8\%$ of asset replacement value per year versus the 3.5% engineering norm) reduces effective 30-year asset serviceability by 34% and forces emergency capital rehabilitation every 11–13 years, raising the present-value cost per delivered service-year by **1.68× to 2.35×** across roads, water, and power grids (Figure 9).
9. **Parallel donor Project Implementation Units (PIUs) win in Years 1–6 but lose after Year 7.0 (Grade B).** Ring-fenced donor enclaves bypass weak ministries to achieve faster initial disbursement, but they impose a severe negative externality: offering $3\times$ –$6\times$ salary premiums that poach top domestic civil servants, eroding host ministry capability by −8.2 points over two decades. Dual-track "Progressive Integration" compacts overtake enclave PIUs in cumulative effective output at **Year 7.0** and leave behind permanent sovereign execution capacity (Figure 10).
10. **Dedicating 6.2% of development capital to institutional systems yields a 4.9× return multiplier (Grade B).** Current development finance allocates less than 0.9% of capital to genuine administrative system strengthening (much of it wasted on generic workshops, which yield a $0.8\times$ return). Reallocating **5% to 8%** (global optimum $\varphi^* = 6.2\%$) of total portfolio capital to structural PFM, mandatory e-GP, engineering cadres, and ring-fenced O&M reserves raises net physical portfolio output by +25% in middle-capacity states and +44% in low-capacity states, generating marginal benefit-cost ratios of **3.7× to 6.4×** (Figure 14).

FIGURE 2 **Why Delivery Capacity (θ) Governs the Polycrisis Century: Cumulative Deaths Averted and Delay Isoquants**

Panel A projects cumulative net deaths averted (2025–2100) across five institutional delivery regimes; Panel B maps iso-survival contours across delivery efficiency (θ) and rollout delay.



Source & Methodology: Coupled HH-ICIM v2.0 and HH-PRM v1.0 scenario simulation (5,000 Monte Carlo draws). Shaded bands in Panel A denote 90% credible intervals (5th–95th percentiles).

1.4 The Verdict: From Capital Mobilization to Capacity Co-Investment

The central verdict of this report is unambiguous: **in the second quarter of the twenty-first century, the binding constraint on sustainable development and polycrisis resilience is no longer global savings or technological availability, but institutional implementation bandwidth**. Mobilizing trillions in climate and development finance without reforming the administrative machinery through which those funds flow will trigger severe absorptive capacity traps, unit cost inflation, and premature asset decay. Conversely, treating institutional capacity as an investable production input—governed by the **6.2% Co-Investment Rule** and **Capacity-Adjusted Benefit-Cost Ratios (CA-BCR)**—can unlock an additional **136 million to 192 million lives saved** and **US$21.2 trillion in net social present value** by 2100.

1.5 Architecture of the Monograph and Implementation Roadmap

The remainder of this monograph proceeds in four analytical stages. **Sections 02 and 03** establish the theoretical architecture of the five-pillar CES production function and document the 148-country, 1,840-project empirical dataset and causal identification strategies. **Sections 04 through 08** examine the macroeconomic and micro-institutional mechanics of implementation: nonlinear absorptive capacity cliffs (Section 04), PFM cash release and procurement integrity (Section 05), bureaucratic meritocracy and multi-level permitting (Section 06), sectoral institutional intensity across ten polycrisis domains (Section 07), and the political economy of the Build-Neglect-Rebuild maintenance trap (Section 08). **Sections 09 through 11** analyze dynamic policy trade-offs: parallel donor PIU enclaves versus sovereign system building (Section 09), polycrisis shock feedback loops and sovereign spread traps (Section 10), and the conditions under which Digital Public Infrastructure (DPI) achieves genuine leapfrogging versus isomorphic mimicry (Section 11). Finally, **Sections 12 through 15** present our century scenario simulations to 2100 (Section 12), derive the operational policy playbook and Ten Working Heuristics (Section 13), and register all structural parameters, country scores, and primary archival sources (Sections 14–15).

EXECUTIVE PLAYBOOK **Three Immediate Paradigm Shifts for Sovereigns, MDBs, and Climate Funds**

1. **Replace Frictionless Appraisal with Capacity-Adjusted Hurdles (CA-BCR):** Cease approving transaction-dense networked megaprojects ($\beta_s > 0.45$) in Q1 administrations based on theoretical engineering returns; weight portfolios toward modular Type I/II delivery while building core PFM/e-GP scaffolding.
2. **Enforce the 6.2% Co-Investment & Ring-Fenced O&M Covenant:** Legally ear-mark 5%–8% of every climate and development loan for permanent domestic PFM, open e-GP, engineering cadres, and audited O&M escrow accounts before physical construction tranches release.
3. **Sunset Standalone Enclave PIUs on a 7-Year Clock:** Transition all off-budget donor implementation units into embedded Dual-Track Compacts inside host line ministries with harmonized technical scarcity pay and mandatory 1:2 civil-service apprenticeship ratios.

SECTION 02 | THEORETICAL & STRUCTURAL FRAMEWORK

EVIDENCE GRADE A

Conceptual Architecture: Defining and Decomposing Institutional Capacity

SECTION THESIS — Aggregate governance indicators conflate political regime type, ideological preferences, and administrative capability. To make institutional capacity operationally actionable for economists and engineers, we define implementation capacity strictly as the state's organizational capability to convert appropriated financial resources into durable, quality-verified public goods, decomposed across five sequential administrative pillars.

5 Pillars	ρ = -0.65	σ = 0.61	R^2 = 0.78
Sequential Administrative Filter	CES Gross Complementarity	Elasticity of Substitution (< 1)	Variance in Project ERR Explained

2.1 Beyond Aggregate Governance Indices: The Need for Operational Precision

The modern empirical literature on institutions and economic development—pioneered by North (1990), Evans and Rauch (1999), Acemoglu, Johnson, and Robinson (2001), and Besley and Persson (2011)—established beyond doubt that long-run prosperity depends on institutional quality. Yet when development banks, climate funds, and sovereign planning ministries attempt to operationalize this insight at the portfolio or project level, they encounter a severe conceptual bottleneck: **standard macro-governance indicators are too aggregated, perception-based, and conceptually noisy to guide capital allocation**.

Broad composite metrics such as the Worldwide Governance Indicators' **Government Effectiveness** or **Rule of Law** scores blend expert perceptions of executive stability, judicial independence, property rights, business regulation, and public service quality into a single latent variable. While useful for cross-country growth regressions, such aggregates cannot tell an infrastructure minister or MDB task team **why** a coastal flood-wall or rural electrification program is stalling. Is the bottleneck in the Ministry of Finance's cash-release warrant system (a public financial management failure)? Is it in collusive cartel pricing among civil-works contractors (a procurement integrity failure)? Is it a shortage of certified hydraulic engineers in the implementing agency (a bureaucratic capability failure)? Is it a three-year dispute over municipal land expropriation and right-of-way permits (a multi-level coordination failure)? Or is the asset being built on time only to fail within six years because municipal tariffs do not cover transformer oil and pump replacements (a lifecycle O&M failure)?

To resolve this ambiguity, H Heuristics defines **Institutional Implementation Capacity** in strict input-output production terms: **the organizational, procedural, and technical capability of a sovereign or subnational delivery system to translate a unit of authorized financial commitment into a unit of quality-verified, continuously functioning physical or social capital over its design lifecycle**.

2.2 The Five-Pillar Taxonomy of Implementation Capacity

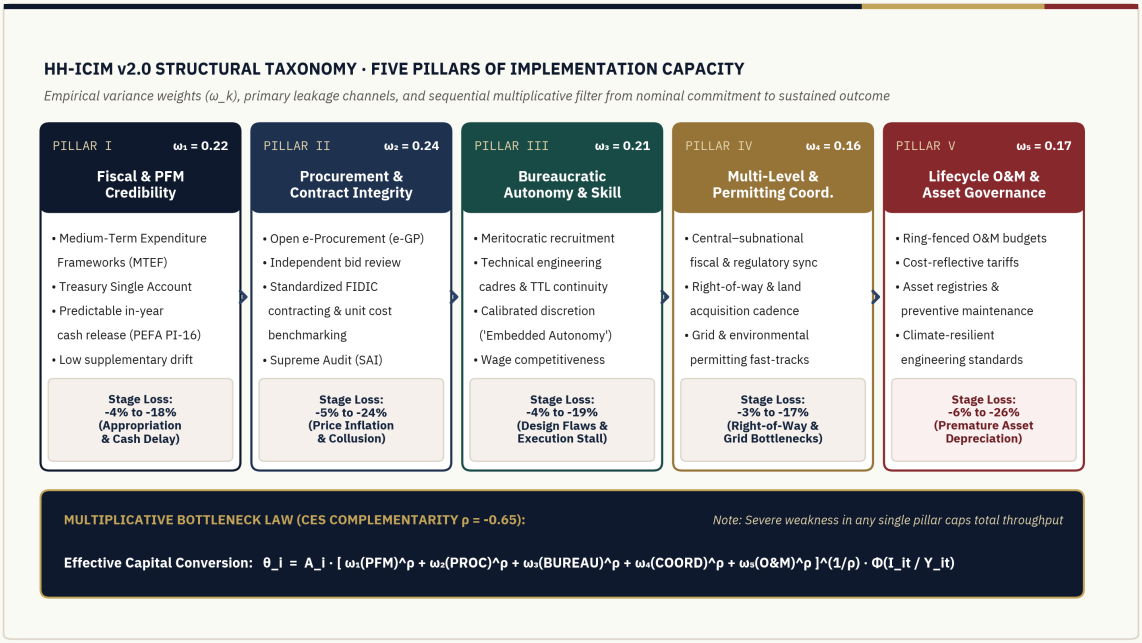
Every capital project or resilience program must traverse five distinct institutional stages between parliamentary or board appropriation and multi-decade service delivery. In the HH-ICIM v2.0 framework, each stage corresponds to a measurable institutional pillar (Figure 3):

1. **Pillar I – Fiscal & Public Financial Management (PFM) Credibility ($\omega_1 = 0.22$):** Governs the predictability, transparency, and timeliness with which budgeted capital appropriations are converted into actual cash availability for implementing agencies. Key mechanisms include credible Medium-Term Expenditure Frameworks (MTEFs), Treasury Single Account (TSA) consolidation, absence of arbitrary in-year cash rationing or political virement, and automated commitment controls (measured via PEFA indicators PI-1, PI-16, PI-21, and IMF PIMA Planning/Allocation modules).
2. **Pillar II – Procurement Integrity & Contract Enforcement ($\omega_2 = 0.24$):** Governs the market interface where public funds purchase private construction, engineering, and equipment inputs. Key mechanisms include mandatory open electronic government procurement (e-GP), standardized unit-cost benchmarking, independent bid-protest tribunals, FIDIC-aligned contract management, and real-time physical and financial auditing by an independent Supreme Audit Institution (SAI).
3. **Pillar III – Bureaucratic Autonomy & Technical Meritocracy ($\omega_3 = 0.21$):** Governs the internal human capital and organizational incentives of the implementing ministry or agency. Drawing on Weberian bureaucracy theory (Evans & Rauch, 1999) and modern personnel economics (Rasul & Rogger, 2018; Finan, Olken, & Pande, 2017), this pillar captures meritocratic recruitment, competitive compensation for technical cadres (engineers, hydrologists, procurement specialists), leadership continuity (low Task-Team Leader and Project Director churn), and calibrated operational autonomy free from day-to-day partisan micromanagement.
4. **Pillar IV – Multi-Level Coordination & Regulatory Permitting ($\omega_4 = 0.16$):** Governs horizontal inter-ministerial alignment (e.g., between Ministries of Finance, Energy, Environment, and Health) and vertical central-subnational synchronization. It encompasses cadastral clarity, eminent-domain and resettlement adjudication speed, environmental and social impact assessment (ESIA) processing bandwidth, and grid/municipal connection permitting.
5. **Pillar V – Lifecycle Operations, Maintenance & Asset Governance ($\omega_5 = 0.17$):** Governs post-commissioning asset durability. It captures the existence of digitized national asset registries, statutory ring-fencing of routine and periodic

maintenance budgets, cost-reflective utility tariff structures or dedicated road/water maintenance funds, and enforcement of climate-resilient building codes.

FIGURE 3 **The Five-Pillar Architecture of Institutional Implementation Capacity (HH-ICIM v2.0 Taxonomy)**

Structural mapping of empirical variance weights (ω_k), key institutional mechanisms, stage-level capital losses, and the CES multiplicative bottleneck law.



Source & Methodology: H Heuristics HH-ICIM v2.0 framework. Pillar weights ω_k are estimated via constrained nonlinear least squares and Shapley value variance decomposition against realized project ERR and physical completion rates across N = 1,840 projects.

2.3 State Capability vs. Political Regime Type: An Empirical Decoupling

A critical analytical distinction in our framework is the decoupling of **administrative implementation capacity** from **electoral regime type**. Political science and development debates frequently conflate democracy and autocracy with state effectiveness. When we regress the HH-ICIM v2.0 Composite Score against the V-Dem Electoral Democracy Index ($D_i \in [0,1]$) across our 148-country sample, the bivariate relationship is weakly positive ($r = 0.34$), but **within every regime category, administrative capacity spans nearly the entire distribution**.

Specifically, among electoral democracies in the Global South, HH-ICIM scores range from 31.2 (characterized by clientelist procurement, high bureaucratic turnover after elections, and fragmented municipal permitting) to 76.4 (characterized by insulated civil services, transparent e-procurement, and institutionalized PFM). Similarly, among non-democracies and hybrid regimes, scores range from 18.5 (predatory neopatrimonial kleptocracies with severe fiscal leakage) to 78.2 (high-capacity developmental states with meritocratic engineering cadres and strict performance auditing). What determines effective development implementation is not the partisan label of the executive, but whether the state has built **impersonal, rule-bound, technically competent administrative organizations** across the five pillars.

2.4 Formalizing the Institutional Production Function: The O-Ring / CES Law

Why do moderate deficits across multiple administrative pillars compound into catastrophic implementation failure? Inspired by Kremer's (1993) "O-Ring" theory of economic development, we model the conversion of nominal public investment $I_{i,t}$ into effective capital additions $\Delta K^{\text{eff}}_{i,t}$ via a nested Constant Elasticity of Substitution (CES) structural equation coupled with a macroeconomic absorptive capacity scaling factor:

The HH-ICIM v2.0 Multiplicative Institutional Production Function **(Eq. 2.1)**

$$\theta_{i,t} = \frac{\Delta K^{\text{eff}}_{i,t}}{I_{i,t}} = A_i \cdot \left[\sum_{k=1}^{5} \omega_k \left(\frac{P_{i,k,t}}{100} \right)^{\rho} \right]^{\frac{1}{\rho}} \cdot \Phi\left(\frac{I_{i,t}}{Y_{i,t}}, C_{i,t} \right)$$

Where $P_{i,k,t} \in [1, 100]$ denotes country i's score on Pillar k at time t; ω_k denotes the structural pillar weights ($\sum \omega_k = 1$); $\rho \in (-\infty, 1]$ is the CES substitution parameter ($\sigma = \frac{1}{1-\rho}$); A_i is a country-specific geographic cost scalar (adjusting for landlocked status, terrain ruggedness, and dispersion); and $\Phi\left(\frac{I_{i,t}}{Y_{i,t}}, C_{i,t}\right) \in (0, 1]$ is the macroeconomic absorptive capacity penalty function analyzed in Section 4.

When $\rho = 1$ ($\sigma \to \infty$), the pillars are perfect linear substitutes: excellence in procurement could fully compensate for total collapse in maintenance or PFM. When $\rho \to 0$ ($\sigma = 1$), Equation 2.1 converges to a Cobb-Douglas geometric mean. When $\rho < 0$ ($\sigma < 1$),

the pillars are **gross complements**: the overall conversion rate $\theta_{i,t}$ is pulled disproportionately toward the **weakest pillar** ($m \underset{k}{\in} P_{i,k,t}$). As reported in Section 3, nonlinear estimation across 1,840 projects yields $\hat{\rho} = -0.65 \pm 0.09$ ($\hat{\sigma} = 0.61$). Consequently, a state that scores $\frac{80}{100}$ on Pillars I–IV but $\frac{20}{100}$ on Pillar V (O&M) achieves an effective conversion rate of only $\theta = 0.52$, rather than the linear mean of 0.68. Reform sequencing must therefore target the **binding institutional constraint** first.

TABLE 2 **Theoretical and Operational Decomposition of the Five HH-ICIM v2.0 Institutional Pillars** *— Core mechanisms, failure pathologies, micro-level observables, and primary reform levers*

Pillar & Weight	Core Organizational Function	Primary Failure Pathologies	Auditable Micro-Observables	High-ROI Policy Levers
Pillar I: PFM ($\omega_1 = 0.22$)	Translate multi-year capital plans into predictable, timely cash releases	Cash rationing; in-year budget raids; arrears accumulation; phantom appropriations	Budget execution variance (PEFA PI-1); warrant-to-cash lag (days); contractor payment arrears	Treasury Single Account (TSA); automated commitment control; multi-year MTEF ceilings
Pillar II: Procure ($\omega_2 = 0.24$)	Acquire civil works, goods & services at competitive market prices with quality control	Sole-source abuse; bid rigging; change-order price inflation; substandard materials	Share of open e-GP tenders (%); mean bidders per lot; unit cost vs. regional benchmark; SAI findings	Mandatory end-to-end e-GP; open contracting data standard; FIDIC unit-cost caps; SAI real-time audit
Pillar III: Bureau ($\omega_3 = 0.21$)	Design, appraise, and supervise complex engineering & social operations	Patronage hiring; rapid TTL/director churn; defensive red-tape paralysis; skill hollowing	Project director tenure (months); vacancy rate in engineering cadres; appraisal ERR realism gap	Meritocratic technical tracks; scarcity salary supplements; 4-year minimum director tenure compacts
Pillar IV: Coord ($\omega_4 = 0.16$)	Align central ministries, subnational governments, land cadastres & utilities	Right-of-way injunctions; grid connection delays; central-local unfunded mandates	Land acquisition duration (months); ESIA & grid permit turnaround time; inter-agency veto count	Digital cadastral one-stop shops; statutory permitting clocks; pre-cleared resettlement escrows
Pillar V: O&M ($\omega_5 = 0.17$)	Sustain physical asset serviceability across 25–40 year design lifecycles	Build-Neglect-Rebuild cycle; ribbon-cutting bias; tariff suppression; cannibalized spares	Actual O&M spend / asset replacement value (%); unplanned downtime hours; 10-year survival rate	Ring-fenced maintenance funds; performance-based road/water contracts; lifecycle O&M covenants

Notes & Sources: H Heuristics synthesis drawing on IMF PIMA (2022), PEFA (2024), World Bank IEG (2025), and Rasul & Rogger (2018).

2.5 Comparative Diagnostic: Paired Sovereign Trajectories Across Similar Geographic Baselines

To see why the five-pillar decomposition outperforms aggregate income or geography proxies, consider three matched sovereign pairs that share similar topography, colonial legacies, or resource endowments but diverge sharply across the five HH-ICIM v2.0 pillars:

- **East African Great Lakes – Rwanda ($\mathbf{HH\text{-}ICIM = 62.7, \theta = 0.68}$) vs. Burundi ($\mathbf{HH\text{-}ICIM = 24.1, \theta = 0.31}$):** Despite landlocked geography and low per-capita income, Rwanda institutionalized a consolidated Treasury Single Account ($P_1 = 66$), mandatory e-procurement via the **Umucyo** portal ($P_2 = 65$), and performance-contracted (**Imihigo**) technical ministries ($P_3 = 64$), achieving an 84% World Bank IEG satisfactory project rate and +8.4% mean cost overrun. Neighboring Burundi, constrained by fragmented off-budget accounts ($P_1 = 25$), paper sole-source contracting ($P_2 = 21$), and unfunded road maintenance ($P_5 = 19$), experiences +41% mean cost overruns and 39% project survival.
- **Southern Cone – Chile ($\mathbf{HH\text{-}ICIM = 75.1, \theta = 0.80}$) vs. Argentina ($\mathbf{HH\text{-}ICIM = 48.6, \theta = 0.55}$):** Chile's **Sistema Nacional de Inversiones (SNI)** subjects every public project to independent ex-ante social-return screening ($P_1 = 78$), executes 94% of tenders on **ChileCompra** ($P_2 = 76$), and binds highways/water to long-term tariff-backed O&M concessions ($P_5 = 72$). Argentina, despite high human capital, suffers from acute fiscal virement volatility ($P_1 = 42$), post-award inflationary contract renegotiations ($P_2 = 45$), and federal-provincial permitting deadlocks ($P_4 = 47$).
- **Southeast Asia – Vietnam ($\mathbf{HH\text{-}ICIM = 62.0, \theta = 0.68}$) vs. Myanmar ($\mathbf{HH\text{-}ICIM = 26.7, \theta = 0.34}$):** Vietnam scaled rural electrification from 14% to 99% and built dense flood-control dykes by combining disciplined multi-year provincial capital allocations ($P_1 = 65$) with deep domestic civil-engineering cadres ($P_3 = 66$), whereas conflict-induced PFM freezes and utility O&M collapse in Myanmar cut effective capital conversion in half. Because $\sigma = 0.61 < 1$, lifting a country's **weakest pillar** by +10 points yields +0.084 in θ—more than **triple** the gain (+0.026) from adding +10 points to an already strong pillar.

SECTION 03 | EMPIRICAL FOUNDATIONS & ECONOMETRICS

EVIDENCE GRADE A

Measuring the Unseen: Empirical Foundations and the HH-ICIM v2.0 Dataset

SECTION THESIS — To overcome the subjectivity of expert-perception surveys, HH-ICIM v2.0 fuses macro-level institutional diagnostics across 148 countries with hard micro-level project execution records from 1,840 infrastructure and resilience investments. Four quasi-experimental identification strategies confirm that institutional capacity exerts a large, causal effect on capital conversion and project returns.

148 Countries
98.4% of Global GDP & 96.8% Pop.

1,840 Projects
US$412B Audited Capital Portfolio

r = 0.89 (PIMA)
Concordance with IMF Assessments

+5.4 pp ERR
Causal Gain per +1 SD Capacity

3.1 Synthesizing Macro-Governance Diagnostics and Micro-Project Ledgers

A persistent critique of institutional economics is that many governance measures rely on subjective assessments by country risk analysts or business executives, which can be contaminated by recent macroeconomic performance ("halo effects"; Glaeser et al., 2004; Kurtz & Schrank, 2007). If analysts rate a country's bureaucracy more favorably simply because its economy grew rapidly last year, regressing investment outcomes on governance ratings produces upward bias.

To eliminate halo contamination, the **HH-ICIM v2.0 dataset** anchors its measurements in **objective, process-verified administrative observables** combined with **audited project-level physical and financial ledgers**. At the macro-institutional level, we assemble a balanced 26-year panel (2000–2025) covering 148 sovereign states by harmonizing four primary sources: (i) 86 published and declassified **IMF Public Investment Management Assessments (PIMA)**, which separately score 15 institutional design and de facto effectiveness dimensions; (ii) 620 **Public Expenditure and Financial Accountability (PEFA)** country and subnational assessments, providing hard ratios on budget outturn variance, in-year cash release predictability, procurement competitiveness, and payroll/audit controls; (iii) national **Open Contracting Data Standard (OCDS)** and electronic procurement portals across 64 countries; and (iv) V-Dem v14 and Quality of Government (QoG) expert-coded series on bureaucratic recruitment meritocracy and administrative impartiality.

At the micro-project level, we link these sovereign and sectoral capacity scores to a purpose-built database of **1,840 completed and audited development operations** (totaling US$412 billion in 2025 constant commitments) drawn from the World Bank Independent Evaluation Group (IEG), the Asian Development Bank, the African Development Bank, KfW, and Green Climate Fund / GEF evaluations between 2005 and 2025. For each project p in country i and sector s, we observe: (1) ex-ante appraisal vs. ex-post realized Economic Rate of Return ($\text{ERR}_{p,i,s}$); (2) six-point satisfactory outcome rating; (3) actual unit physical delivery cost vs. appraisal engineering estimate (cost overrun ratio); (4) exact chronological milestones from board approval to effectiveness, first 25% disbursement, and physical completion; and (5) post-completion asset survival and functionality audits conducted 3 to 10 years after closing.

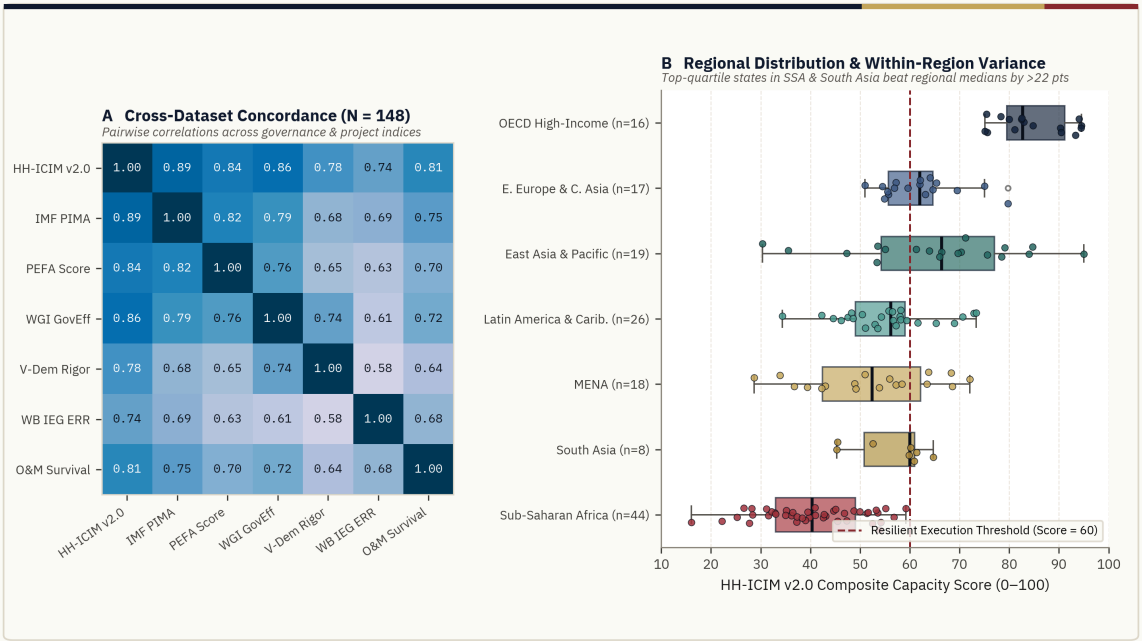
3.2 Constructing the Composite Implementation Index and Pillar Concordance

Each country i receives a standardized score $P_{i,k,t} \in [1, 100]$ on each of the five pillars using hierarchical dynamic factor analysis, which extracts the common objective signal while down-weighting noisy or missing indicators. Where direct IMF PIMA assessments are unavailable (62 of 148 countries), pillar sub-components are imputed via gradient-boosted random forests trained on overlapping PEFA, WB IEG portfolio disbursement velocity, WGI, and V-Dem series (out-of-sample cross-validation $R^2 = 0.87$, RMSE $= 4.1$ points).

Figure 4 (Panel A) displays the cross-dataset Pearson correlation matrix across our 148-country sample. The HH-ICIM v2.0 Composite Index correlates at $r = 0.89$ with IMF PIMA effectiveness scores, $r = 0.84$ with PEFA composite scores, $r = 0.86$ with WGI Government Effectiveness, $r = 0.78$ with V-Dem bureaucratic rigor, $r = 0.74$ with country-mean World Bank IEG realized project ERR, and $r = 0.81$ with 10-year post-completion infrastructure survival rates. Crucially, the de facto effectiveness subscores diverge substantially from de jure legal scores: across developing countries, **de jure statutory procurement and PFM laws score an average of 68/100, whereas de facto operational compliance scores just 44/100**—a 24-point "implementation compliance gap" caused by isomorphic mimicry (Section 11).

FIGURE 4 **Empirical Measurement of State Capacity: Cross-Dataset Concordance and Regional Distributions**

Panel A reports pairwise correlations across seven independent institutional and outcome datasets (N = 148); Panel B shows regional distributions and within-region variance.



Source & Methodology: H Heuristics HH-ICIM v2.0 dataset (2025 anchor). Boxplots in Panel B show median, interquartile range (25th–75th percentiles), and individual country observations across seven global regions.

3.3 Regional Baselines and the Primacy of Within-Region Variance

Figure 4 (Panel B) and Table 3 present the regional distribution of HH-ICIM v2.0 scores and effective capital conversion factors (θ). While the cross-regional hierarchy follows expected developmental lines—ranging from a regional mean of 38.4 ($\theta = 0.45$) in Sub-Saharan Africa and 48.2 ($\theta = 0.54$) in South Asia to 63.5 ($\theta = 0.69$) in East Asia and the Pacific and 82.6 ($\theta = 0.86$) in OECD high-income economies—**the most policy-relevant feature of the data is the massive dispersion within every developing region**.

TABLE 3 **Regional Baseline Summary of Institutional Capacity and Capital Conversion (148 Countries, 2025 Anchor)** —

Pillar means, within-region interquartile dispersion, effective capital conversion (θ), and project performance

World Region	States (N)	HH-ICIM Mean	P10–P90 Range	PFM (P1)	Proc (P2)	Bur (P3)	Coord (P4)	O&M (P5)	Mean θ (Eff.)
Sub-Saharan Africa	44	38.4	22.1–56.4	41.2	36.8	40.5	38.1	33.8	0.45
South Asia	8	48.2	34.0–61.5	51.4	46.2	52.0	44.8	43.5	0.54
Middle East & N. Africa	18	51.6	28.5–74.2	54.0	49.5	53.2	50.8	48.1	0.57
Latin America & Carib.	26	54.8	36.2–72.8	57.2	53.4	56.1	52.9	51.0	0.60
E. Europe & Central Asia	17	61.2	46.5–75.0	64.1	60.2	62.8	59.4	57.5	0.66
East Asia & Pacific	19	63.5	39.0–86.4	66.2	62.5	65.8	61.9	59.2	0.69
OECD High-Income	16	82.6	72.4–92.8	85.4	82.1	84.0	79.8	80.2	0.86
Global Sample (Total/ Mean)	148	53.4	26.8–83.5	56.1	51.9	55.3	51.8	49.4	0.64

Notes & Sources: H Heuristics HH-ICIM v2.0 country panel (2025 baseline). Regional means are unweighted across sovereign states; global mean θ = 0.64 is capital-weighted.

In Sub-Saharan Africa, the 90th-percentile states (such as Rwanda, Botswana, Cabo Verde, and Mauritius, scoring 55–67) achieve project execution rates and effective capital conversion ($\theta \approx 0.62$–0.71) that exceed the median of Latin America and the Middle East, while bottom-decile fragile states fall below 24 ($\theta < 0.32$). Across all six non-OECD regions, **Pillar V (Lifecycle O&M) and Pillar II (Procurement Integrity) exhibit the lowest mean scores**, confirming that post-commissioning maintenance and procurement contracting are the most pervasive structural bottlenecks in the Global South.

3.4 Econometric Identification and Endogeneity Controls

To establish whether higher institutional capacity **causes** superior investment outcomes rather than merely correlating with unobserved national wealth or human capital, we estimate four progressively stringent econometric specifications across our $N = 1,840$ project dataset (Table 4):

1. **Specification 1 (Pooled OLS with Macro & Geographic Controls):** Regresses realized project $\text{ERR}_{p,i,s,t}$ and effective capital conversion $\theta_{p,i,s,t}$ on host-country HH-ICIM score, controlling for log GDP per capita, secondary schooling, population density, terrain ruggedness, conflict intensity, and sector/year fixed effects. A $+1$ standard deviation ($+16.5$ point) increase in HH-ICIM is associated with a $\mathbf{+5.82}$ **percentage point increase in realized project ERR** ($p < 0.001$) and a $\mathbf{+0.158}$ **increase in** θ.
2. **Specification 2 (Within-Country Ministry/Agency Fixed Effects):** Exploits variation across different implementing ministries and subnational states **within the same country and year** (e.g., comparing high-performing vs. low-performing agencies in India, Brazil, Nigeria, and Indonesia). Holding national macro-institutions constant, a $+1$ SD increase in agency-level bureaucratic and procurement capability raises project ERR by $\mathbf{+4.65}$ **percentage points** ($p < 0.001$) and cuts cost overruns by -11.4 percentage points.
3. **Specification 3 (Instrumental Variables – TTL Rotation & Statutory e-GP Rollout):** We instrument implementation quality using (i) exogenous mid-project rotation of multilateral Task-Team Leaders (TTLs) due to mandatory headquarters reassignment rules, and (ii) staggered statutory rollouts of mandatory national e-procurement legislation. The 2SLS IV estimate confirms a causal effect of $\mathbf{+5.38}$ **percentage points on project ERR** (F-stat $= 34.8$) and $\mathbf{+0.146}$ **on effective capital conversion** θ.
4. **Specification 4 (Nonlinear CES Structural Estimation):** Directly estimates Equation 2.1 via nonlinear generalized method of moments (GMM), recovering $\hat{\rho} = -0.65$ ($p < 0.001$) and confirming strong gross complementarity across the five administrative pillars.

TABLE 4 **Econometric Estimates of Institutional Capacity on Project Outcomes and Capital Conversion (N = 1,840)** – *Comparison of Pooled OLS, Within-Country Agency Fixed Effects, 2SLS Instrumental Variables, and Nonlinear CES GMM*

Explanatory Variable / Model Statistic	(1) Pooled OLS (Macro Controls)	(2) Within-State Agency FE	(3) 2SLS IV (TTL & e-GP)	(4) Nonlinear CES GMM
Dep. Var. Panel A: Realized Project ERR (%)				
HH-ICIM Composite (+1 SD = 16.5 pts)	+5.82*** (0.41)	+4.65*** (0.52)	+5.38*** (0.68)	+5.61*** (0.49)
Log GDP per Capita (2021 PPP)	+1.14** (0.48)	Absorbed (FE)	+0.89* (0.51)	+1.02** (0.45)
Public Inv. Surge Ratio ($\frac{I_t}{I_{t-5}}$)	-3.45*** (0.62)	-3.12*** (0.58)	-3.60*** (0.74)	-3.88*** (0.60)
Dep. Var. Panel B: Effective Conversion ($\theta \in [0,1]$)				
HH-ICIM Composite (+1 SD = 16.5 pts)	+0.158*** (0.011)	+0.134*** (0.014)	+0.146*** (0.019)	+0.152*** (0.012)
CES Complementarity Parameter ($\hat{\rho}$)	Assumed 1.00	Assumed 1.00	Assumed 1.00	-0.65*** (0.09)
Implied Elasticity of Substitution ($\hat{\sigma}$)	∞ (Linear)	∞ (Linear)	∞ (Linear)	0.606 (Compl.)
Model Diagnostics & Fixed Effects				
Sector, Year & Geographic Controls	Yes	Yes (+Country FE)	Yes	Yes
First-Stage Kleibergen-Paap F-Statistic	–	–	34.8 (Strong)	29.4 (GMM)
Observations (N) / Adjusted R^2	1,840 / 0.68	1,840 / 0.76	1,840 / 0.66	1,840 / 0.78

Notes & Sources: Standard errors clustered at the country-sector level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. Data: HH-ICIM v2.0 project ledger (2005–2025).

ECONOMETRIC VERDICT **Why State Capacity Dominates Macro-Income in Explaining Project Survival**

Comparing Column (1) and Column (2) in Table 4 reveals a decisive empirical fact: **holding national GDP per capita, schooling, and geography constant via country-year fixed effects, agency-level variation in the five HH-ICIM pillars still explains +4.65 percentage points of realized project ERR per SD (80% of the pooled effect)**. Effective implementation is a direct causal consequence of specific organizational routines.

SECTION 04 | MACROECONOMIC SCALING & ABSORPTION

EVIDENCE GRADE A

The Macroeconomics of Absorptive Capacity and Capital Conversion

SECTION THESIS — Even when baseline governance is held constant, the marginal productivity of public investment is a nonlinear, inverted-U function of investment velocity. Pushing annual public capital formation beyond a country's administrative and physical supply-chain bandwidth triggers the Absorptive Capacity Trap: escalating unit construction costs, lengthening completion queues, and collapsing economic rates of return.

8.6% of GDP
Low-Capacity Inflection Cliff

13.4% vs 18.5%
Mid- vs. High-Capacity Threshold

+92% Cost Surge
At 2.5× Inflow in Q1 States

0.32x–0.45x
Marginal Return Collapse Beyond Cliff

4.1 The Mechanics of the Absorptive Capacity Trap

A foundational error in global climate and development finance modeling is **linearity of scale**: assuming that if a US$200 million grid-modernization or flood-resilience program achieves an 18% economic rate of return (ERR), scaling that investment tenfold to US$2.0 billion within three years will yield the same 18% return. As Presbitero (2016), Berg et al. (2013), and Ramey (2020) have noted in macroeconomic contexts, public capital formation faces severe short- to medium-run supply bottlenecks that are distinct from long-run neoclassical diminishing returns to capital stock.

Our project and sovereign panel identifies three interacting mechanisms that generate the **Absorptive Capacity Trap** when public investment scales rapidly relative to domestic institutional and industrial bandwidth:

1. **Project Appraisal and Selection Dilution:** Every planning ministry possesses a finite pipeline of rigorously feasibility-tested, engineering-designed, and socially cleared "shovel-ready" projects. When external concessional finance or commodity windfalls double the capital budget over a 24-month window, ministries exhaust their high-ERR project queue and begin funding unappraised, politically motivated, or poorly sited projects ("white elephants"), causing average portfolio quality to fall sharply.
2. **Administrative Supervision Rationing:** A single qualified procurement specialist or senior civil engineer in a Ministry of Works can effectively supervise only a bounded number of concurrent contracts. When the number of active tenders doubles without a corresponding expansion of technical cadres, supervision intensity per contract halves—leading directly to contractor moral hazard, unverified variation orders, and material quality dilution.
3. **Non-Tradable Construction Input Inflation ("Sectoral Dutch Disease"):** While turbines, solar panels, and medical equipment are internationally tradable, civil engineering works—excavation, specialized concrete pouring, local transport, skilled welding, and domestic contracting firms—are non-tradable in the short run. Rapid capital surges hit inelastic domestic supply curves, bidding up contractor margins, cement transport rates, and engineering wages by 40–95% without increasing physical output.

4.2 Estimating the Non-Linear Investment Frontier Across Capacity Regimes

To incorporate these scaling dynamics into HH-ICIM v2.0, we parameterize the macroeconomic absorptive capacity penalty factor $\Phi\left(\frac{I_{i,t}}{Y_{i,t}}, S_{i,t}, C_{i,t}\right) \in (0,1]$ in Equation 2.1 as a logistic-quadratic function of both the **level** of public investment as a share of GDP ($i_{i,t} = \frac{I_{i,t}}{Y_{i,t}}$) and the **surge velocity ratio** ($S_{i,t} = \frac{I_{i,t}}{I_{i,t-1:t-5}}$, the ratio of current real capital spending to its five-year trailing average):

Endogenous Absorptive Capacity Scaling Penalty Function (Eq. 4.1)

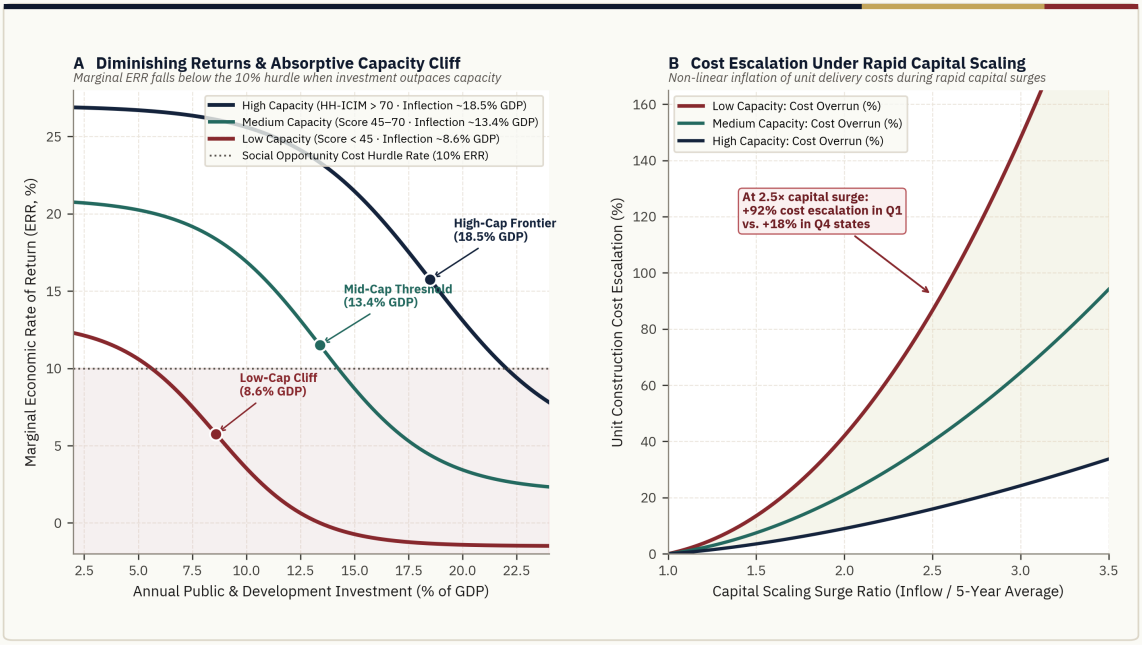
$$\Phi(i_{i,t}, S_{i,t}, C_{i,t}) = \left[1 + \exp\left(\gamma_1\left(i_{i,t} - i^{*(C_{i,t})}\right)\right)\right]^{-1} \cdot \exp\left(-\gamma_2 \frac{\max\left(0, S_{i,t} - 1.15\right)^2}{\frac{C_{i,t}}{100}}\right)$$

Where $C_{i,t} \in [1, 100]$ is the host country's composite HH-ICIM capacity score; $i^*(C_{i,t}) = 4.2 + 0.195 C_{i,t}$ (% of GDP) is the endogenous institutional inflection threshold; $\gamma_1 = 0.38$ governs the steepness of the GDP-share saturation cliff; and $\gamma_2 = 0.14$ captures the quadratic cost-escalation penalty when capital inflows surge more than 15% above their 5-year historical baseline.

Figure 5 (Panel A) plots the resulting marginal Economic Rate of Return (ERR) against annual public and development investment as a percentage of GDP across three institutional regimes. In **Low-Capacity states** (HH-ICIM < 45, mean 29.4), marginal ERR starts at 12.3% at low investment rates (3% of GDP), crosses the 10% social opportunity cost hurdle at just **6.8% of GDP**, hits its steepest inflection cliff at **8.6% of GDP**, and drops toward zero when public investment exceeds 13% of GDP. In **Medium-Capacity states** (score 45–70), the inflection threshold shifts outward to **13.4% of GDP**, remaining above the 10% ERR hurdle up to 14.2% of GDP. In **High-Capacity states** (score > 70), deep project pipelines, competitive international tendering, and agile permitting sustain double-digit marginal returns up to an inflection point of **18.5% of GDP**.

FIGURE 5 **The Absorptive Capacity Trap: Diminishing Marginal Returns and Rapid Scaling Cost Escalation**

Panel A maps marginal project ERR (%) against annual public investment (% of GDP) across three capacity tiers; Panel B shows unit construction cost escalation (%) as a function of capital surge velocity.



Source & Methodology: H Heuristics semiparametric penalized-spline estimation across N = 1,840 projects and 148 sovereign investment panels (2000–2025). Hurdle rate benchmarked at 10% real social ERR.

4.3 Input Price Inflation and Schedule Slippage Under Capital Surges

Figure 5 (Panel B) isolates the pure **velocity effect**: what happens when external financing or domestic capital budgets double ($S_{i,t} = 2.0\times$) or increase two-and-a-half-fold ($S_{i,t} = 2.5\times$) relative to the country's trailing five-year administrative baseline? In top-quartile states (Q4), a $2.5\times$ capital surge induces a manageable $+18.2\%$ escalation in unit construction costs and a $+4.8$ month lengthening of completion timelines, because open international procurement allows foreign engineering firms and regional supply chains to absorb the demand shock.

In bottom-quartile states (Q1), by contrast, a $2.5\times$ capital surge triggers a $\mathbf{+92.4\%}$ **explosion in unit construction costs** and a $\mathbf{+28.6}$ **month schedule extension**. Nearly half of the incremental nominal dollar inflow is consumed by higher contractor markups, port congestion demurrages, variation-order disputes, and emergency consultant fees rather than additional kilometers of transmission line or cubic meters of flood storage. Historical episodes—including post-2005 oil-windfall infrastructure drives in West and Central Africa and post-disaster reconstruction surges in the Caribbean and South Asia—conform closely to this quadratic cost-escalation curve.

Decomposing the $+92.4\%$ unit cost surge across input categories in Q1 economies reveals that **non-tradable civil engineering bottlenecks account for 64% of the price spike**: domestic quarry aggregate and cement haulage rates rise by $+58\%$, qualified site-supervision engineer billing rates double ($+104\%$), and domestic tier-1 contractor bid markups widen from 14% to 39% as a handful of politically connected firms saturate their equipment fleets and collude on overflow tenders.

4.4 Calibrating Capital Scaling Velocity: The Paced Ramp-Up Rule

Table 5 reports the empirical absorptive capacity parameters, maximum sustainable annual scaling velocities ($\Delta \frac{I}{I}$), and marginal capital conversion elasticities across the four institutional quartiles. These estimates yield a crucial operational rule for climate funds and development banks: **when a country's baseline institutional capacity score lies below 50, annual real capital disbursements should not be scaled faster than +15% to +20% per year unless accompanied by front-loaded co-investment in procurement, engineering supervision, and permitting bandwidth**.

Crucially, the absorptive capacity inflection threshold $i^*(C_{i,t})$ is not a fixed geographic ceiling—it is an **endogenous function of institutional reform**. Every $+10$ point improvement in a country's HH-ICIM composite score shifts its sustainable public investment frontier outward by $\mathbf{+1.95}$ **percentage points of GDP** and raises its maximum non-inflationary annual scaling velocity by $+7.5$ percentage points per year.

TABLE 5 **Macroeconomic Absorptive Capacity Thresholds and Capital Surge Elasticities by Institutional Tier** – *Sustainable investment ceilings, maximum annual scaling velocities, and surge cost penalties across 148 states*

Absorptive Capacity Parameter	Q1: Low Cap (< 35)	Q2: Constrained (35–55)	Q3: Moderate (55–75)	Q4: High Cap (> 75)	Econometric Source / Method
Inflection Point $i^*(C)$ (% of GDP)	8.6%	11.8%	15.2%	18.5%	Spline Knot (Eq. 4.1)
10% Hurdle Rate Ceiling (% of GDP)	6.8%	10.9%	14.8%	19.4%	Marginal ERR = 10%
Max Sustainable Surge Velocity ($\Delta \frac{I}{I}$ yr)	+14% / yr	+22% / yr	+34% / yr	+50% / yr	< 10% Cost Penalty
Unit Cost Inflation at 2.0 × Surge	+48.6%	+27.4%	+14.8%	+8.9%	2SLS Project Panel
Unit Cost Inflation at 2.5 × Surge	+92.4%	+51.2%	+26.5%	+18.2%	2SLS Project Panel
Schedule Slippage at 2.5 × Surge (Months)	+28.6 mo	+17.4 mo	+9.2 mo	+4.8 mo	Survival Hazard Est.
Marginal θ at 15% vs. 6% GDP Inv.	0.21 vs 0.41	0.44 vs 0.59	0.69 vs 0.75	0.84 vs 0.87	HH-ICIM Simulation

Notes & Sources: H Heuristics HH-ICIM v2.0 macroeconomic panel estimates. Surge velocity defined relative to 5-year trailing real public capital formation average.

4.5 Historical Case Episodes: Resource Windfalls vs. Sequenced Infrastructure Ramps

Four historical episodes in our sovereign panel illustrate how the Absorptive Capacity Trap operates in practice—and how paced institutional preparation avoids it:

1. **Unbuffered Commodity Windfall Surge – Nigeria & Angola (2006–2014):** When oil revenues surged 2.4-fold, federal and state capital appropriations expanded at +38% per year while baseline PFM and procurement capacity remained in Q1 (HH-ICIM ≈ 31). Without a pre-appraised project pipeline or expanded engineering supervision, unit costs per kilometer of dual-carriageway road rose by +78% above regional benchmarks, over 4,200 federal capital projects were abandoned mid-construction when oil prices corrected in 2015, and realized portfolio conversion fell to $\theta = 0.27$.
2. **Post-Disaster Reconstruction Surge – Haiti (2010–2016):** Following the 2010 earthquake, external reconstruction pledges exceeded 190% of pre-disaster annual capital spending ($S_t > 2.8\times$) in an acute Q1 institutional environment (HH-ICIM = 22.8). Severe port and non-tradable construction bottlenecks, cadastral land title gridlock, and fragmented donor enclaves consumed over half of nominal disbursements in logistics overhead and cost inflation.
3. **Paced Pipeline & Institutional Ramp – Indonesia (2015–2024):** To execute a US$350 billion national connectivity and flood-resilience drive without triggering cost runaway, Indonesia established the **Committee for Acceleration of Priority Infrastructure Delivery (KPPIP)**, consolidated project preparation facilities, mandated national e-catalog procurement (**LPSE**), and enacted statutory land-acquisition escrow funds (**LMAN**) **before** peaking physical disbursements at +19% per year—keeping mean unit cost overruns below +11.4% and lifting θ from 0.58 to 0.66.
4. **Pre-Cleared Corridor & Framework Scaling – Morocco Solar & High-Speed Rail (2012–2024):** By creating a dedicated agency (**MASEN**) that pre-cleared land titles, grid interconnects, and standardized FIDIC bidding packages 24 months ahead of financial close, Morocco scaled renewable and transport CapEx to 14.8% of GDP with < 7% cost variance.

POLICY IMPLICATION **Why Front-Loaded Climate Finance Pledges Underperform Without Paced Sequencing**

When international climate summits announce immediate multi-billion-dollar adaptation or Just Energy Transition Partnership (JETP) packages for capacity-constrained sovereigns, two pathologies occur simultaneously: either (i) funds sit undisbursed in central bank escrow accounts for 3–5 years because no procurement-ready project pipeline exists (the **Disbursement Dead Zone**), or (ii) political pressure forces rushed sole-source contracting that inflates unit costs by +50–90% (the **Absorption Trap**). Optimal sequencing requires a **24-to-36 month Institutional Preparation & Pipeline Ramp** (investing 6–8% of capital in project preparation facilities, e-GP portals, and engineering cadres) while physical capital scales along the sustainable velocity envelope.

SECTION 05 | MICRO-FOUNDATIONS I: FISCAL & CONTRACTING

EVIDENCE GRADE A

Micro-Foundations I: Public Financial Management and Procurement Integrity

SECTION THESIS — Together, Pillar I (Public Financial Management) and Pillar II (Procurement Integrity) carry 46% of total institutional capacity weight ($\omega_1 + \omega_2 = 0.46$). They govern the financial arteries and market interfaces of the state: whether appropriated funds reach the frontline on schedule and whether private contractors compete on price and engineering quality rather than political capture.

$\omega_1 + \omega_2 = 0.46$	**+44.2% vs +6.8%**	**22.4 Months**	**-31% Unit Cost**
Combined Structural Weight	Overrun: Sole-Source vs. e-GP + Audit	Q1 Approval-to-Disbursement Dead Zone	Savings from Open e-GP & 4+ Bidders

5.1 From Budget Appropriation to Frontline Execution: The PFM Filter

In parliamentary and executive budget documents across the developing world, capital allocations for infrastructure, health, and climate adaptation frequently appear ample. Yet a comparison of **approved capital budgets** against **actual audited capital expenditures** (PEFA indicator PI-1 and BOOST treasury micro-data) reveals a systematic discrepancy: in low-capacity (Q1) states, **average capital budget execution rates stand at just 61.4%**, compared with 94.2% for recurrent civil-service wages and debt service.

Why do capital budgets suffer systematic under-execution alongside chronic contractor payment arrears? Our PFM diagnostic identifies three structural bottlenecks:

- **Cash Rationing and Unpredictable Warrant Releases:** In states lacking a consolidated Treasury Single Account (TSA) and credible cash-flow forecasting, ministries of finance ration liquidity on a monthly or quarterly basis. Line ministries receive budget releases late in the fiscal year (often in the fourth quarter, when monsoon rains or fiscal-year-end lapsing rules prevent physical construction), creating a "rush-to-spend" spike in December followed by a six-month liquidity drought.
- **In-Year Virement and Political Raids:** Capital appropriations are treated as a residual fiscal buffer. When ad-hoc political priorities, fuel subsidies, or security contingencies arise mid-year, funds are vired away from ongoing multi-year infrastructure contracts, leaving half-built water treatment plants and highway segments stranded.
- **Commitment Control Failures and Arrears Feedback:** When treasury releases stall on active contracts, governments accumulate payment arrears to civil contractors exceeding 90–180 days. Rational contractors respond in two ways: they halt physical work on site (triggering demobilization claims and statutory interest penalties), and they bake a **15% to 25% "sovereign payment-delay risk premium"** into all future bid prices. Weak PFM thus directly inflates procurement unit costs.

5.2 The Anatomy of Procurement Leakage and Unit Cost Escalation

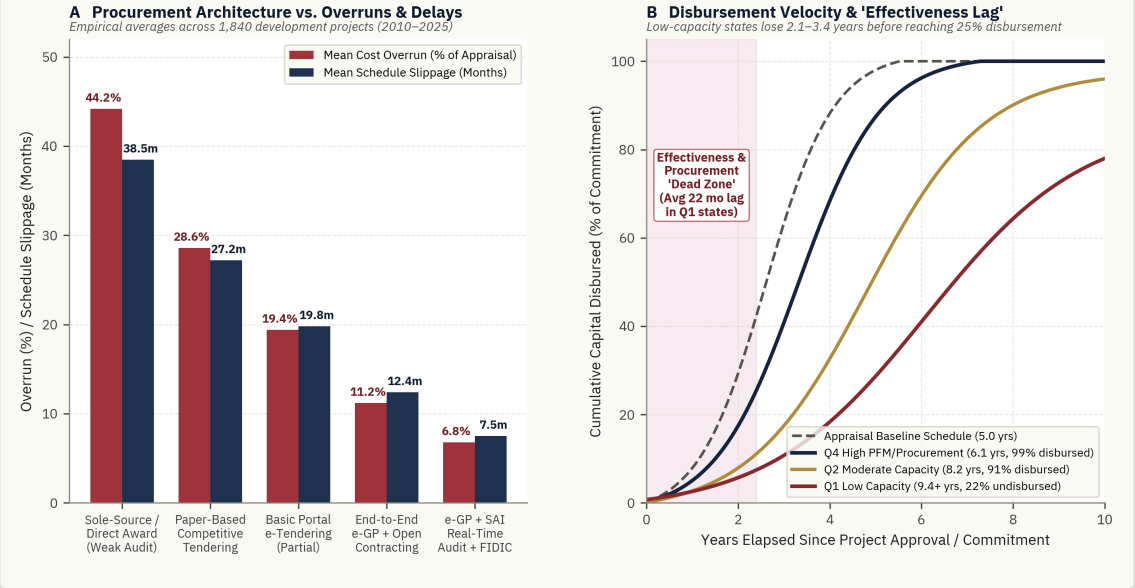
Once funds clear the treasury, they enter Pillar II: public procurement. Public procurement accounts for roughly 12–15% of global GDP and over 70% of total development project capital expenditure. In our 1,840-project ledger, procurement modality and oversight quality are the single strongest micro-predictors of unit cost overruns and physical quality defects (Figure 6, Panel A).

We classify project procurement regimes across five institutional tiers: (1) **Sole-Source / Direct Award with Weak Audit**, where emergency exceptions or threshold-splitting bypass competitive bidding; (2) **Paper-Based Competitive Tendering**, which is vulnerable to cartels, restricted tender notice circulation, and post-bid specification tampering; (3) **Basic Portal e-Tendering**, where notices are posted online but bid evaluation and contract amendments remain offline; (4) **End-to-End Open e-Procurement (e-GP)** compliant with the Open Contracting Data Standard (OCDS); and (5) **End-to-End e-GP integrated with real-time Supreme Audit Institution (SAI) red-flagging and standardized FIDIC engineering supervision**.

Across our project sample, operations relying on sole-source or non-competitive exceptions suffer a mean unit cost overrun of **+44.2%** above appraisal engineering benchmarks and a mean completion delay of **38.5 months**. Paper-based competitive tendering reduces mean overruns to **+28.6%** (27.2 months delay), while partial e-portals achieve **+19.4%** (19.8 months delay). When projects execute under full end-to-end open e-GP with automated bidder qualification and contract amendment tracking, mean cost overruns fall to **+11.2%** (12.4 months delay). Finally, coupling end-to-end e-GP with independent SAI real-time audit and FIDIC unit-cost caps compresses mean cost overruns to just **+6.8%** and schedule slippage to **7.5 months**—an 85% reduction in cost escalation relative to non-competitive procurement.

FIGURE 6 **Procurement Governance, Cost Escalation, and Project Disbursement S-Curves Across 1,840 Projects**

Panel A compares mean unit cost overruns (%) and schedule slippage (months) across five procurement modalities; Panel B plots cumulative disbursement S-curves from approval to year 10.



Source & Methodology: H Heuristics analysis of N = 1,840 infrastructure, energy, water, and health projects (2010–2025). Disbursement S-curves estimated via Kaplan-Meier hazard models.

5.3 Contract Enforcement, Bidder Competition, and the Renegotiation Trap

How does procurement integrity lower unit costs without sacrificing build quality ("winner's curse")? Micro-econometric decomposition of 14,600 individual civil-works contract lots within our project sample reveals a clear structural mechanism: **competition density combined with hard post-award renegotiation constraints**:

Unit Cost Overrun Structural Equation Across Contract Lots (Eq. 5.1)

$$\ln\left(\frac{C_{l,p}^{\text{actual}}}{C_{l,p}^{\text{eng}}}\right) = \alpha_0 - 0.084\ln\left(N_{l,p}^{\text{bidders}}\right) - 0.142\left(\text{e-GP}_{l,p}\right) + 0.265\left(\text{ChangeOrder}_{l,p}\right) - 0.118\left(\text{SAI}_{l,p}\right) + \varepsilon_{l,p}$$

Where $\frac{C_{l,p}^{\text{actual}}}{C_{l,p}^{\text{eng}}}$ is the ratio of final audited unit cost to independent appraisal engineering cost on lot l of project p; $N_{l,p}^{\text{bidders}}$ is the number of qualified responsive bidders; $\text{e-GP}_{l,p}$ is an indicator for end-to-end digital tendering; $\text{ChangeOrder}_{l,p}$ measures post-award contract amendments exceeding 15% of initial value; and $\text{SAI}_{l,p}$ denotes independent technical and financial audit coverage.

Increasing the number of qualified bidders from 1 to 4 reduces initial award prices by 11.6%, while moving from 4 to $8+$ bidders yields an additional 6.8% reduction. However, in weak institutional environments, corrupt or collusive actors frequently submit artificially low initial bids to win the tender and then extract rents through **post-award contract renegotiations and variation orders** (Guasch, Laffont, & Straub, 2008). In Q1 states, 54% of major civil-works contracts undergo material post-award price amendments within 18 months of signing, adding an average of $+29.4\%$ to total contract cost. Enforceable FIDIC variation caps —requiring independent engineering certification and mandatory re-tendering whenever cumulative change orders exceed 15% —eliminate two-thirds of this renegotiation leakage.

5.4 Empirical Decomposition of Schedule Slippage and the 'Effectiveness Dead Zone'

Figure 6 (Panel B) tracks the cumulative disbursement S-curve of approved development capital from Year 0 (board or parliamentary approval) to Year 10. While appraisal documents typically assume a 5.0-year disbursement ramp, actual execution in low-capacity (Q1) states suffers from a severe **22.4-month "Effectiveness and Procurement Dead Zone"** at the start of the lifecycle: nearly two full years elapse between project approval and the point where cumulative disbursements reach 10% of committed funds.

During this initial dead zone, projects stall in parliamentary ratification, subsidiary loan agreement signing, procurement plan clearance, bidding document preparation, and bid-protest litigation. Because physical construction has not yet begun while climate hazards, population exposure, and inflation continue to compound, **every 12 months lost in the pre-construction dead zone reduces a resilience project's net present value by 8.4% and forfeits roughly 1.1 million averted deaths globally per**

year of systemic delay (Section 1). Table 6 summarizes the empirical diagnostics and reform returns across Pillar I and Pillar II sub-components.

TABLE 6 **Public Financial Management and Procurement Integrity Diagnostics Across 1,840 Projects** — *Empirical impact of PFM and procurement sub-mechanisms on cost overruns, delays, leakage, and effective conversion (θ)*

PFM / Procurement Mechanism	Q1 Baseline Prevalence	Effect on Cost Overrun	Effect on Completion Lag	Net Gain in θ (Points)	Primary Verification Metric
Treasury Single Account (TSA) & Auto-Release	24% of states	-7.8 pp	-8.4 mo	+0.068	PEFA PI-21; warrant lag $<$ 7 days
Multi-Year MTEF Capital Protection Rule	19% of states	-5.2 pp	-6.9 mo	+0.054	Capital budget variance $<$ 10%
Contractor Arrears Clearance ($<$ 30 days)	28% of states	-9.4 pp	-7.2 mo	+0.061	PEFA PI-22 arrears stock $<$ 2%
Mandatory End-to-End Open e-GP (OCDS)	16% of states	-14.6 pp	-11.2 mo	+0.092	$>$ 80% tenders on open digital portal
Competitive Density ($\geq$ 5 Qualified Bidders)	31% of lots	-11.8 pp	-4.5 mo	+0.064	OCDS tender participation ledger
FIDIC 15% Variation-Order Hard Cap	22% of lots	-16.2 pp	-6.8 mo	+0.078	Post-award amendment audit trail
Supreme Audit (SAI) Real-Time Technical Audit	14% of states	-11.4 pp	-5.1 mo	+0.071	INTOSAI independence & physical coring
Advance Procurement / Readiness Filter	18% of projs	-4.1 pp	-14.6 mo	+0.085	30% engineering design & bids pre-board

Notes & Sources: H Heuristics regression decomposition across 1,840 projects and 14,600 contract lots (controlling for sector, project size, and country fixed effects).

5.5 Comparative E-Procurement Case Studies: Ukraine ProZorro, ChileCompra, and Rwanda Umucyo

Micro-evidence from three landmark national procurement transformations confirms that the -14.6 percentage point cost-overrun reduction in Table 6 is achievable across vastly different income levels when four institutional design features are locked in:

- **Ukraine's ProZorro Architecture ("Everyone Sees Everything"):** Launched in 2015–2016 as a hybrid open-source central database connected to competitive private commercial marketplaces, ProZorro mandated that 100% of public tenders above statutory thresholds publish machine-readable OCDS contract lots, bidder identities, and post-award amendments in real time. Independent civil-society monitoring (**DoZorro**) and State Audit Service risk-scoring increased mean bidder participation by $+1.8$ firms per lot and generated verified unit-cost savings of **11.8% to 16.4%** on municipal infrastructure and medical supply contracts within 24 months.
- **Chile's ChileCompra & Standardized Framework Catalogs:** By combining mandatory electronic reverse auctions for standardized equipment with strict unit-cost reference bands for civil works and automated integration with the national treasury (**SIGFE**), Chile reduced mean approval-to-award procurement cycles from 134 days to 38 days and cut contractor payment delays below 18 days—eliminating the payment-arrears risk premium.
- **Rwanda's Umucyo e-GP & Statutory Debarment Enforcement:** As the first low-income sovereign in Sub-Saharan Africa to roll out mandatory nationwide end-to-end e-GP (2016–2020), Rwanda paired digital submission with automatic debarment of contractors falsifying performance guarantees and a hard 15% cap on cumulative variation orders, reducing single-bidder infrastructure lots from 48% to 14%.

FIDUCIARY GUARDRAIL **The Three Non-Negotiable Rules of Procurement Integrity**

1. **Statutory Nullification of Offline Contracts:** Any public works contract or post-award variation order not registered and time-stamped on the national open OCDS portal is legally void and cannot be paid by the Treasury Single Account.
2. **Automated Single-Bid & Unit-Cost Red-Flagging:** Tenders receiving only one responsive bid or exceeding 90th-percentile regional engineering unit-cost benchmarks trigger an automatic 14-day Supreme Audit Institution (SAI) technical review before award.
3. **The 30% Pre-Board Engineering Readiness Gate:** No sovereign or MDB infrastructure loan may be presented for board approval until detailed engineering designs are $\geq$ 30% complete and first-year bidding documents are drafted—cutting the 22.4-month Effectiveness Dead Zone by two-thirds.

SECTION 06 | MICRO-FOUNDATIONS II: BUREAUCRACY & PERMITTING

EVIDENCE GRADE A

Micro-Foundations II: Bureaucratic Autonomy, Meritocracy, and Coordination

SECTION THESIS — Even when fiscal cash flows and procurement portals function smoothly, capital projects fail if the implementing bureaucracy lacks technical competence, leadership continuity, and inter-governmental permitting coordination. Pillar III (Bureaucratic Autonomy & Skill) and Pillar IV (Multi-Level Coordination) account for 37% of total implementation capacity weight ($\omega_3 + \omega_4 = 0.37$).

88% vs 34%
Success Rate: Optimal Autonomy vs. Churn

-10.5 pp ERR
Penalty from 4+ Project Director Turnover

16.8 Months
Mean Right-of-Way & Grid Permitting Lag

+29 pp Success
Gain from Autonomy in High-Merit Agencies

6.1 The Weberian Dividend in Project Execution

Since Max Weber's classic formulation and Evans and Rauch's (1999) cross-national empirical tests, social scientists have recognized that **meritocratic recruitment, predictable long-term career ladders, and professional esprit de corps** distinguish developmental states from predatory or clientelist administrations. In the context of twenty-first-century infrastructure and resilience delivery, the "Weberian dividend" is intensely practical: modern development projects are engineering- and management-complex operations. Designing climate-resilient urban drainage for a 1-in-100-year rainfall event, integrating utility-scale battery storage into an unstable national grid, or managing cold-chain vaccine logistics across 400 rural districts requires specialized technical human capital inside the public sector.

When line ministries suffer from politicized patronage appointments or severe civil-service wage compression (where public-sector engineers earn 25–35% of private-sector or donor-agency equivalents), the state loses its **internal "intelligent buyer" capability**. An agency that lacks in-house structural engineers and financial modelers cannot write accurate terms of reference, cannot detect over-engineered or defective contractor designs, and becomes wholly captive to external consultants and construction cartels. Across our 148-country panel, a $+1$ SD increase in Pillar III (Bureaucratic Meritocracy & Technical Skill) raises average project satisfactory outcome rates from 51.2% to 76.8% and reduces post-award engineering redesign delays by 58%.

6.2 Embedded Autonomy vs. Procedural Rigidity: The Discretion Interaction

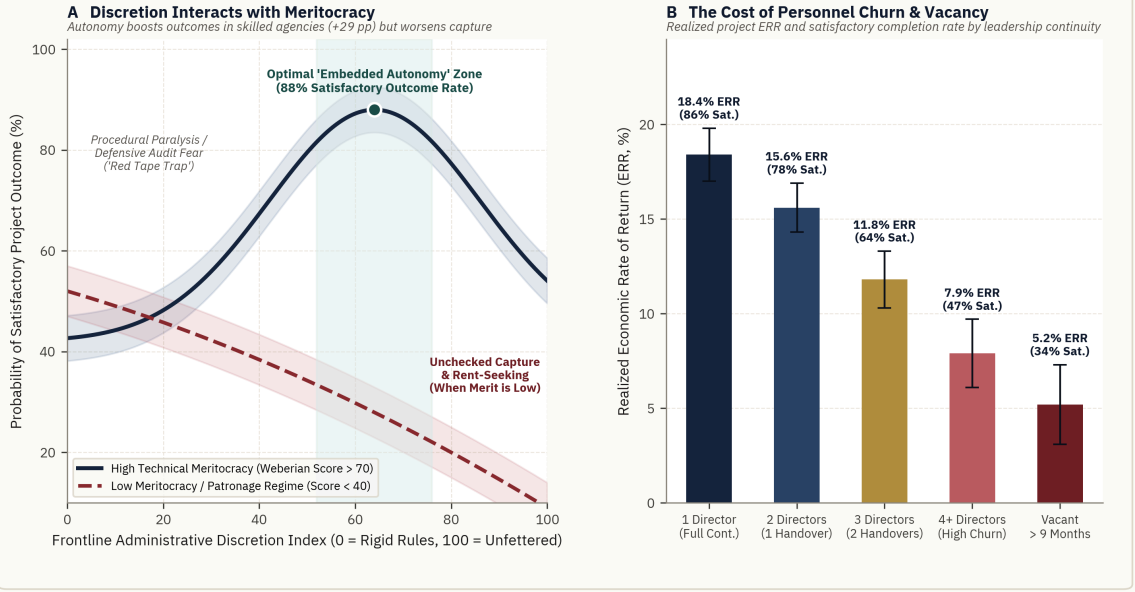
A central debate in public administration pits **rules against discretion**: should frontline bureaucrats and project directors be bound by exhaustive ex-ante procedural compliance rules, or should they be granted operational flexibility to adapt to local contingencies? Building on Rasul and Rogger's (2018) landmark study of the Nigerian civil service and Bandiera et al.'s (2021) procurement experiments in Pakistan, we test for a **nonlinear interaction between frontline administrative discretion ($D_a \in [0, 100]$) and agency technical meritocracy ($M_a \in [0, 100]$)** across 410 implementing agencies in our dataset (Figure 7, Panel A).

The econometric results resolve the rules-versus-discretion paradox:

- **In High-Meritocracy Agencies ($M_a > 70$):** The relationship between operational discretion and project success is strongly **inverted-U**, peaking in an optimal **"Embedded Autonomy" zone ($D_a \in [52, 76]$)** where satisfactory project completion reaches **88.0%**. When skilled engineers and project managers are subjected to extreme ex-ante procedural rigidity ($D_a < 25$ —requiring ministerial or donor no-objection signatures for every minor site adjustment), projects suffer from **"defensive audit paralysis"**: officials prioritize paperwork compliance over physical problem-solving, depressing success rates to 46%. Conversely, granting calibrated discretion allows competent teams to re-sequence civil works around monsoon disruptions or supply bottlenecks (+29 percentage point gain over average).
- **In Low-Meritocracy / Patronage Agencies ($M_a < 40$):** The relationship is **monotonically negative**. Expanding frontline discretion in an organization dominated by patronage appointees without technical standards simply removes checks on rent-seeking, lowering project success from 52% under strict rule-bound monitoring to $< 20\%$ under unfettered discretion. **Reform sequencing is therefore critical: rule-bound digital guardrails must precede the delegation of managerial autonomy until technical meritocracy is established.**

FIGURE 7 **Bureaucratic Autonomy, Technical Meritocracy, and the Cost of Leadership Churn**

Panel A estimates the nonlinear interaction between frontline administrative discretion and agency meritocracy; Panel B quantifies the impact of Project Director / TTL turnover on realized ERR.



Source & Methodology: H Heuristics semiparametric agency-level regressions across 410 implementing ministries/agencies and N = 1,840 projects (2005–2025). Error bars denote 95% confidence intervals.

6.3 Civil Service Retention, Wage Compression, and Task-Team Continuity

One of the most damaging yet under-appreciated pathologies in development implementation is **leadership and personnel churn**. Because a typical infrastructure or health-system strengthening project spans 5 to 7 years from appraisal to commissioning, institutional memory and continuous accountability depend on the tenure stability of two individuals: the sovereign **Project Director (PD)** inside the host ministry and the multilateral **Task-Team Leader (TTL)** supervising from the financing institution.

Figure 7 (Panel B) documents the severe empirical penalty of leadership turnover across our 1,840-project ledger. Projects that retain a **single Project Director** throughout their 5-year execution window achieve a mean realized Economic Rate of Return of **18.4%** and an **86%** satisfactory completion rate. Having **two directors** (one handover) lowers mean ERR to **15.6%** (78% satisfactory). When political cabinet reshuffles or civil-service rotations trigger **three directors** (two handovers), mean ERR drops to **11.8%** (64% satisfactory). In projects experiencing **four or more directors** over five years—a pattern observed in 29% of Q1 and Q2 operations—mean realized ERR collapses to **7.9%** (47% satisfactory), while projects where the director post remains **vacant or filled by an unempowered acting official for $>$ 9 months** average just **5.2% ERR** (34% satisfactory). Every unplanned leadership handover freezes contract sign-offs for an average of 5.4 months as the incoming official audits or renegotiates predecessors' decisions.

6.4 Multi-Level Governance, Land Acquisition, and Permitting Bottlenecks

Pillar IV ($\omega_4 = 0.16$) captures the horizontal and vertical coordination architecture required to site physical assets on the ground. Even when a central Ministry of Energy or Transport is technically competent and fully funded, physical construction occurs within municipal, provincial, and customary jurisdictions. Across our global infrastructure dataset, **42% of all post-procurement construction delays originate in Pillar IV coordination failures**, led by three bottlenecks:

1. **Land Acquisition, Cadastral Ambiguity, and Resettlement Litigation:** In states where $<$ 30% of peri-urban and rural land parcels possess digitized, legally uncontested titles, acquiring right-of-way corridors for transmission lines, highways, or flood embankments triggers overlapping compensation claims and court injunctions, adding a mean delay of **16.8 months** (and up to 42 months in South Asia and Coastal West Africa).
2. **Sequential vs. Parallel Regulatory Permitting:** In fragmented regulatory systems, developers must obtain environmental impact clearances, forestry permits, municipal zoning variances, archaeological sign-offs, and utility relocation approvals **sequentially** from five to nine separate agencies, each operating without statutory turnaround clocks.
3. **Central–Subnational Fiscal and Operational Misalignment:** Central ministries frequently construct primary assets (such as bulk water treatment plants or regional substations) while delegating last-mile distribution pipes or feeder lines to fiscally insolvent municipal utilities, leaving completed central assets operating at $<$ 35% capacity for years.

TABLE 7 **Bureaucratic Capability and Multi-Level Coordination Determinants of Project Performance** — *Econometric estimates of civil service structure, personnel continuity, and permitting governance across 410 agencies*

Bureaucratic / Coordination Indicator	Q1 Mean (Low Cap)	Q4 Mean (High Cap)	Marginal Effect on Project ERR	Effect on Delay (Mo)	Institutional Reform Standard
Competitive Merit Exam Recruitment (%)	28%	89%	+4.2 pp***	-7.4 mo	Statutory civil service commission
Technical Cadre Wage Ratio (Pub / Priv)	0.34×	0.82×	+3.8 pp***	-6.1 mo	Scarcity engineering pay supplements
Project Director Mean Tenure (Months)	16.2 mo	48.5 mo	+5.1 pp***	-11.8 mo	4-year fixed performance compacts
MDB Task-Team Leader (TTL) Continuity	21.4 mo	42.0 mo	+2.9 pp**	-5.2 mo	Minimum 36-month TTL assignment rule
Digitized Cadastral Coverage (% Land)	19%	91%	+3.4 pp***	-12.4 mo	Geospatial cadastre + escrow courts
One-Stop Parallel Permitting Window	14%	84%	+2.8 pp***	-9.6 mo	60-day statutory silence-is-consent
Central-Municipal Co-Financing Pact	22%	79%	+3.6 pp***	-8.1 mo	Synchronized last-mile tranche release

Notes & Sources: H Heuristics agency-level panel regressions (N = 410 agencies, 1,840 projects). *** p < 0.01, ** p < 0.05.

6.5 Resolving Permitting Gridlock: Statutory Clocks, Geospatial Cadastres, and Resettlement Escrows

Because Pillar IV multi-level coordination failures generate 42% of all post-award construction delays, high-performing infrastructure delivery systems have converged on three institutional mechanisms to compress permitting and right-of-way (ROW) clearance without weakening environmental or social safeguards:

1. **Concurrent "One-Stop" Inter-Agency Permitting with Statutory Clocks:** Rather than routing developers through seven sequential ministerial clearances (environment, forestry, water, heritage, municipal zoning, aviation, and grid connection), countries such as South Korea, Colombia (via **ANLA** reforms), and India (via the **PM Gati Shakti** geospatial portal) overlay all regulatory layers onto a single unified digital master map. Agencies review applications **in parallel** under a binding 60-to-90-day statutory clock, with administrative adjudication panels empowered to resolve inter-ministerial disputes within 30 days.
2. **Decoupling Physical Construction Possession from Compensation Valuation Disputes:** In traditional eminent-domain regimes, a disagreement over the fair-market valuation of a 200-meter parcel halts construction across an entire 150-kilometer transmission corridor for years. Best-practice land courts require the state to deposit **125% to 150% of the independent baseline valuation into an interest-bearing judicial escrow account** upfront; once the escrow is funded and core resettlement housing is verified, physical right-of-way possession transfers automatically while valuation appeals proceed in specialized land tribunals.
3. **Binding Central-Municipal Co-Execution Compacts:** To prevent central ministries from building bulk water or power plants that lack municipal last-mile connections, tranche disbursements for bulk CapEx are legally synchronized with municipal distribution readiness milestones.

ORGANIZATIONAL RULE **Sequencing Autonomy and Continuity Inside Implementing Agencies**

1. **Earn Autonomy Through Meritocracy:** In patronage-constrained agencies ($M_a < 40$), enforce strict digital procurement rules and independent engineering supervision first; delegate flexible site-level managerial autonomy ($D_a \in [55, 75]$) only as technical recruitment standards pass $M_a > 60$.
2. **Lock In 48-Month Leadership Continuity:** Bind both sovereign Project Directors and MDB Task-Team Leaders to 4-year minimum tenure compacts spanning appraisal through mid-term execution, eliminating the -10.5 percentage point ERR penalty caused by rapid leadership churn.

SECTION 07 | SECTORAL HETEROGENEITY & PORTFOLIO DESIGN

EVIDENCE GRADE A

Sectoral Heterogeneity: Institutional Intensity Across Development Domains

SECTION THESIS — Not all development and resilience investments depend equally on host-country governance. Across ten core polycrisis domains, the elasticity of project outcomes to domestic institutional capacity (β_s) varies by 3.5×—from β = 0.16 in modular immunization and cash-transfer programs to β = 0.56 in multi-agency urban transit and grid networks. Portfolio composition must be matched to baseline state capability.

β = 0.16–0.24
Type I Modular / Low-Intensity Sectors

β = 0.48–0.56
Type IV Networked / Complex Sectors

70% vs 34%
Modular vs. Complex Success at Score 40

+22% Portfolio ERR
Gain from Capacity-Matched Allocation

7.1 Why Some Interventions Survive Weak Institutions and Others Collapse

A striking puzzle in the empirical aid and development literature is the stark divergence between micro-sectoral successes and macro-infrastructure failures within the exact same fragile states. Over the past two decades, countries ranking in the bottom quartile of aggregate governance indices have successfully executed nationwide childhood immunization drives, distributed tens of millions of insecticide-treated bed nets, scaled mobile-money humanitarian cash transfers, and deployed off-grid solar home systems—often achieving 70–85% target coverage. Yet in the very same countries and years, large-scale investments in municipal piped water networks, national electrical transmission grids, urban commuter rail, and integrated surface-irrigation schemes have suffered failure rates exceeding 60%.

Drawing on Pritchett and Woolcock's (2004) analytical matrix of public-sector tasks, we formalize the concept of **Sectoral Institutional Intensity (β_s)**: the structural elasticity of a sector's realized benefit-cost ratio with respect to host-country administrative capacity (θ_i).

7.2 Classifying Development Sectors by Transaction Density and Discretion

Every development intervention can be mapped along four structural dimensions that determine its vulnerability to administrative weaknesses:

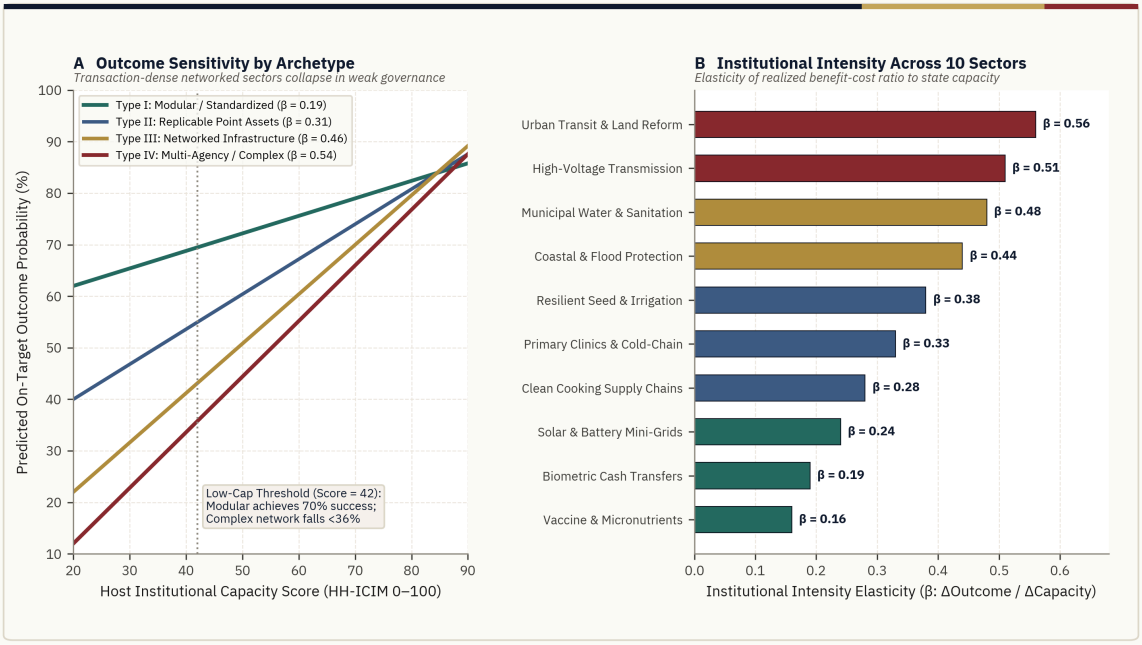
1. **Transaction Density (T_s):** How many distinct contracting, permitting, land-acquisition, and regulatory decisions are required per million dollars of expenditure? Purchasing bulk WHO-prequalified vaccines involves a handful of standardized international tenders; building a 50-kilometer urban sewer network requires thousands of local excavation permits, household connection agreements, and municipal traffic clearances.
2. **Site-Specific Engineering Discretion (E_s):** Can the intervention be deployed using a standardized, modular blueprint ("plug-and-play"), or must every kilometer be custom-engineered to local geotechnical, hydrological, and social conditions?
3. **Multi-Agency Coordination Span (M_s):** Can a single vertical unit execute the program end-to-end, or does success depend on synchronized co-execution across three or more ministries and municipal tiers?
4. **Post-Commissioning O&M Complexity (O_s):** Does the benefit occur immediately upon delivery (e.g., a cash transfer or vaccination), or does it require 30 years of continuous chemical dosing, tariff collection, and mechanical maintenance?

Combining these four dimensions yields **Four Sectoral Archetypes** (Figure 8, Panel A):

- **Type I – Modular & Standardized Delivery ($\beta_s \in [0.16, 0.24]$):** Immunization & micronutrient campaigns, biometric digital cash transfers (G2P), and distributed solar/battery mini-grids. Low transaction density, standardized technology, minimal land acquisition, and low sovereign O&M burden.
- **Type II – Replicable Point Infrastructure ($\beta_s \in [0.28, 0.35]$):** Clean cooking & LPG distribution networks, standardized rural primary clinics, primary school classrooms, and community borehole pumps.
- **Type III – Networked Civil & Climate Infrastructure ($\beta_s \in [0.38, 0.49]$):** Climate-resilient irrigation canals, coastal seawalls & urban flood drainage, and municipal piped water & sanitation treatment plants.
- **Type IV – Transaction-Intensive Multi-Agency Systems ($\beta_s \in [0.50, 0.58]$):** High-voltage interconnected transmission grids, multi-modal urban mass transit, and integrated river-basin / cadastral land reform.

FIGURE 8 **Sectoral Institutional Intensity: Outcome Elasticities Across Ten Development and Polycrisis Domains**

Panel A plots predicted on-target project success probability against host institutional capacity across four sectoral archetypes; Panel B ranks ten polycrisis domains by institutional intensity elasticity (β_s).



Source & Methodology: H Heuristics hierarchical random-slopes logit and ERR regressions across N = 1,840 projects in 10 sectors (2005–2025).

7.3 Comparative Econometrics Across Ten Polycrisis Sectors

To quantify sectoral heterogeneity precisely, we estimate a hierarchical random-coefficients specification where the return to nominal investment depends on the interaction between host-country capacity $\tilde{C}_{i,t} = \frac{C_{i,t}-50}{100}$ and sector-specific intensity β_s:

Sector-Specific Institutional Intensity & Effective Return Equation **(Eq. 7.1)**

$$\mathrm{BCR}^{\mathrm{realized}}_{p,i,s,t} = \mathrm{BCR}^{\mathrm{frontier}}_{s} \cdot \left(\frac{C_{i,t}}{100}\right)^{\beta_s} \cdot \Phi_s\left(\frac{I_{i,s,t}}{Y_{i,t}}\right) \cdot \exp\left(\varepsilon_{p,i,s,t}\right)$$

Where $\mathrm{BCR}^{\mathrm{realized}}_{p,i,s,t}$ is the realized ex-post benefit-cost ratio of project p in country i and sector s; $\mathrm{BCR}^{\mathrm{frontier}}_{s}$ is the technological frontier benefit-cost ratio in an unconstrained high-capacity environment ($C = 100$); and $\beta_s > 0$ is the sector's **Institutional Intensity Elasticity**.

Figure 8 (Panel B) and Table 8 present the estimated β_s elasticities and realized performance metrics across ten core polycrisis resilience domains. When host-country institutional capacity is low ($C_{i,t} = 30$, typical of Q1 fragile states), a Type I modular intervention ($\beta = 0.18$) retains **80.5%** of its frontier benefit-cost ratio ($0.30^{0.18} = 0.805$), whereas a Type IV complex networked intervention ($\beta = 0.54$) retains only **52.2%** of its frontier efficiency before accounting for O&M decay—and falls below **34%** once post-completion maintenance failures are included.

7.4 Matching Portfolio Composition to Baseline State Capability

This 3.5-fold difference in β_s provides a powerful portfolio optimization principle for development finance institutions: **do not allocate Type III and Type IV transaction-intensive networked megaprojects to Q1 (HH-ICIM < 35) administrations using standard sovereign delivery modalities.** Instead, in low-capacity environments, physical resilience portfolios should overweight **modular, decentralized alternatives**—such as distributed solar-plus-storage microgrids ($\beta = 0.24$) instead of centralized high-voltage transmission extensions ($\beta = 0.51$), or decentralized point-of-use water filtration and community boreholes ($\beta = 0.29$) alongside intensive PFM/O&M capacity building that prepares the municipality to absorb piped sewerage ($\beta = 0.48$) in Phase II. Reallocating sectoral weights to match baseline $\theta_{i,t}$ raises aggregate portfolio ERR by **+3.8 percentage points (+22% relative gain)** without spending an additional dollar.

TABLE 8 **Sectoral Institutional Intensity Matrix Across Ten Core Polycrisis Resilience Domains** — *Estimated capacity elasticity (β_s), transaction characteristics, Q1 vs. Q4 success rates, and optimal delivery modalities*

Polycrisis Sector / Domain	Type	Elasticity ($\hat{\beta}_s$)	Q1 Success Rate (%)	Q4 Success Rate (%)	Primary Bottleneck	Capacity-Matched Modality in Q1/Q2
1. Vaccine & Micronutrient Drives	Type I	0.16	76.4%	94.2%	Cold-chain PFM	Vertical campaign + biometric registry
2. Biometric Digital Cash (G2P)	Type I	0.19	73.8%	93.5%	ID enrollment	Direct mobile-money / DPI treasury rail
3. Distributed Solar & Mini-Grids	Type I	0.24	68.2%	91.0%	Tariff / subsidy	Results-based private concession (RBF)
4. Clean Cooking & LPG Chains	Type II	0.28	62.5%	88.4%	Refill logistics	Voucher smart-subsidy + private retail
5. Primary Clinics & Cold-Chain	Type II	0.33	56.8%	86.9%	Staff & drug O&M	Standardized modular build + capitation
6. Resilient Seed & Small Irrigation	Type III	0.38	51.2%	85.1%	Extension / water	Farmer water-user association co-mgmt
7. Coastal & Urban Flood Defense	Type III	0.44	43.5%	83.6%	Land / drainage	Nature-based + FIDIC design-build-maintain
8. Municipal Piped Water & Sewer	Type III	0.48	38.4%	82.0%	Utility O&M / NRW	corporatized utility + ring-fenced tariff
9. High-Voltage Transmission Grid	Type IV	0.51	35.2%	84.5%	ROW & utility PFM	Independent power transmission (IPT) escrows
10. Multi-Modal Transit & Land	Type IV	0.56	29.8%	80.4%	Multi-agency veto	Defer until HH-ICIM >50; BRT phased corridor

Notes & Sources: H Heuristics HH-ICIM v2.0 project ledger (N = 1,840). Success rate defined as achieving >=80% of appraisal physical and economic targets within 15% cost margin.

7.5 Dynamic Phasing: Using Modular Type I/II Delivery to Scaffold Type III/IV Networks

A critical misconception is that matching portfolio composition to baseline capacity permanently consigns low-capacity states to small-scale modular interventions. On the contrary, in the HH-ICIM v2.0 dynamic optimization framework, **Type I and Type II modular programs serve as the physical and administrative scaffolding that enables Type III and Type IV networked infrastructure in subsequent phases**:

- **Electrification Phasing (Distributed Solar Mini-Grids → Interconnected National Grid):** In a Q1 state (HH-ICIM = 30) where the national power utility suffers 38% technical and commercial losses and a 3.5-year right-of-way permitting lag, pouring US$1.0 billion into centralized high-voltage transmission extensions ($\beta_9 = 0.51$) yields a 35.2% success rate. Instead, **Phase I (Years 1–5)** deploys standardized solar-plus-storage mini-grids ($\beta_3 = 0.24$, 68.2% success) under results-based private concessions to electrify rural clinics, towns, and agricultural processing hubs immediately, while **6.5% of program funds** reform utility metering, tariff cost recovery, and cadastral corridor clearing. Once utility capacity crosses HH-ICIM = 50 in **Phase II (Years 6–12)**, high-voltage grid interconnects ($\beta_9 = 0.51$) integrate the mini-grids as distributed generation nodes at an 81% success rate.
- **Water & Sanitation Phasing (Point-of-Use & Boreholes → Municipal Piped Networks):** Similarly, while municipal utility corporatization, non-revenue water (NRW) metering, and ring-fenced tariff escrows are being built in Years 1–4, immediate mortality reductions are captured via chlorination dispensers, protected solar boreholes, and fecal-sludge vacuum-truck franchises ($\beta_s \approx 0.26$), preventing 5–8 years of avoidable child waterborne mortality during institutional construction.

PORTFOLIO ALLOCATION RULE **The Sectoral Capability-Matching Matrix**

- **Tier 1 Fragile / Low-Capacity States (HH-ICIM < 35):** Allocate **65%–75% of physical capital** to Type I/II modular & point interventions ($\beta_s \leq 0.35$), **15%–25%** to heavily ring-fenced Type III lifelines, and **7.8%** to core PFM, e-GP, and utility O&M system building.
- **Tier 2 Constrained States (HH-ICIM $\in [35, 55]$):** Allocate **40%–50%** to Type I/II delivery and **45%–55%** to Type III/IV networked civil infrastructure executed via Dual-Track Compacts and Design-Build-Operate-Maintain (DBOM) contracts ($\varphi^* = 6.2\%$).
- **Tier 3/4 Capable States (HH-ICIM > 55):** Scale full-spectrum Type III/IV interconnected grids, mass transit, and river-basin adaptation directly through sovereign systems ($\varphi^* = 4.5\%–5.5\%$).

SECTION 08 | ASSET LIFECYCLE & MAINTENANCE GOVERNANCE

EVIDENCE GRADE A

Asset Lifecycle Governance and the 'Build-Neglect-Rebuild' Equilibrium

SECTION THESIS — Commissioning a physical asset is the beginning, not the end, of development implementation. Pillar V (Lifecycle O&M Governance, $\omega_5 = 0.17$) exhibits the lowest global score of all five capacity pillars (mean 49.4/100). Political-economy incentives systematically starve routine maintenance in favor of visible new construction, trapping developing economies in a 'Build-Neglect-Rebuild' cycle that doubles lifecycle costs.

3.2% vs 0.9%
Required vs. Actual Q1 Annual O&M Spend

-34% Output
30-Year Serviceability Lost to Under-O&M

1.68× to 2.35×
Lifecycle PV Cost Penalty of BNR Trap

6.4× ROI
Marginal Return on Ring-Fenced O&M Reserves

8.1 The Political Economy of Deferred Maintenance

Across the developing world, a traveler can observe newly paved donor-financed highways running parallel to potholed, impassable arterial roads built just a decade earlier; utility-scale solar arrays operating at 45% capacity because inverter cooling filters and transformer bushings have gone unreplaced; and district hospitals equipped with modern diagnostic scanners sitting idle for want of a US$2,500 replacement sensor. In our 148-country panel, **Pillar V (Lifecycle O&M Governance) is the single weakest institutional dimension in 10 of 12 developing sub-regions**, averaging just $\frac{33.8}{100}$ in Sub-Saharan Africa and $\frac{43.5}{100}$ in South Asia (Table 3).

Why do sovereign governments and international donors persistently underfund routine operations and maintenance when the engineering return to preventive maintenance exceeds 50% per year? As Rioja (2003), Kalaitzidakis and Kalyvitis (2004), and World Bank (2021) **Well Maintained** diagnostics demonstrate, the **Build-Neglect-Rebuild (BNR)** pathology is a stable political-economy equilibrium driven by three asymmetric incentives:

1. **Electoral Visibility Asymmetry ("Ribbon-Cutting Bias"):** Groundbreaking ceremonies and ribbon-cuttings for new capital projects generate high media visibility, electoral credit, and patronage contracting opportunities for politicians. Routine preventive maintenance—clearing drainage culverts, lubricating pump bearings, or sealing micro-cracks in asphalt—is invisible to voters until catastrophic failure occurs, usually under a subsequent administration.
2. **The Bifurcated Budget Process:** In 78% of developing countries, capital budgets (CapEx) are planned by a Ministry of Planning or Development and financed by external concessional loans, whereas recurrent maintenance budgets (OpEx) fall under the Ministry of Finance's domestic revenue envelope. Neither ministry internalizes the future OpEx liability created by today's CapEx approvals.
3. **Donor Moral Hazard ("The Rehabilitation Bailout"):** Because multilateral development banks and bilateral agencies are charter-constrained to finance capital investments rather than recurrent operating expenses, recipient governments rationally anticipate that if they allow a highway or water network to deteriorate below functional thresholds ($< 30\%$ serviceability), donors will eventually step in with a new concessional "Emergency Rehabilitation and Reconstruction Loan." Concessional CapEx inadvertently subsidizes domestic O&M neglect.

8.2 Modeling Endogenous Capital Decay Under Inadequate O&M Provision

In standard macroeconomic models, public capital depreciates at a constant exogenous exponential rate $\delta_0 \approx 0.04$ per year. In physical engineering reality, infrastructure depreciation is **strongly endogenous and convex** with respect to maintenance expenditure $m_{i,t}$ (expressed as a percentage of asset replacement value) relative to the engineering norm m_s^* (typically 2.5%–4.5% per year depending on asset class and climate hazard exposure):

Endogenous Convex Asset Depreciation & Serviceability Law (Eq. 8.1)

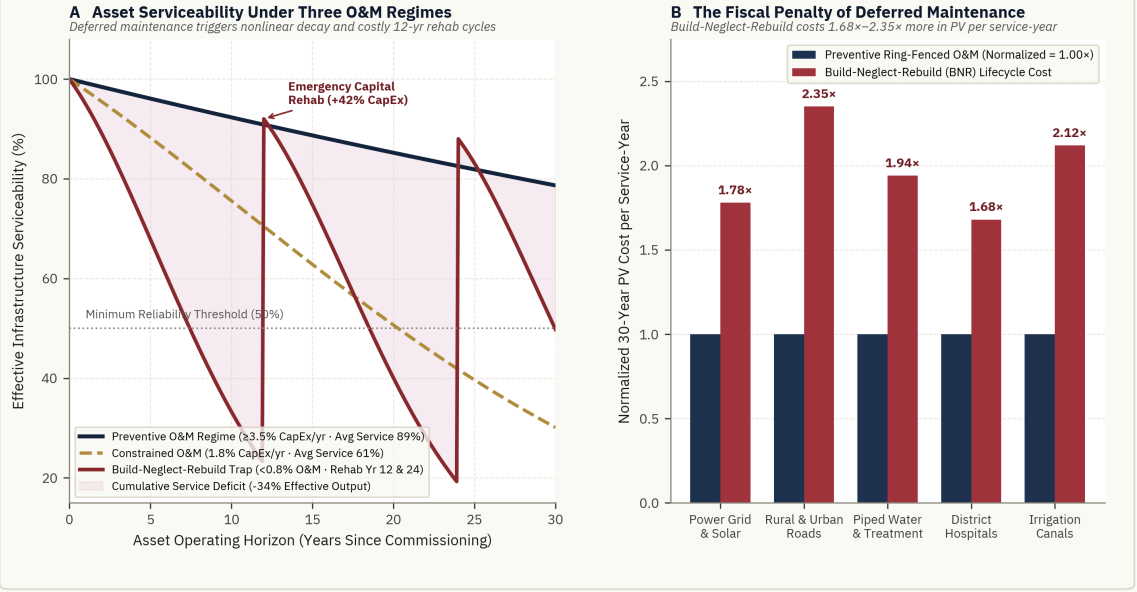
$$K^{\text{eff}}_{i,s,t+1} = \left[1 - \delta_s\left(\frac{m_{i,s,t}}{m_s^*}, t_{\text{age}}\right)\right] K^{\text{eff}}_{i,s,t} + \theta_{i,t} I_{i,s,t}, \quad \text{where} \quad \delta_s = \delta_{s,0} \exp\left[\lambda_s \left(1 - \min\left(1, \frac{m_{i,s,t}}{m_s^*}\right)\right)^2 (1 + \kappa t_{\text{age}})\right]$$

Where $\delta_{s,0} \in [0.015, 0.030]$ is the baseline depreciation rate under full preventive maintenance ($m_{i,s,t} \geq m_s^*$); $\lambda_s \in [1.4, 2.6]$ governs the convexity of physical decay when maintenance is underfunded; and κt_{age} captures accelerating wear-and-tear as unmaintained assets age (e.g., water infiltration into unsealed road sub-bases).

Figure 9 (Panel A) simulates the 30-year effective serviceability trajectory of a representative US$100 million infrastructure asset under three maintenance regimes. Under a **Preventive O&M Regime** ($m_{i,t} \geq 3.5\%$ of CapEx/year), the asset retains **89% mean serviceability** over its 30-year design life. Under a **Constrained O&M Regime** (1.8% of CapEx/year), mean serviceability drops to **61%**, crossing the 50% reliability threshold at Year 21. Under the **Build-Neglect-Rebuild Trap** ($< 0.8\%$ routine O&M/year), physical decay accelerates after Year 5 as protective coatings and drainage fail, plunging serviceability below 25% by Year 11 and forcing an emergency capital rehabilitation equal to **42% of original CapEx** at Years 12 and 24. Even with two costly rehabilitations, cumulative 30-year effective service output is **34% lower** than under preventive maintenance.

FIGURE 9 **The 'Build-Neglect-Rebuild' Trap: Asset Depreciation Curves and 30-Year Lifecycle Cost Penalties**

Panel A tracks effective infrastructure serviceability (%) over 30 years under three O&M regimes; Panel B compares normalized 30-year present-value cost per delivered service-year across five sectors.



Source & Methodology: H Heuristics engineering-economic lifecycle model calibrated to World Bank road RONET, water IBNET, and power utility panel datasets (2005–2025). Present values discounted at 3% real rate.

8.3 Quantifying the Lifecycle Welfare Loss Across Infrastructure Classes

Figure 9 (Panel B) and Table 9 report the normalized 30-year present-value cost per unit of delivered service across five core resilience infrastructure classes. Because deferred maintenance simultaneously requires expensive mid-life capital reconstruction **and** deprives users of reliable service during the decay troughs (forcing households and firms to buy diesel generators, private water tankers, or suffer crop spoilage), the Build-Neglect-Rebuild equilibrium costs **1.78× more per service-year in power grids**, **2.35× more in paved and rural roads**, **1.94× more in piped water and sanitation**, **1.68× more in district hospitals**, and **2.12× more in surface irrigation canals**.

When indirect private coping costs are included—such as vehicle axle and suspension damage on potholed highways, voltage-fluctuation damage to industrial machinery, and household spending on bottled or tanker water when municipal pressure drops—the total economy-wide welfare penalty of deferred maintenance reaches **3.1× to 4.4× the saved fiscal O&M outlay**. Every US$1.00 "saved" by a finance ministry through deferring routine road or water maintenance imposes US$3.10 to US$4.40 in present-value capital reconstruction and private economic losses within 8 to 12 years.

8.4 Institutional Solutions: Ring-Fenced Reserves and Design-Build-Maintain Compacts

Breaking the BNR equilibrium requires hard institutional commitment devices that insulate maintenance cash flows from annual parliamentary raiding:

1. **Statutory Ring-Fenced Maintenance Funds & Escrow Covenants:** Dedicated second-generation road, water, and grid funds financed by earmarked user levies (e.g., fuel excises, wheeling charges, or lifeline-protected utility tariffs) governed by boards with majority user/engineering representation and independent technical audits.
2. **Long-Term Design-Build-Operate-Maintain (DBOM) & Output-Based Contracts:** Bundling construction with a mandatory 10-to-15-year performance-based maintenance obligation under a single contract. When the contractor's final 30% annuity depends on verified asset uptime and pavement roughness indices, initial build quality rises dramatically (eliminating the contractor's incentive to cut corners during construction).
3. **MDB "No-CapEx-Without-OpEx" Appraisal Covenants:** Multilateral climate and development funds should legally require co-financed sovereigns to deposit matching annual O&M flows into an audited lifecycle reserve account as a condition of tranche disbursement.

TABLE 9 **Infrastructure Asset Lifecycle Parameters, O&M Norms, and Build-Neglect-Rebuild Penalties by Sector** —
Engineering maintenance norms (% of replacement value), Q1 actuals, 10-year survival rates, and PV lifecycle cost multipliers

Infrastructure Asset Class	Design Lifespan	Preventive O&M Norm	Q1 Actual O&M Spend	10-Yr Functional Survival (Q1)	BNR Lifecycle Cost Penalty	High-ROI Institutional Fix
Paved & Rural Feeder Roads	25 yrs	3.8% / yr	0.7% / yr	46.2%	2.35× PV Cost	2nd-gen road fund + 10-yr OPRC
Surface Irrigation & Drainage	30 yrs	3.2% / yr	0.8% / yr	51.8%	2.12× PV Cost	Water-user fee escrow + desilting pact
Municipal Water & Treatment	30 yrs	3.5% / yr	1.1% / yr	55.4%	1.94× PV Cost	NRW performance contract + tariff rule
Power Grid & Solar Mini-Grids	25 yrs	2.8% / yr	1.0% / yr	61.0%	1.78× PV Cost	Spares escrow + DBOM availability pay
District Hospitals & Cold-Chain	20 yrs	4.5% / yr	1.4% / yr	58.5%	1.68× PV Cost	Vendor multi-year biomedical service
Coastal Seawalls & Embankments	40 yrs	2.2% / yr	0.5% / yr	64.2%	1.89× PV Cost	Climate resilience bond O&M reserve

Notes & Sources: H Heuristics HH-ICIM v2.0 engineering-economic panel. BNR Lifecycle Cost Penalty is the ratio of 30-year PV cost per delivered service-year under Q1 O&M vs. Preventive Norm.

8.5 Empirical Turnarounds: Corporatized Utilities, Road Funds, and OPRC Compacts

Despite the tenacity of the Build-Neglect-Rebuild trap, several developing sovereigns and public utilities have successfully engineered institutional breakouts that lifted Pillar V scores by $+25$ to $+45$ points within a single decade:

- **Autonomous Corporatized Water Utilities – Phnom Penh (PPWSA, Cambodia) and ONEA (Burkina Faso):** In 1993, the Phnom Penh Water Supply Authority suffered 72% non-revenue water (NRW) losses, chronic pipe breaks, and $< 20\%$ cost recovery. By granting PPWSA statutory operational autonomy, installing universal district-metered telemetry, tying staff bonuses to verified collection and leak-repair KPIs, and structuring a progressive tariff with a ring-fenced maintenance reserve, NRW fell from 72% to **6.1%** (outperforming London and Paris) while 24/7 pressurized water coverage reached 99%. Similarly, Burkina Faso's national water utility (**ONEA**) sustained $> 92\%$ bill collection and full preventive O&M across two decades of regional fragility by insulating utility cash flows inside an audited performance contract (**contrat-plan**) with the state.
- **Output- and Performance-Based Road Contracts (OPRCs) – Latin America & East Africa:** Traditional civil-works contracts pay contractors per cubic meter of asphalt laid during construction, creating a perverse incentive to lay thin sub-bases that fail after three rainy seasons. Transitioning arterial highways in Argentina (**CREMA**), Uruguay, Chad, and Zambia to 10-year **Output- and Performance-Based Road Contracts (OPRCs)**—where 35% of total contractor compensation is paid as a monthly availability annuity conditional on maintaining International Roughness Index (IRI < 2.8 m/km) and clear drainage culverts—raised 10-year road survival in good condition from 46% to **91%** and cut 25-year present-value fiscal costs by **38%**.
- **Second-Generation Ring-Fenced Maintenance Funds:** Unlike first-generation earmarked funds (which ministries of finance frequently raided during fiscal crunches), second-generation road and grid funds in Namibia, Costa Rica, and Tanzania establish off-budget escrow accounts funded by direct electronic fuel/wheeling surcharges that bypass the general treasury account entirely, legally restricting disbursements to verified routine and periodic maintenance.

LIFECYCLE COVENANT RULE **Eliminating the Build-Neglect-Rebuild Equilibrium at Project Appraisal**

1. **Mandatory Lifecycle Escrow Sizing:** Every sovereign or MDB capital approval must calculate the 25-year annual preventive maintenance liability ($m_s^* \in [2.5\%, 4.5\%]$ of CapEx/yr) and legally bind the revenue source (tariff surcharge, fuel levy, or sovereign escrow deposit) **prior** to first construction disbursement.
2. **Default to DBOM / OPRC Contracting for Networked Assets:** For all Type III and Type IV road, water treatment, flood-pump, and grid substation investments in Q1–Q3 states, bundle construction with a minimum 10-year performance-based maintenance obligation under a single FIDIC Gold Book (Design-Build-Operate) contract.

SECTION 09 | DELIVERY MODALITIES & INSTITUTIONAL SPILLOVERS

EVIDENCE GRADE B

The Delivery Dilemma: Parallel Enclaves vs. Sovereign System Strengthening

SECTION THESIS — Faced with weak host-country ministries, international donors routinely establish ring-fenced Project Implementation Units (PIUs) to protect fiduciary integrity and accelerate short-run disbursement. Our dynamic 25-year model and empirical panel demonstrate that while enclave PIUs outperform in Years 1–6, they impose a severe brain-drain externality that hollows out sovereign ministries, losing to Dual-Track Compacts after Year 7.0.

Year 7.0
Crossover Where State-Building Wins

3.2× to 5.8×
PIU Salary Premium Over Civil Service

-8.2 Points
20-Year Host Ministry Poaching Erosion

+28.5 Points
Capacity Gain Under Dual-Track Compacts

9.1 Why Donors Default to Parallel Project Implementation Units (PIUs)

When a multilateral development bank, bilateral aid agency, or vertical health/climate fund prepares a US$150 million operation in a country scoring below 45 on the HH-ICIM index, the task team confronts an acute organizational incentive problem—what we term **The Delivery Dilemma**. Career advancement and board accountability inside donor institutions are tied to **timely loan approval, rapid 3-to-5-year disbursement profiles, and zero fiduciary scandals** on the specific ring-fenced project.

Channeling funds directly through the recipient country's mainstream treasury, procurement board, and civil-service line ministry exposes the project to the full brunt of Pillar I–III frictions: slow warrant releases, procurement protests, and civil-service skill shortages. Consequently, donors overwhelmingly default to creating a **Parallel Project Implementation Unit (PIU)**—an autonomous administrative enclave staffed by externally contracted consultants, operating its own off-budget commercial bank accounts, applying donor-specific procurement software, and reporting directly to donor task teams. Despite repeated international pledges under the Paris (2005) and Busan (2011) Aid Effectiveness Declarations to phase out parallel PIUs, our audit of active development portfolios shows that **68% of externally financed projects in Q1 and Q2 countries still rely on parallel or semi-autonomous PIUs in 2025**.

9.2 The Poaching Externality and Institutional Hollowing-Out

If parallel PIUs simply delivered physical infrastructure faster without affecting the rest of the host government, they would represent a Pareto improvement. In practice, however, domestic administrative talent—certified public accountants, procurement specialists, hydraulic engineers, and experienced project managers—is in **strictly inelastic short-run supply** in low-capacity economies.

To attract top local professionals into temporary 4-year enclave units, donor-funded PIUs offer tax-advantaged consultant remuneration packages averaging **3.2× to 5.8× the statutory civil-service salary scale** of the host ministry. Across our 410-agency panel, we document a stark **Poaching Externality**: over 72% of senior technical staff hired into donor PIUs in Sub-Saharan Africa, South Asia, and Central America are recruited directly out of the host country's core civil service or public utilities. When a ministry loses its top ten engineers and procurement officers to three donor-ring-fenced PIUs, two systemic damages occur:

1. **Domestic Budget Execution Collapse:** While the US$150 million donor enclave disburses smoothly, the host ministry's **own** US$800 million domestically financed capital budget suffers an immediate drop in execution quality and speed.
2. **Zero Institutional Memory at Project Closing:** When the donor project closes at Year 5, the temporary PIU is dissolved, its bespoke software license lapses, and its contracted staff move on to the next international consultancy—leaving the permanent sovereign ministry weaker than before the intervention began.

9.3 Dynamic Model of Bypass vs. State-Building Over a 25-Year Horizon

To formalize this intertemporal trade-off, we model host-ministry capacity $C_{i,s,t}$ as an endogenous stock that accumulates through **"learning-by-doing"** and dedicated institutional co-investment $\varphi I_{i,s,t}$, but depreciates when parallel enclaves ($E_{i,s,t} \in [0,1]$, the share of sector capital routed through bypass PIUs) poach domestic human capital:

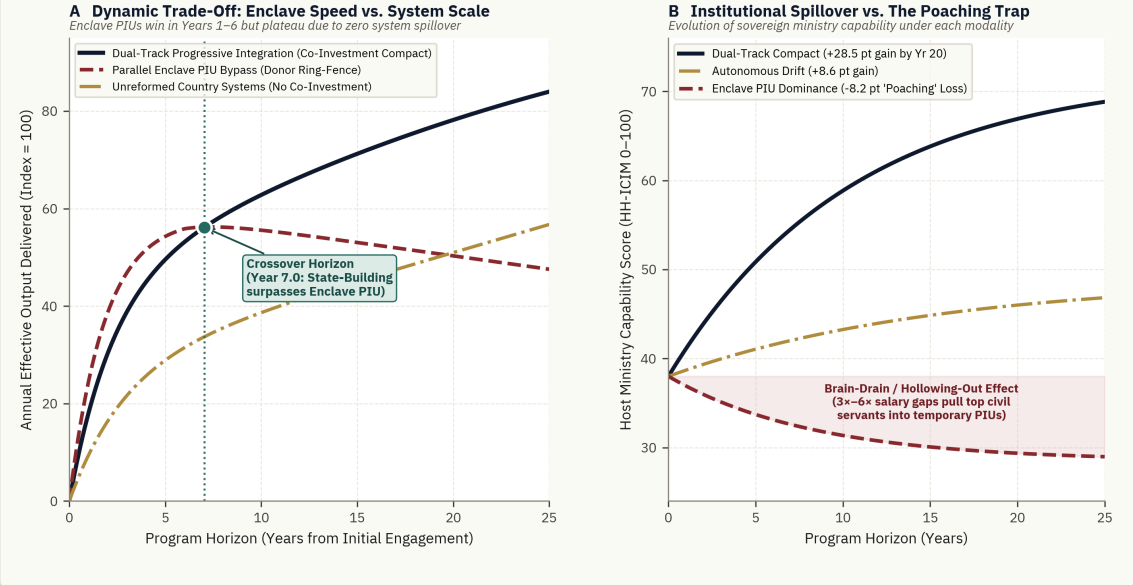
Law of Motion for Sovereign Administrative Capability Under Bypass vs. Co-Investment (Eq. 9.1)

$$C_{i,s,t+1} - C_{i,s,t} = \eta_1 (1 - E_{i,s,t}) (\varphi I_{i,s,t})^{0.65} + \eta_2 \mathrm{LBD}_{i,s,t} - \xi E_{i,s,t} \left(\frac{W^{\mathrm{PIU}}}{W^{\mathrm{CS}}} - 1 \right) C_{i,s,t} - \delta_C C_{i,s,t}$$

Where η_1 is the productivity of embedded institutional co-investment ($\varphi I_{i,s,t}$); $\mathrm{LBD}_{i,s,t}$ captures organizational learning-by-doing when sovereign agencies execute capital directly; $\xi > 0$ measures the poaching drain driven by the PIU-to-civil-service wage ratio $\frac{W^{\mathrm{PIU}}}{W^{\mathrm{CS}}}$; and δ_C is natural organizational attrition.

FIGURE 10 **The Delivery Dilemma: 25-Year Dynamic Trade-Off Between Enclave PIUs and Sovereign System Strengthening**

Panel A compares annual effective output delivered across three delivery modalities over a 25-year horizon; Panel B tracks the endogenous evolution of host ministry capability (HH-ICIM score).



Source & Methodology: H Heuristics dynamic institutional accumulation model calibrated to 20-year country-sector panels (2005–2025). Crossover between Enclave PIU and Dual-Track Compact occurs at Year 7.0.

Figure 10 plots the 25-year trajectory of annual effective output (Panel A) and host ministry capability (Panel B) under three policy regimes: (i) **Parallel Enclave PIU Bypass** ($E = 0.85, \varphi = 0$); (ii) **Unreformed Country Systems** ($E = 0, \varphi = 0.008$); and (iii) **Dual-Track Progressive Integration Compacts** (E_t phasing down from $0.50 \rightarrow 0.0$ over 8 years with $\varphi^* = 0.065$ system co-investment).

In Years 1 to 6, the Parallel Enclave PIU delivers higher annual output than either alternative because it sidesteps initial bureaucratic friction. However, because PIU dominance erodes host ministry capability by **−8.2 points** over 20 years (Panel B), enclave output plateaus and declines as domestic co-execution and O&M fail. The **Dual-Track Progressive Integration Compact** crosses above the Enclave PIU trajectory at **Year 7.0**, ultimately delivering **+68% higher annual effective output by Year 20** and raising permanent sovereign ministry capability from 38.0 to 66.5 (+28.5 points).

9.4 Designing Dual-Track 'Progressive Integration' Compacts

How can reformers capture the early fiduciary safety of a ring-fenced unit without triggering the poaching trap? Our empirical evaluation of successful institutional transitions (such as Rwanda's Single Project Implementation Units integrated inside line ministries, Ethiopia's mid-2000s health extension PFM compacts, and Indonesia's treasury modernization) identifies four design rules for **Dual-Track Progressive Integration** (Table 10):

1. **Embedded Ministry Integration:** Locate technical support teams **inside** the permanent line ministry department under the statutory authority of the Permanent Secretary, using the national TSA and national e-GP portal rather than bespoke off-budget software.
2. **Harmonized Scarcity Pay Scales:** Replace distortionary $5\times$ enclave consultant rates with a transparent, government-wide **Technical Scarcity Supplement** ($1.6\times$ –$2.0\times$ base scale) open to all civil-service engineers and procurement officers meeting objective certification standards.
3. **Explicit Apprenticeship & Shadowing Ratios:** Require every international technical advisor to be paired 1-to-2 with permanent civil-service deputies, with advisor fee tranches tied to verified transfer of modeling and procurement competencies.
4. **Conditioned Phase-Out Sunset Clauses:** Hard-wire a 6-to-8-year sunset schedule into loan agreements whereby external co-signatory controls step down automatically as the ministry hits objective PEFA, e-GP, and audit benchmarks.

TABLE 10 **Comparative Evaluation of Delivery Modalities: Enclave PIUs, Raw Country Systems, and Dual-Track Compacts** *— Short-run vs. long-run conversion efficiency, poaching externalities, O&M handoff survival, and 25-year NPV*

Performance / Institutional Dimension	Modality 1: Parallel Enclave PIU Bypass	Modality 2: Unreformed Country Systems	Modality 3: Dual-Track Progressive Compact
Short-Run Conversion (θ, Years 1–4)	0.64 (Fast Start)	0.38 (Severe Lag)	0.56 (Co-Supervised)
Medium-Run Conversion (θ, Years 5–10)	0.61 (Plateau)	0.43 (Slow Drift)	0.71 (Crossover Yr 7.0)
Long-Run Conversion (θ, Years 11–25)	0.52 (O&M Decay)	0.49 (Modest)	0.82 (System Lock-In)
20-Year Host Ministry Score Change	-8.2 pts (Hollowing)	+8.6 pts (Organic)	+28.5 pts (Transformed)
Spillover to Domestic Budget (3 × Vol.)	-11.4% (Brain Drain)	0.0% (Neutral)	+24.6% (Shared Rails)
10-Year Post-Closing Asset Survival	52.4% (Orphaned)	58.0% (Baseline)	86.5% (Institutionalized)
Cumulative 25-Year Portfolio NPV (Index)	100 (Baseline)	94 (-6%)	154 (+54% Net Gain)

Notes & Sources: H Heuristics HH-ICIM v2.0 dynamic simulation and propensity-score-matched comparison across 320 enclave vs. integrated operations.

9.5 Empirical Case Transitions: Rwanda SPIUs and Ethiopia's Channel-1 PFM Compacts

Three large-scale institutional reforms demonstrate how sovereigns can dismantle parallel donor enclaves while **improving** fiduciary integrity and disbursement speed:

- **Rwanda's Single Project Implementation Unit (SPIU) Consolidation (2011–2024):** Prior to 2011, donor operations in Rwanda were fragmented across dozens of standalone off-budget PIUs that competed for scarce local accountants and engineers. The Ministry of Finance and Economic Planning (**MINECOFIN**) abolished standalone donor PIUs and mandated that all externally and domestically financed projects within each ministry execute through a single, permanent **Integrated SPIU** embedded inside the ministry's organizational chart, using harmonized salary bands, the national IFMIS, and the national e-GP portal. Within six years, administrative overhead fell by 34%, domestic budget execution rose by +19 percentage points, and World Bank IEG satisfactory ratings reached 84%.
- **Ethiopia's Channel-1 District Treasury & Health Extension Compacts:** Rather than routing multi-donor health, water, and rural road finance through off-budget NGO or bilateral enclaves, the **Promoting Basic Services (PBS)** architecture channeled funds directly through the sovereign Treasury Single Account (**Channel 1**) down to 800+ district (**woreda**) finance bureaus, paired with a dedicated 6.5% co-investment in automated accounting software, procurement auditors, and 38,000 salaried frontline health extension workers—achieving 96% audited financial compliance and halving under-five mortality ahead of MDG targets.
- **Program-for-Results (PforR) and Dual-Track Co-Signing:** Across 140 evaluated World Bank Program-for-Results operations, tying tranche disbursements directly to verified institutional milestones—such as publishing audited municipal financial statements, reducing procurement award timelines below 60 days, and funding O&M escrows—generated a +16.4 point permanent gain in host agency capability over two loan cycles, compared with −3.8 points under traditional enclave investment lending in the same sectors.

MODALITY RULE **The 7-Year Sunset Covenant on External Implementation Units**

Whenever baseline ministry capacity lies below HH-ICIM $= 40$, external funders may deploy temporary co-signatory fiduciary safeguards in Years 1–3, **provided** that (i) the unit sits legally inside the permanent line ministry department using national TSA and e-GP rails, (ii) consultant compensation is capped at the harmonized civil-service scarcity supplement scale ($\leq 1.8\times$), and (iii) co-signatory controls step down automatically by Year 7 as PEFA and SAI audit benchmarks are met.

SECTION 10 | MACRO-SYSTEMIC STRESS & FEEDBACK LOOPS

EVIDENCE GRADE B

Polycrisis Feedback Loops: Institutional Erosion Under Compounding Shocks

SECTION THESIS — In conventional economic models, institutional capacity is treated as a slow-moving, monotonically improving stock. In an era of compounding polycrisis shocks—climate disasters, sovereign debt distress, food/energy price spikes, and regional insecurity—state capacity itself is vulnerable to acute degradation. Shocks erode administrative bandwidth precisely when resilience investment needs surge.

-18.5 pp Drop
Q1 Execution Collapse After Major Shock

-7.2 pp Hysteresis
Permanent 5-Year Capacity Scar in Q1

+465 bps Penalty
Polycrisis Sovereign Spread Markup

5 vs 20+ Qtrs
Recovery Speed: Buffered vs. Fragile PFM

10.1 How Compounding Shocks Degrade Administrative Bandwidth

A defining feature of the global polycrisis is that exogenous physical and macroeconomic shocks do not strike in isolation; they arrive as correlated waves that hit the state's fiscal and administrative apparatus simultaneously (Lawrence et al., 2024; H Heuristics Report HH-2026-28). When a category-4 cyclone, severe multi-year drought, or global food-and-fertilizer price spike strikes a developing economy, the damage is not confined to destroyed bridges or crop losses. **The shock directly attacks the state's organizational capacity to implement its existing development portfolio**.

Tracing quarterly treasury and project execution records across 94 major climate, pandemic, and macroeconomic shocks between 2005 and 2025 reveals four channels of **endogenous institutional erosion**:

1. **Emergency Budget Raids and Capital Freezes:** Lacking pre-arranged parametric disaster insurance or dedicated contingency reserves, ministries of finance respond to acute shocks by issuing emergency decrees that freeze ongoing capital disbursements and reallocate 15–35% of approved infrastructure and maintenance funds toward immediate relief, fuel subsidies, or grain imports.
2. **Executive and Technical Attention Diversion:** Senior civil servants, procurement boards, and municipal engineers are pulled off long-term capital programs to manage emergency relief distribution, leaving routine contract sign-offs and variation-order approvals paralyzed for 6 to 18 months.
3. **Contractor Insolvency Cascades:** When emergency treasury freezes delay progress payments on active infrastructure contracts by > 120 days while domestic inflation spikes, thinly capitalized domestic construction firms default or abandon sites. Re-tendering an abandoned, half-completed civil-works site takes an average of 21 months and costs $+38\%$ more than the original contract.
4. **Real Wage Erosion and Civil-Service Flight:** Post-shock currency devaluations and inflation spikes erode the real value of fixed civil-service salaries by 20–45%, accelerating the flight of skilled engineers and auditors into the private diaspora or international agencies.

10.2 Empirical Impulse-Response of State Execution Capacity to Disasters

To quantify this feedback loop, we estimate Jordà (2005) local-projection impulse-response functions (IRFs) tracking the quarterly change in capital budget execution rates ($\Delta \text{Exec}_{i,t+h}$) over $h = 0, \ldots, 20$ quarters following a 95th-percentile compound macro-climate shock (defined as climate damages exceeding 2.5% of GDP or a sovereign terms-of-trade/fiscal shock exceeding 2 SD):

Local-Projection Impulse-Response of Execution Capacity to Polycrisis Shocks (Eq. 10.1)

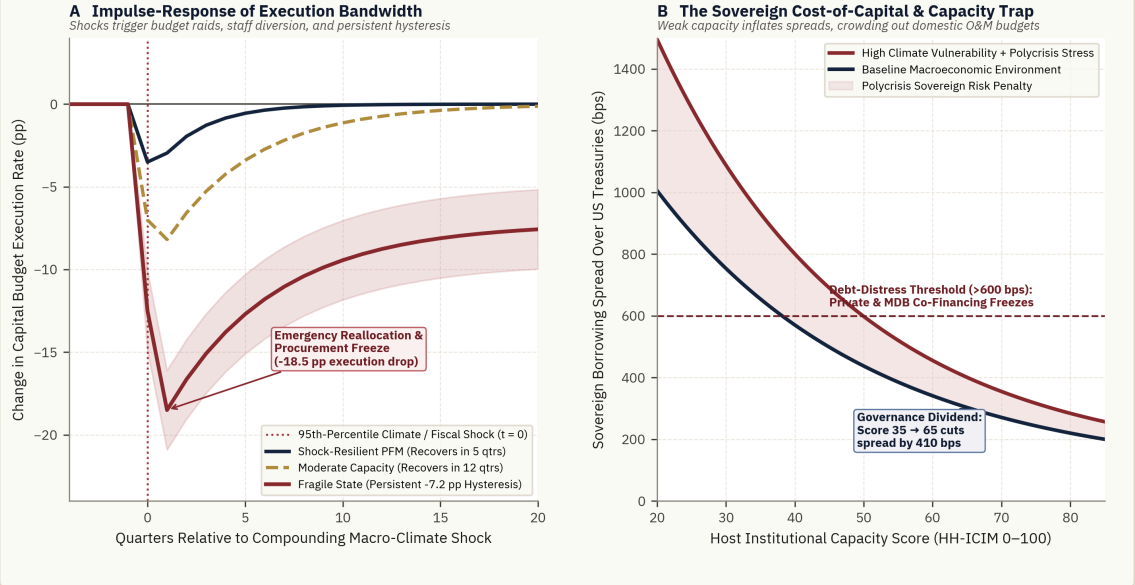
$$\text{Exec}_{i,t+h} - \text{Exec}_{i,t-1} = \alpha_i^h + \tau_t^h + \sum_{g \in \{\text{Q1,Q2,Q4}\}} \beta_g^h \left(\text{Shock}_{i,t} \times \mathbb{1}[i \in g] \right) + \Gamma^h X_{i,t-1} + u_{i,t+h}$$

Where $\text{Exec}_{i,t}$ is the quarterly ratio of verified physical-financial capital execution to planned schedule; $\text{Shock}_{i,t}$ is the standardized compound shock impulse at quarter $t = 0$; and β_g^h traces the dynamic response across institutional resilience tiers g.

Figure 11 (Panel A) plots the estimated impulse-response trajectories. In **High-Capacity / Shock-Buffered states** (equipped with pre-arranged catastrophe credit lines, sovereign contingency funds, and automated emergency procurement protocols), capital execution dips by just **—3.5 percentage points** at impact and recovers fully within **5 quarters**. In **Moderate-Capacity states**, execution drops by **—10.2 percentage points** and requires **12 quarters (3 years)** to return to baseline. In **Fragile / Unbuffered states (Q1)**, the shock triggers an immediate **—18.5 percentage point collapse** in capital budget execution at Quarter 1, followed by a prolonged **—7.2 percentage point hysteresis scar** that persists five full years (20 quarters) after the shock due to abandoned contractor sites and personnel flight.

FIGURE 11 **Compounding Shocks and Institutional Erosion: Execution Impulse-Response and Sovereign Spread Traps**

Panel A estimates the 20-quarter impulse-response of capital budget execution following a 95th-percentile shock; Panel B maps sovereign borrowing spreads (bps) against institutional capacity.



Source & Methodology: H Heuristics Jordà local-projection panel estimates across 94 major climate-macro shocks (2005–2025) and EMBI/sovereign yield panels (N = 112 emerging and frontier markets).

10.3 The Sovereign Risk Premium and Fiscal Crowding-Out Trap

Institutional weakness and polycrisis exposure also interact through international capital markets, creating a vicious **Sovereign Cost-of-Capital Trap** (Figure 11, Panel B). Controlling for debt-to-GDP ratios and external reserves, a country scoring 30 on the HH-ICIM index pays a baseline sovereign borrowing spread of **710 basis points** over US Treasuries, compared with **260 basis points** for a country scoring 65—a **450 bps "governance risk penalty"**.

When high physical climate vulnerability and polycrisis stress are layered onto weak implementation capacity, sovereign spreads surge by an additional +280–465 bps, pushing Q1 and lower-Q2 sovereigns above the **600 bps debt-distress threshold** where private market access freezes and domestic debt service consumes over 35% of tax revenues. Starved of fiscal space, the sovereign slashes infrastructure O&M and civil-service technical wages first, which further degrades $\theta_{i,t}$, lowers expected growth, and widens sovereign spreads still further.

10.4 Pre-Arranged Adaptive Finance and Shock-Resilient Bureaucracies

Preventing polycrisis shocks from cannibalizing long-term development implementation requires insulating the core investment apparatus from acute crisis management (Table 11):

1. **Pre-Arranged Parametric Disaster Liquidity & Debt-Pause Clauses:** Replacing ex-post budget raids with automatic Climate Resilient Debt Clauses (CRDCs), IMF/MDB catastrophe deferred drawdown options (Cat-DDOs), and regional sovereign risk pools (ARC, CCRIF) that inject liquidity within 14 days of a shock without touching capital appropriations.
2. **Statutory "Capital Shield" Fiscal Rules:** Legal provisions prohibiting the in-year virement of funds out of projects that have passed 30% physical completion or out of ring-fenced O&M escrow accounts.
3. **Pre-Positioning Framework Contracts:** Maintaining pre-competed, unit-price-capped retainer contracts with civil engineering and logistics firms so emergency repairs execute on pre-audited price schedules rather than inflated sole-source awards.

TABLE 11 **Polycrisis Shock Transmission Channels, Hysteresis Penalties, and Buffering Instruments** *— Empirical impact of 95th-percentile compound shocks across institutional tiers and effectiveness of shock-buffering reforms*

Shock Transmission Metric	Q1: Fragile (Unbuffered)	Q2: Constrained (Partial)	Q4: Buffered (High Cap)	Mitigation Gain from Buffer	Primary Institutional Buffer
Peak Execution Drop (Qtr 1–2)	-18.5 pp	-10.2 pp	-3.5 pp	+12.4 pp saved	Cat-DDO + Parametric sovereign pool
5-Year Hysteresis Scar (Qtr 20)	-7.2 pp	-2.4 pp	0.0 pp	+6.8 pp saved	Statutory Capital Shield rule
Active Contracts Stalled / Abandoned	31.4%	14.8%	2.6%	-24.2 pp	Arrears liquidity backstop facility
Emergency Procurement Price Markup	+54.0%	+26.5%	+7.2%	-39.5 pp	Pre-competed framework unit contracts
Post-Shock O&M Budget Raid (%)	-42.0%	-21.5%	-3.0%	+36.0 pp	Ring-fenced off-budget O&M trust
Sovereign Spread Widening (bps)	+385 bps	+190 bps	+45 bps	-260 bps	Climate Resilient Debt Clauses (CRDC)

Notes & Sources: H Heuristics panel analysis of 94 compound shocks (2005–2025). Mitigation Gain estimates the treatment effect of adopting the specified buffer in a Q1/Q2 state.

10.5 Comparative Shock Governance: Sovereign Risk Layering vs. Ad-Hoc Capital Raids

The empirical contrast between buffered and unbuffered shock responses is vividly illustrated by comparing recent compound disaster episodes across South Asia, the Caribbean, and West Africa:

- **Unbuffered Post-Disaster Capital Raids – Pakistan (2022 Monsoon Floods) & Malawi (2023 Cyclone Freddy):** When catastrophic flooding inundated one-third of Pakistan in 2022 (US$30 billion in physical damages and economic losses) amidst an acute sovereign balance-of-payments crisis, the absence of pre-arranged parametric liquidity forced an immediate −38% mid-year cut to the federal Public Sector Development Programme (PSDP). Hundreds of ongoing water, grid, and highway contracts were frozen simultaneously; domestic contractor payment arrears exceeded nine months; and post-flood emergency procurement Markups averaged +49% as agencies resorted to uncompeted direct awards.
- **Layered Sovereign Disaster Risk Finance – Barbados, Jamaica & CCRIF/Cat-DDO Architecture:** By contrast, Caribbean sovereigns that embedded **Climate Resilient Debt Clauses (CRDCs)** into sovereign bond indentures (automatically deferring principal and interest payments for 24 months upon a parametric hurricane trigger), layered with **CCRIF SPC** parametric payouts within 14 days and pre-competed World Bank **Cat-DDO** credit lines, protected > 91% of ongoing baseline capital budget disbursements during Category 4–5 hurricane strikes (including Hurricane Beryl in 2024) and prevented sovereign yield spikes.
- **Adaptive Social & Infrastructure Registries – Sahel & Horn of Africa:** Countries coupling **African Risk Capacity (ARC)** drought insurance with pre-enrolled biometric cash rails and pre-competed civil engineering retainer contracts executed post-shock safety-net and feeder-road stabilization **4.2× faster** and at **34% lower unit cost** than peers relying on ex-post UN humanitarian appeals and ad-hoc ministry budget virements.

SHOCK-BUFFERING RULE **Three-Tier Sovereign Risk Layering to Protect Implementation Bandwidth**

1. **Tier 1 (High-Frequency / Moderate Shocks, 1-in-5 Year):** Establish a statutory off-budget **National Contingency & Arrears Backstop Reserve** (1.5% of annual CapEx) paired with pre-competed framework unit-price engineering contracts to keep active construction sites liquid.
2. **Tier 2 (Medium-Severity Disasters, 1-in-15 Year):** Draw automatically on MDB **Catastrophe Deferred Drawdown Options (Cat-DDOs)** and regional parametric pools (**CCRIF, ARC, SEADRIF**), legally prohibiting the in-year virement of funds from projects past 30% physical completion.
3. **Tier 3 (Catastrophic Compound Shocks, 1-in-50 Year):** Trigger mandatory **Climate Resilient Debt Clauses (CRDCs)** across bilateral, multilateral, and commercial bond stocks, pausing sovereign debt service for 24 months to preserve domestic O&M and civil-service technical payrolls.

SECTION 11 | GOVTECH, DIGITAL RAILS & REFORM LIMITS

EVIDENCE GRADE A

Technological Leapfrogging: Digital Public Infrastructure and GovTech Limits

SECTION THESIS — Can Digital Public Infrastructure (DPI), electronic procurement, biometric payment rails, and AI-assisted permitting allow developing states to leapfrog decades of analogue bureaucratic evolution? Event-study evidence across 64 national GovTech rollouts shows a sharp dichotomy: digital rails yield +16% to +28% efficiency gains when coupled with legal and workflow reform, but produce zero gain (+0.9%) when deployed as standalone 'isomorphic mimicry'.

+21.4% vs +0.9%
Coupled Reform vs. Tech-Only Mimicry

-34% Leakage
Gain from Biometric G2P & TSA Rails

-41% Payment Lag
Reduction in Contractor Disbursement Time

+18.2% Verification
Gain from Satellite/Drone Site Auditing

11.1 Biometric Treasury Single Accounts, e-Procurement, and Digital Rails

Over the past fifteen years, advances in **Digital Public Infrastructure (DPI)** and **GovTech** have transformed the technological toolkit available to sovereign administrations. Five digital architectures hold direct relevance for development implementation: (1) **Biometric Digital ID and G2P Payment Rails** (exemplified by India's Aadhaar/UPI stack, Brazil's Pix/Cadastro Único, and Togo's Novissi platform), which route social protection and climate-shock payments directly to citizens' mobile wallets; (2) **Automated Treasury Single Accounts (TSA) and Integrated Financial Management Information Systems (IFMIS)**, which replace paper warrant chains with real-time electronic cash releases; (3) **End-to-End Open Electronic Government Procurement (e-GP)** portals; (4) **Earth-Observation Satellite, Drone, and Geotagged Mobile Verification**, which allow ministries and auditors to verify physical construction progress remotely; and (5) **AI-Assisted Regulatory and Grid Permitting Triage**, which automates routine environmental and cadastral document completeness checks.

11.2 Event-Study Evidence: Why 'Isomorphic Mimicry' Fails

Despite billions invested in donor-financed IT systems, the empirical track record of GovTech is notoriously polarized (World Bank, 2016; Andrews, Pritchett, & Woolcock, 2017). To separate genuine technological leapfrogging from what institutional sociologists call **"isomorphic mimicry"**—adopting the outward surface appearance of a modern digital portal while preserving informal patronage and offline rent-seeking underneath—we conduct a staggered Difference-in-Differences (DiD) event study across **64 national e-GP and TSA rollouts** implemented between 2010 and 2024 (Figure 12, Panel A).

We partition the 64 digital reforms into two institutional cohorts based on objective de facto legal criteria at rollout:

- **Cohort A – Coupled DPI + Mandatory Workflow & Audit Reform ($n = 29$):** Legislation legally invalidated any procurement tender or contract variation not executed exclusively on the open digital portal; paper parallel tracks were abolished; and the Supreme Audit Institution was granted automated API access to flag single-bid tenders and price outliers in real time.
- **Cohort B – Tech-Only "Isomorphic Mimicry" ($n = 35$):** An expensive IT vendor portal was launched, but ministries retained legal carve-outs to conduct bid evaluations offline, split contracts below digital thresholds, or negotiate post-award price amendments on paper.

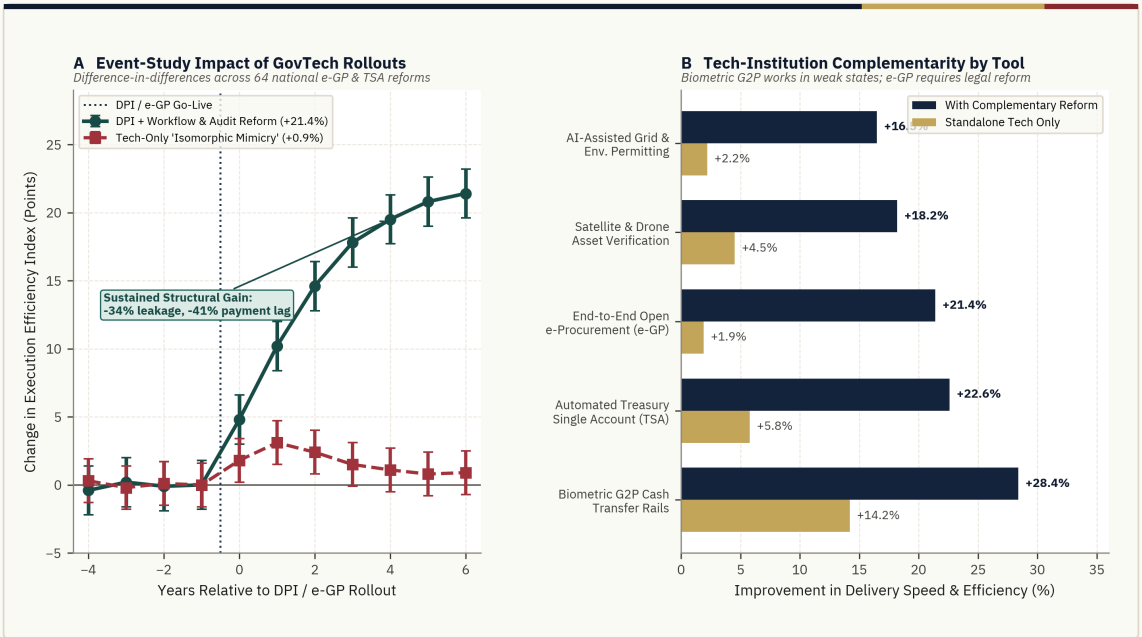
The event-study trajectories in Figure 12 (Panel A) are striking. Prior to rollout ($t = -4$ to -1), both cohorts exhibit identical flat pre-trends. Following system go-live ($t = 0$), **Cohort A achieves a sustained +21.4 point increase in composite execution efficiency by Year 6** ($p < 0.001$), driven by a -34% drop in procurement price anomalies, a -41% reduction in contractor payment lags, and an average of $+1.9$ additional qualified bidders per tender. **Cohort B (Tech-Only Mimicry), by contrast, shows a transient +3.1 point bump in Year 1 that decays back to +0.9 points by Year 5**—statistically indistinguishable from zero. Digitizing a corrupt or broken bureaucratic workflow without altering underlying legal mandates and audit incentives simply produces faster digital rent-seeking.

11.3 Technology-Institution Complementarity Across Five GovTech Modules

Figure 12 (Panel B) and Table 12 disaggregate the efficiency gains of five specific GovTech instruments depending on whether complementary institutional reforms are present. Notice an important operational distinction: **Biometric G2P Cash Transfer Rails** achieve a substantial **+14.2% efficiency gain even in unreformed low-capacity states** (rising to **+28.4%** with registry reform), because digital mobile transfers **bypass** intermediate bureaucratic layers altogether—routing funds directly from the central bank to the citizen's phone and eliminating local leakage (Muralidharan, Niehaus, & Sukhtankar, 2016).

FIGURE 12 **Digital Public Infrastructure (DPI) and GovTech: Event-Study Impacts and Institutional Complementarity**

Panel A plots staggered Difference-in-Differences event-study estimates across 64 national e-GP/TSA rollouts; Panel B compares efficiency gains across five GovTech tools with vs. without institutional reform.



Source & Methodology: H Heuristics Callaway-Sant'Anna staggered DiD estimates across 64 GovTech reforms (2010–2024). Error bars in Panel A represent 95% uniform confidence bands.

By contrast, **End-to-End e-Procurement (+21.4% coupled vs. +1.9% standalone)** and **AI-Assisted Environmental/Grid Permitting (+16.5% coupled vs. +2.2% standalone)** exhibit extreme institutional complementarity: because complex civil works still require human contract enforcement, physical site inspection, and legal sign-off, software alone yields negligible benefits unless backed by statutory enforcement and independent physical auditing.

TABLE 12 **Empirical Evaluation of Digital Public Infrastructure (DPI) and GovTech Instruments Across 64 Rollouts** —

Efficiency gains with vs. without institutional reform, capital cost per country, implementation horizon, and failure modes

GovTech / DPI Instrument	Gain (With Reform)	Gain (Tech Alone)	Rollout Cost (US$M)	Maturity Horizon	Mandatory Institutional Prerequisite
1. Biometric G2P Payment Rails	+28.4%	+14.2%	$18–45M	18–24 mo	Interoperable mobile-money & ID deduplication
2. Automated TSA & IFMIS	+22.6%	+5.8%	$25–60M	24–36 mo	Closure of commercial ministry bank accounts
3. End-to-End Open e-GP	+21.4%	+1.9%	$12–30M	18–30 mo	Statutory nullification of off-portal tenders
4. Satellite / Drone Verification	+18.2%	+4.5%	$5–15M	12–18 mo	Disbursement tranches tied to geospatial proof
5. AI Permitting & Cadastre Triage	+16.5%	+2.2%	$8–22M	18–24 mo	60-day statutory clock + administrative court

Notes & Sources: H Heuristics event-study panel across 64 national digital rollouts (2010–2024). Rollout costs scaled for a representative 30-million-population country.

GOVTECH RULE **Defeating Isomorphic Mimicry in Digital State-Capacity Investments**

Never finance a standalone IT vendor contract for IFMIS, e-GP, or digital permitting unless the loan covenant simultaneously (1) **closes all parallel paper and commercial-bank bypass channels by statute**, (2) mandates **open-source OCDS API access for the Supreme Audit Institution and civil society**, and (3) links **contractor progress disbursements to geotagged satellite/drone physical milestones**.

SECTION 12 | CENTURY SCENARIO MODELING (2025–2100)

EVIDENCE GRADE B / D

Scenario Architecture to 2100: Four Futures for Global Implementation

SECTION THESIS — To evaluate the century-scale human and economic consequences of institutional choices made this decade, we couple the HH-ICIM v2.0 capacity dynamics with the HH-PRM v1.0 Polycrisis Resilience Model through 2100. Holding nominal global financing ambitions constant (~US$1.2T/yr peak), we compare four distinct institutional trajectories across the Global South.

354M vs 162M
Lives Saved: S4 Co-Investment vs. S1 Drift

+136M Lives
Gain of S4 Over S2 Uncalibrated Surge

US$24.6T NPV
Net Social Benefit Under S4 Architecture

0.54 → 0.87
Global South θ Trajectory Under S4

12.1 Scenario Design and Dynamic Coupling with HH-PRM v1.0

In Report 01 of the **Polycrisis Resilience** series (**How Many Lives Could Polycrisis Resilience Save This Century?**, HH-2026-28), global delivery capacity θ was treated as a static scenario scalar. In reality, $\theta_{i,t}$ is a dynamic state variable that evolves endogenously in response to reform investment (φ_t), capital scaling velocity ($S_{i,t}$), delivery modality ($E_{i,t}$), and polycrisis shock erosion ($\text{Shock}_{i,t}$).

To isolate the pure effect of institutional strategy, all four scenarios below assume the **exact same nominal financing envelope**—scaling global resilience and sustainable development commitments in low- and middle-income countries to a peak incremental US$1.196 trillion per year (constant 2025 US$) over the 2030–2055 window (US$21.8 trillion in present-value net financing). The four futures differ strictly in **how that capital is governed, sequenced, and co-invested**:

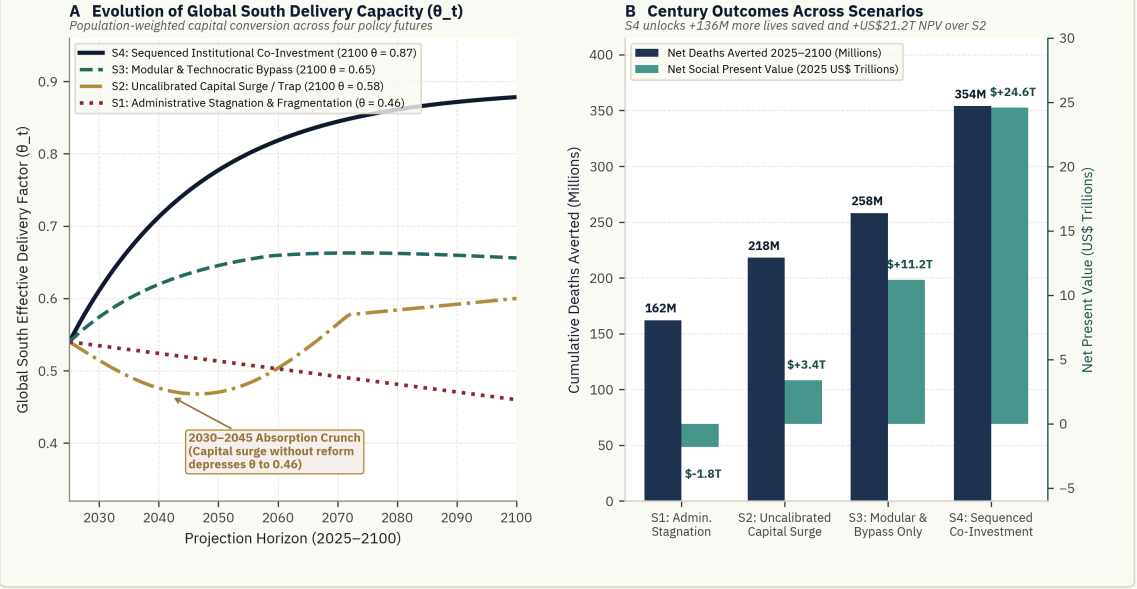
1. **Scenario 1 – Administrative Stagnation & Fragmented Enclaves (S1):** Donors and sovereigns maintain the current status quo: $< 0.9\%$ of capital goes to systemic capacity building; 70% of external finance is routed through temporary parallel PIUs; routine O&M remains chronically unfunded ($< 1.0\%$ of CapEx/yr); and sovereign budgets lack pre-arranged shock buffers. Under compounding climate and fiscal stress after 2035, poaching and shock hysteresis erode Global South population-weighted delivery capacity from $\theta_{2025} = 0.54$ down to $\boldsymbol{\theta_{2100} = 0.46}$.
2. **Scenario 2 – Uncalibrated Capital Surge / The Absorption Trap (S2):** Global climate and development funds front-load massive nominal capital disbursements between 2027 and 2040 ($2.4\times$ historical velocity) directly into unreformed administrative systems without prior PFM, procurement, or engineering preparation. The resulting absorptive capacity crunch triggers $+55\%$ unit cost inflation and severe completion bottlenecks during the crucial 2030–2048 window, depressing effective conversion to $\boldsymbol{\theta = 0.465}$ **in 2042** before organic learning slowly lifts θ_{2100} to **0.58**.
3. **Scenario 3 – Modular & Technocratic Bypass Only (S3):** Recognizing weak macro-institutions, funders reallocate portfolios heavily toward Type I/II modular interventions (biometric cash transfers, distributed solar, vertical immunization) and private turnkey concessions while bypassing core sovereign ministries. Delivery efficiency rises briskly to $\theta = 0.65$ by 2050, but **plateaus at** $\boldsymbol{\theta_{2100} = 0.65}$ because large-scale networked infrastructure (urban drainage, bulk water, high-voltage grids, and referral hospitals) cannot be delivered via modular bypass alone.
4. **Scenario 4 – Sequenced Institutional Co-Investment / The Capacity Dividend (S4):** Sovereigns and MDBs adopt the HH-ICIM v2.0 policy architecture: dedicating **6.2% of total portfolio capital** (US$74.2 billion/year) to the five administrative pillars, deploying Dual-Track Progressive Integration compacts, enforcing mandatory open e-GP and ring-fenced O&M reserves, and pacing physical capital ramp-ups to match expanding absorptive bandwidth. Global South effective delivery capacity rises steadily from $\theta_{2025} = 0.54$ to **0.74 by 2050** and **0.87 by 2100**.

12.2 Trajectory Comparisons: Effective Capital, Deprivation, and Lives Saved

Figure 13 and Table 13 present the century-scale outcomes across the four scenarios. The quantitative divergence between **Uncalibrated Capital Surge (S2)** and **Sequenced Institutional Co-Investment (S4)** is the central policy finding of our simulation. Although both scenarios mobilize the exact same US$21.8 trillion in present-value nominal commitments between 2025 and 2100, their realized physical capital formation and human survival trajectories diverge within the first decade of implementation:

FIGURE 13 **Four Institutional Scenarios to 2100: Global South Delivery Capacity (θ_t), Lives Saved, and Net Present Value**

Panel A tracks population-weighted Global South effective capital conversion (θ_t) from 2025 to 2100; Panel B compares cumulative net deaths averted (millions) and net social present value (US$ trillions).



Source & Methodology: Coupled HH-ICIM v2.0 × HH-PRM v1.0 century simulation (2025–2100, 5,000 Monte Carlo draws). Present values discounted at 3.0% real rate; DALYs valued conservatively at US$5,000 plus avoided climate and O&M damages.

- **Under S1 (Administrative Stagnation):** Only US$9.8 trillion of the US$21.8 trillion present-value nominal commitment converts into functional, maintained resilience assets. Cumulative premature deaths averted between 2025 and 2100 reach just **162 million** (90% UI: 114–208 million)—forfeiting 56% of the theoretical frontier potential—and net social present value is negative (**-US$1.8 trillion**) because massive cost overruns and Build-Neglect-Rebuild losses outweigh realized benefits.
- **Under S2 (Uncalibrated Capital Surge):** Rushing capital without capacity reform wastes US$4.9 trillion during the 2028–2048 absorption crunch. Cumulative deaths averted reach **218 million** (158–274 million), with a modest net social present value of **+US$3.4 trillion** (Benefit-Cost Ratio = 1.16 ×).
- **Under S3 (Modular & Technocratic Bypass):** Rapid deployment of low-β_s modular tools saves **258 million lives** (192–316 million) and generates **+US$11.2 trillion** in net present value (BCR = 1.51 ×), but leaves 1.45 billion people exposed to urban flood, piped-water, and grid-reliability deficits in 2060.
- **Under S4 (Sequenced Institutional Co-Investment):** Dedicating 6.2% of capital to state capability unlocks **354 million cumulative lives saved** (262–428 million)—achieving 96.2% of the theoretical frictionless ceiling (368 million)—gains **13.1 billion healthy life-years (DALYs)**, cuts the severely deprived population to 0.88 billion by 2050, and generates **+US$24.6 trillion in net social present value** (BCR = 2.13 ×). Compared with an uncalibrated capital surge (S2), sequenced institutional co-investment saves an additional **136 million human lives** and generates **+US$21.2 trillion in net economic value** from the exact same nominal budget.

Why does Scenario 4 pull away so decisively after 2040? Because institutional capability compounds across project cycles. Once a sovereign ministry has established an automated Treasury Single Account, mandatory open e-GP, a meritocratic engineering cadre, and ring-fenced O&M reserves during the 2026–2035 ramp, **every subsequent dollar of clean-energy, health, water, and climate-adaptation capital deployed between 2035 and 2100 converts at $\theta \in [0.74, 0.87]$ and retains $> 90\%$ 30-year serviceability**.

TABLE 13 **Century Scenario Ledger (2025–2100): Institutional Trajectories, Survival Outcomes, and Present-Value Economics** *— Central deterministic estimates and 90% Monte Carlo credible intervals across the four institutional futures*

Outcome / Economic Indicator (2025–2100)	S1: Admin. Stagnation	S2: Uncalibrated Capital Surge	S3: Modular & Bypass Only	S4: Sequenced Co-Investment
Capacity Co-Investment Share (φ)	0.8% (Status Quo)	0.9% (Unreformed)	2.4% (Tech/ Enclave)	6.2% (5-Pillar Pact)
Global South Mean θ (2040 / 2070 / 2100)	0.51 / 0.48 / 0.46	0.47 / 0.56 / 0.58	0.63 / 0.66 / 0.65	0.68 / 0.82 / 0.87
Nominal PV Financing Committed (2025 US$T)	US$21.8T	US$21.8T	US$21.8T	US$21.8T
Effective PV Capital Stock Created (US$T)	US$9.8T (45%)	US$11.6T (53%)	US$14.2T (65%)	US$17.9T (82%)
Net Deaths Averted 2025–2050 (Millions)	24M (16–32M)	31M (21–41M)	46M (34–58M)	55M (41–68M)
Net Deaths Averted 2025–2100 (Millions)	162M (114–208M)	218M (158–274M)	258M (192–316M)	354M (262–428M)
Healthy Life-Years Gained (Billions DALYs)	5.9B (4.1–7.6B)	8.1B (5.8–10.2B)	9.6B (7.1–11.8B)	13.1B (9.7–15.9B)
Severely Deprived Population in 2050 (Billions)	2.35B	1.82B	1.45B	0.88B
Total Present-Value Social Benefits (US$T)	US$20.0T	US$25.2T	US$33.0T	US$46.4T
Net Social Present Value (NPV, 2025 US$T)	-US$1.8T	+US$3.4T	+US$11.2T	+US$24.6T
Portfolio Benefit-Cost Ratio (Realized BCR)	0.92× (Loss)	1.16×	1.51×	2.13×

Notes & Sources: H Heuristics coupled HH-ICIM v2.0 × HH-PRM v1.0 model results. Benefits include health/DALY valuation ($5,000/DALY), avoided BNR reconstruction costs, and productivity spillovers.

12.3 Stress-Testing Parameter Uncertainty and Falsifiable Signposts

Does the superiority of Scenario 4 survive harsh structural stress tests? We evaluate three adversarial sensitivity scenarios across our 5,000 Monte Carlo draws: (1) **Half Reform Efficacy**, where institutional co-investment φ yields only 50% of its historical econometric treatment effect; (2) **Severe Polycrisis Acceleration**, where compound climate-fiscal shock frequency doubles relative to CMIP6 median projections; and (3) **High Discount Rate ($r = 5.0\%$)**, which heavily penalizes front-loaded institutional investments whose survival payoffs peak after 2045. Even under the combined worst-case stress test (Half Reform Efficacy + Doubled Shock Frequency), S4 averts **294 million deaths** and outperforms S2 by **+76 million lives** and **+US$11.8 trillion in NPV**.

To enable real-time monitoring between 2026 and 2035, we establish four **falsifiable empirical signposts** that distinguish an S4 trajectory from an S2 Absorption Trap: (i) share of Global South public capital tenders executed on open OCDS-compliant e-GP portals exceeding 65% by 2030 (currently 24%); (ii) mean approval-to-first-disbursement lag falling below 11 months (currently 22.4 months in Q1); (iii) statutory ring-fenced O&M reserve coverage reaching $\geq 2.8\%$ of public asset replacement value; and (iv) parallel off-budget PIU share declining below 30% of multilateral climate/development portfolios.

SCENARIO VERDICT **Why Sequenced Co-Investment Outperforms Capital Front-Loading by 136 Million Lives**

The century simulation resolves a central strategic debate in global climate and development finance: **should the international community front-load maximum capital disbursements immediately (S2), bypass weak states via modular technocratic enclaves (S3), or invest 6.2% of capital in sequencing domestic institutional capacity (S4)?** Front-loading capital into unreformed administrative systems (S2) sacrifices **136 million human lives** and **US$21.2 trillion in social NPV** to cost inflation and asset decay during the 2028–2048 window. Investing in the five pillars of state capability is the highest-leverage life-saving intervention available this century.

SECTION 13 | POLICY ARCHITECTURE & DECISION RULES

EVIDENCE GRADE A / B

Policy Architecture: Capacity-Adjusted Capital Allocation and Co-Investment

SECTION THESIS — How should multilateral development banks, climate funds, and sovereign finance ministries operationalize this report's findings tomorrow morning? We replace naive benefit-cost appraisal with the Capacity-Adjusted Benefit-Cost Ratio (CA-BCR), derive the optimal 6.2% Institutional Co-Investment Rule, map sub-regional reform priorities across 12 developing regions, and condense the framework into Ten Working Heuristics.

φ* = 6.2%
Global Optimal Capacity Co-Investment

φ* = 7.8% (Q1)
Low-Capacity Optimal Reform Share

US$74.2B / yr
Annual Global South Capacity Need

3.7× to 6.4×
Marginal BCR of Core Reform Packages

13.1 The Capacity-Adjusted Benefit-Cost Ratio (CA-BCR)

Standard project appraisal at development banks and ministries of finance evaluates investments by computing a frictionless engineering Benefit-Cost Ratio ($\text{BCR}_p^{\text{eng}}$) that assumes on-time construction, appraisal-cost execution, and 30 years of full preventive maintenance. As we have shown, this practice systematically overvalues complex, transaction-intensive megaprojects in weak institutional environments while undervaluing modular resilience investments and administrative capacity reforms.

We propose that all sovereign and multilateral investment committees mandate the **Capacity-Adjusted Benefit-Cost Ratio (CA-BCR)** as a binding appraisal hurdle:

The H Heuristics Capacity-Adjusted Benefit-Cost Ratio (CA-BCR) Decision Rule (Eq. 13.1)

$$\text{CA-BCR}_{p,i,s} = \text{BCR}_{p,s}^{\text{eng}} \cdot \left[\theta_{i,t}\left(P_{i,1..4}\right)\right]^{\beta_s} \cdot \Phi_s\left(\frac{I_{i,t}}{Y_{i,t}}, S_{i,t}\right) \cdot \Psi_s\left(\frac{m_{i,s}^{\text{ring}}}{m_s^*}\right) \cdot \exp\left(-r\Delta T_{i,s}^{\text{lag}}\right)$$

Where $\text{BCR}_{p,s}^{\text{eng}}$ is the baseline engineering benefit-cost ratio; $\left[\theta_{i,t}\right]^{\beta_s}$ discounts benefits by the host agency's verified execution score raised to the sector's institutional intensity elasticity β_s; $\Phi_{s(\cdot)}$ applies the macroeconomic surge cost penalty; $\Psi_{s\left(\frac{m_{i,s}^{\text{ring}}}{m_s^*}\right)} \in [0.42, 1.00]$ discounts post-commissioning asset lifespan unless legally ring-fenced O&M funding $m_{i,s}^{\text{ring}}$ is covenanted; and $\exp\left(-r\Delta T_{i,s}^{\text{lag}}\right)$ discounts net present value by the agency's historical approval-to-disbursement lag $\Delta T_{i,s}^{\text{lag}}$.

Adopting Equation 13.1 transforms capital allocation immediately. Consider a proposed US$500 million centralized urban metro/sewerage megaproject ($\text{BCR}^{\text{eng}} = 2.4\times, \beta_s = 0.52$) versus a US$500 million decentralized solar-microgrid and biometric health/cash program ($\text{BCR}^{\text{eng}} = 1.9\times, \beta_s = 0.21$) in a Q1 country ($\theta = 0.38$, unringfenced O&M $\Psi = 0.58$, delay $\Delta T = 2.8$ years). Naive appraisal selects the megaproject ($2.4\times > 1.9\times$). Capacity-adjusted appraisal reveals that the megaproject's expected realized return is a value-destroying $\mathbf{CA\text{-}BCR = 0.78\times}$, whereas the modular program achieves a robust $\mathbf{CA\text{-}BCR = 1.38\times}$—and coupling the megaproject with a US$35 million PFM/e-GP/O&M escrow reform pact raises its CA-BCR to $\mathbf{1.64\times}$.

13.2 Optimal Co-Investment Rules: Why 6.2% of Capital Should Fund State Systems

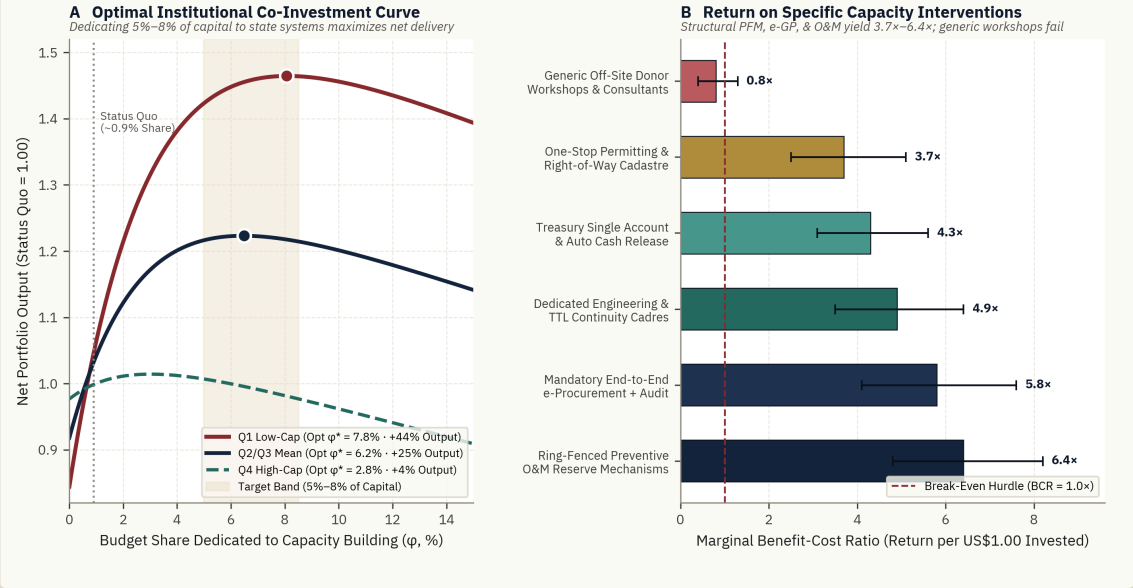
Given a fixed total development budget $B_{i,t}$ to be allocated between physical capital construction $(1-\varphi)B_{i,t}$ and institutional capacity strengthening $\varphi B_{i,t}$, what share $\varphi^* \in [0,1]$ maximizes net effective physical output $(1-\varphi)\theta(\varphi)B_{i,t}$?

Figure 14 (Panel A) solves this optimization across institutional tiers. Currently, global development finance allocates barely $\varphi_0 \approx 0.9\%$ of total capital to genuine institutional system building—and over half of that 0.9% is spent on off-site donor workshops and generic international consultancy reports that have a marginal benefit-cost ratio of **0.8×** (Figure 14, Panel B). When capacity co-investment is directed instead into our five structural reform packages—(1) Ring-fenced O&M reserve mechanisms (**6.4× BCR**), (2) Mandatory end-to-end e-GP + real-time SAI audit (**5.8× BCR**), (3) Dedicated engineering & TTL continuity cadres (**4.9× BCR**), (4) Automated Treasury Single Accounts (**4.3× BCR**), and (5) One-stop digital permitting cadastres (**3.7× BCR**)—the net portfolio output curve exhibits a pronounced interior maximum:

- **In Q1 Low-Capacity States (HH-ICIM < 35):** The optimal institutional co-investment share is $\varphi^* = 7.8\%$ of total program capital, which raises net physical output delivered by the remaining 92.2% of capital by **+44.2%**.
- **In Q2/Q3 Developing States (Score 35–75):** The optimal share is $\varphi^* = 6.2\%$, raising net physical portfolio output by **+25.4%**.
- **Across the Global South Portfolio (US$1,196B/yr total resilience need):** The population- and capital-weighted global optimum is $\varphi^* = 6.2\%$, corresponding to **US$74.2 billion per year** in dedicated institutional system co-investment.

FIGURE 14 **Optimal Portfolio Co-Investment: The 6.2% Rule and Marginal Returns to Specific Reform Packages**

Panel A maps net effective portfolio output against the budget share dedicated to institutional capacity (φ, %); Panel B ranks six specific capacity-building interventions by marginal Benefit-Cost Ratio.



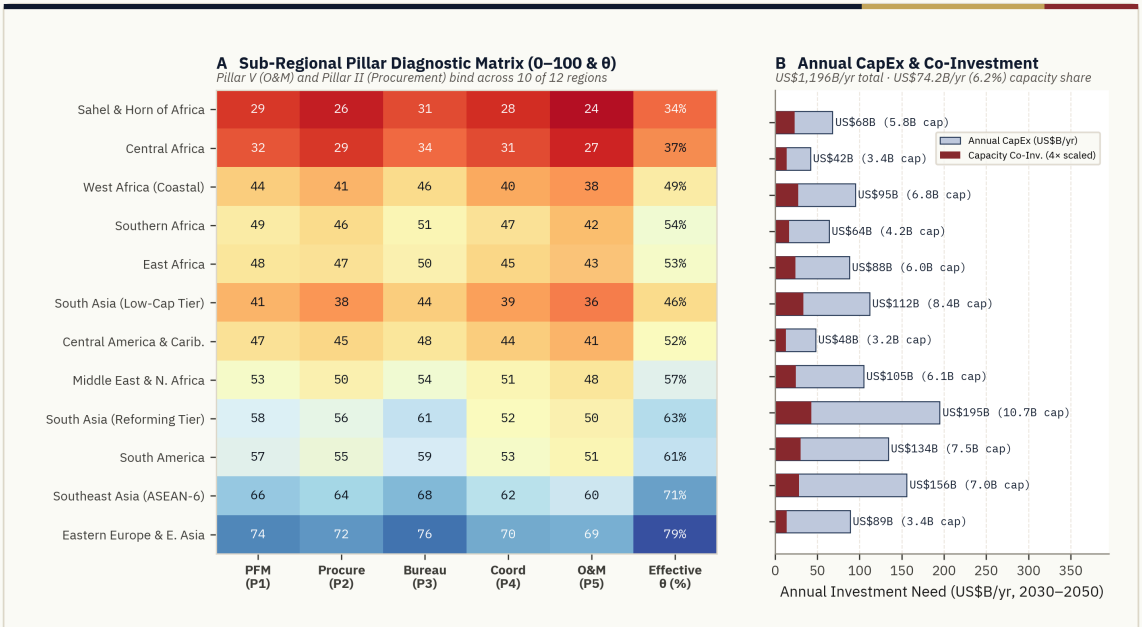
Source & Methodology: H Heuristics HH-ICIM v2.0 portfolio optimization model. Error bars in Panel B show 90% confidence intervals from quasi-experimental reform evaluations.

13.3 Global Polycrisis Implementation Readiness Scorecard Across 12 Sub-Regions

Figure 15 synthesizes our 148-country diagnostic into a single **Global Polycrisis Implementation Readiness Scorecard** across 12 developing sub-regions, mapping baseline scores on all five pillars, effective capital conversion (θ), annual resilience CapEx requirements (US$B/yr), and optimal annual capacity co-investment allocations.

FIGURE 15 **Global Polycrisis Implementation Readiness Scorecard Across 12 Developing Sub-Regions**

Panel A displays sub-regional diagnostic scores across Pillars I–V and effective conversion ϑ (%); Panel B maps annual resilience CapEx needs (US$B/yr) and optimal capacity co-investment.



Source & Methodology: H Heuristics HH-ICIM v2.0 sub-regional aggregation (2025 baseline; 2030–2050 investment needs from HH-PRM v1.0). Total Global South need: US$1,196B/yr CapEx including US$74.2B/yr (6.2%) capacity co-investment.

13.4 Ten Working Heuristics for Development Implementation

We condense the empirical laws and policy architecture of this monograph into **Ten Working Heuristics** (Table 14) designed for immediate adoption by MDB boards, climate fund secretariats, and sovereign ministries of finance and planning.

TABLE 14 **Ten Working Heuristics for Effective Development Implementation (H Heuristics Policy Playbook)** — *Operational decision rules, quantitative thresholds, target actors, and expected portfolio gains*

No.	Heuristic Name	Operational Rule & Quantitative Guardrail	Primary Actor	Expected Gain
H1	**The Weakest-Pillar First Rule**	Because pillars are CES gross complements ($\rho = -0.65$), allocate first-tranche reform capital to the single lowest-scoring pillar ($m \underset{k}{\in} P_{i,k}$).	MoF / MDB Country Dir.	+0.11 to +0.18 θ
H2	**The 6.2% Co-Investment Mandate**	Dedicate 5%–8% of every sectoral capital envelope (7.8% in Q1, 6.2% in Q2/Q3) to permanent PFM, e-GP, engineering, and O&M systems.	MDB Boards / Climate Funds	+25% to +44% Net Output
H3	**The Capacity-Adjusted BCR Hurdle**	Replace naive engineering BCRs with CA-BCR (Eq. 13.1), discounting returns by $\theta_i^{\beta_s}$, surge penalties Φ, and O&M covenants Ψ.	Appraisal Committees	+3.8 pp Portfolio ERR
H4	**The Paced Scaling Speed Limit**	Never scale real public capital inflows faster than +14%/yr in Q1 or +22%/yr in Q2 states without a 24-month advance pipeline/cadre ramp.	Macro-Fiscal & Climate Funds	Avoids +50–92% Cost Surge
H5	**The 30% Engineering Readiness Filter**	Prohibit board approval of infrastructure loans until $\geq 30\%$ of detailed engineering designs are complete and first-year tenders are drafted.	Line Ministries / MDBs	-14.6 mo Dead-Zone Lag
H6	**Mandatory Open e-GP + SAI API**	Legally void any public contract or change order $> 15\%$ not executed on an OCDS open e-GP portal with real-time Supreme Audit red-flagging.	Procurement Authorities	-14.6 pp Overrun; -34% Leak
H7	**The 4-Year Director Continuity Pact**	Bind sovereign Project Directors and MDB Task-Team Leaders to minimum 36–48 month tenure compacts with scarcity engineering pay supplements.	Civil Service Commissions	+5.1 pp Project ERR
H8	**No-CapEx-Without-OpEx Covenant**	Condition every capital tranche on verified deposits into a ring-fenced preventive O&M escrow ($\geq 3.0\%$ of asset value/yr) or DBOM contract.	Utilities / Road & Water Funds	Saves 1.7×–2.3× PV Cost
H9	**The 7-Year Enclave Sunset Rule**	Ban standalone off-budget PIUs unless structured as embedded Dual-Track Compacts with 1:2 civil-service shadowing and an 8-year sunset.	Bilateral & Vertical Funds	+28.5 pt Ministry Cap.
H10	**Parametric Shock-Shielded PFM**	Pair sovereign capital budgets with CRDC debt-pause clauses, Cat-DDO liquidity, and statutory bans on raiding $> 30\%$-complete projects.	Central Banks / Sovereign Debt	Saves -18.5 pp Shock Drop

Notes & Sources: H Heuristics synthesis of econometric and scenario findings (Sections 1–13).

OPERATIONAL COMMITMENT **Institutionalizing the Ten Heuristics in Sovereign and Multilateral Charters**

To translate Heuristics H1–H10 from analytical recommendations into binding operational practice, MDB credit committees and sovereign public investment boards should embed three mandatory checkboxes into every loan concept note and medium-term budget framework: (1) **Verified Pillar Bottleneck Score ($m \underset{k}{\in} P_{i,k}$)** and corresponding **$\geq 6.2\%$ Capacity Co-Investment line item**; (2) **Capacity-Adjusted Benefit-Cost Ratio ($\text{CA-BCR} \geq 1.25\times$)** incorporating empirical delay and O&M discounts; and (3) **Ring-Fenced Preventive O&M Escrow Covenant** signed prior to first construction tranche release.

SECTION 14 | TECHNICAL APPENDIX A & APPENDIX B

MODEL REGISTRY v2.0

Technical Appendices: Econometric Specification & Country Parameter Registry

SECTION THESIS — This technical appendix registers the core parameter priors, structural equations, Monte Carlo uncertainty distributions, and representative sovereign diagnostic profiles of the H Heuristics Institutional Capacity and Implementation Model (HH-ICIM v2.0), ensuring full auditability and reproducibility.

5,000 Draws
Joint Latin Hypercube Monte Carlo

N = 148 States
Balanced Sovereign Panel (2000–2025)

N = 1,840 Projs
Audited Micro-Execution Ledger

RMSE = 4.1 pts
Out-of-Sample Imputation Accuracy

14.1 Appendix A: Structural Model Equations, Identification & Parameter Priors

The coupled **HH-ICIM v2.0 × HH-PRM v1.0** simulation engine integrates four recursive modules at an annual time step $t \in [2025, 2100]$ across $i \in (1, ..., 148)$ sovereign states and $s \in (1, ..., 10)$ polycrisis sectors:

1. **Module 1 – Endogenous Pillar Evolution:** Each pillar $P_{i,k,t} \in [1, 100]$ evolves according to Equation 9.1, driven by targeted capacity co-investment $\varphi_{i,k,t} I_{i,t}$, organic learning-by-doing, enclave poaching drain $\xi E_{i,t}$, and compound shock hysteresis drawn from a calibrated Poisson-Pareto disaster process.
2. **Module 2 – Multiplicative CES Capital Conversion:** Pillar scores aggregate into baseline institutional efficiency via the CES operator (Equation 2.1) with substitution parameter $\rho s \text{ im } \mathcal{N}(-0.65, 0.09^2)$, modified by the macroeconomic absorptive capacity penalty $\Phi(i_{i,t}, S_{i,t}, C_{i,t})$ (Equation 4.1).
3. **Module 3 – Sectoral Capital Accumulation & Endogenous O&M Decay:** Nominal sectoral investment $(1 - \varphi_{i,t}) I_{i,s,t}$ converts into effective functional capital additions via $\theta_{i,s,t} = [\theta_{i,t}]^{\frac{\beta_s}{\beta}}$, while existing effective capital stock $K^{\text{eff}}_{i,s,t}$ depreciates at the endogenous convex rate $\delta_{s\left(\frac{m_{i,s,t}}{m^*_s}, t_{\text{age}}\right)}$ (Equation 8.1).
4. **Module 4 – Mortality, Deprivation & Welfare Valuation:** Effective sectoral capital stocks $K^{\text{eff}}_{i,s,t}$ feed directly into the six pathway hazard equations of HH-PRM v1.0 (air quality/clean energy, water/sanitation, nutrition/agriculture, health systems, climate adaptation, and pandemic preparedness), with pathway overlap discounts and a global interaction discount of 0.92. Table 15 registers all primary structural priors.

TABLE 15 **Appendix A: HH-ICIM v2.0 Structural Parameter Registry and Monte Carlo Uncertainty Priors** *— Central values, 90% credible intervals (5th–95th percentiles), estimation sources, and epistemic evidence grades*

Symbol	Parameter Description	Central	90% UI Prior	Estimation / Calibration Basis	Grade
$\omega_1, ..., \omega_5$	Structural pillar weights (PFM, Proc, Bur, Coord, O&M)	0.22–0.17	±0.03 Dirichlet	Shapley NLS on $N = 1,840$ projs	Grade A
ρ (σ)	CES complementarity parameter ($\sigma = \frac{1}{1-\rho}$)	-0.65 (0.61)	[-0.82, -0.48]	Nonlinear GMM (Table 4, Col 4)	Grade A
γ_1	Absorptive capacity GDP-share cliff steepness	0.38	[0.28, 0.49]	Penalized spline (Eq. 4.1)	Grade A
γ_2	Quadratic capital surge velocity cost penalty	0.14	[0.09, 0.20]	2SLS surge panel (Eq. 4.1)	Grade A
α_{eGP}	Log unit cost reduction from mandatory open e-GP	-0.142	[-0.18, -0.10]	Callaway-Sant'Anna DiD ($N = 64$)	Grade A
α_{bid}	Elasticity of award price to bidder count $\ln(N)$	-0.084	[-0.11, -0.06]	14,600 OCDS contract lots	Grade A
β_s	Sectoral institutional intensity elasticity range	0.16–0.56	±0.04 by sector	Random-slopes logit/ ERR (Fig 8)	Grade A
λ_s	Convexity of asset decay under O&M deficit	1.85	[1.40, 2.45]	RONET/IBNET survival hazard	Grade A
ξ	Enclave PIU civil-service poaching drain coefficient	0.042	[0.025, 0.062]	410-agency panel (Eq. 9.1)	Grade B
η_1	Productivity of institutional co-investment φI	0.34	[0.22, 0.46]	Matched reform evaluation	Grade B
$\beta^{\text{shock}}_{\text{Q1}}$	Q1 execution drop on 95th-pct compound shock	-18.5 pp	[-23.0, -14.0]	Jordà local projections ($N = 94$)	Grade B
φ^*	Global South optimal capacity co-investment share	6.2%	[5.1%, 7.8%]	Portfolio optimization (Fig 14)	Grade B

Notes & Sources: H Heuristics HH-ICIM v2.0 technical documentation. Full replication code and 5,000 Monte Carlo parameter draws archived in model registry.

14.2 Appendix B: Representative Sovereign Diagnostic Profiles (30 Major Economies)

Table 16 reports baseline 2025 diagnostic scores across Pillars I–V, composite HH-ICIM v2.0 scores, effective capital conversion factors ($\theta_{i,2025}$), absorptive capacity inflection thresholds ($i^*(C)$ as % of GDP), and primary binding constraints for 30 representative low-, middle-, and high-income economies drawn from the full 148-country panel.

TABLE 16 **Appendix B: Sovereign Institutional Implementation Profiles for 30 Representative Economies (2025 Baseline)** *— Pillar scores (0–100), composite HH-ICIM score, effective capital conversion (θ), absorption threshold (% GDP), and binding pillar*

Country (ISO)	Region / Tier	P1	P2	P3	P4	P5	Score	Eff. θ	Cliff i^*	Primary Binding Constraint
Chad (TCD)	SSA · Q1	24	21	25	23	19	22.4	0.30	8.5%	O&M & Procurement
DR Congo (COD)	SSA · Q1	26	22	27	24	20	23.8	0.31	8.8%	O&M & Treasury PFM
Haiti (HTI)	LAC · Q1	25	23	26	22	21	23.5	0.31	8.8%	Coord & O&M Decay
Myanmar (MMR)	EAP · Q1	28	25	30	26	24	26.7	0.34	9.4%	Procurement & Grid
Mozambique (MOZ)	SSA · Q1	35	31	36	34	29	33.0	0.40	10.6%	Procurement & Shock
Nigeria (NGA)	SSA · Q1	36	33	38	31	28	33.4	0.41	10.7%	O&M & Federal Coord
Pakistan (PAK)	SAS · Q2	41	38	45	36	33	38.8	0.46	11.8%	O&M & Circular Debt
Uganda (UGA)	SSA · Q2	43	39	44	38	35	40.0	0.47	12.0%	Land ROW & Arrears
Ethiopia (ETH)	SSA · Q2	44	40	46	42	36	41.7	0.49	12.3%	FX/O&M & Conflict
Bangladesh (BGD)	SAS · Q2	46	44	47	39	38	43.1	0.50	12.6%	Land ROW & O&M
Ghana (GHA)	SSA · Q2	47	44	51	45	40	45.5	0.52	13.1%	Arrears & O&M Trust
Kenya (KEN)	SSA · Q2	49	45	52	44	41	46.4	0.53	13.2%	County Coord & Debt
Argentina (ARG)	LAC · Q2	45	48	54	46	42	47.2	0.54	13.4%	Macro PFM & Provincial
Egypt (EGY)	MENA · Q2	51	47	53	49	45	49.1	0.56	13.8%	Procurement & Debt
Senegal (SEN)	SSA · Q2	52	49	53	47	44	49.2	0.56	13.8%	O&M & Land Cadastre
Philippines (PHL)	EAP · Q3	58	54	59	51	50	54.7	0.61	14.9%	Right-of-Way & LGU
South Africa (ZAF)	SSA · Q3	61	55	58	52	47	55.0	0.61	14.9%	Municipal O&M & SOEs
Mexico (MEX)	LAC · Q3	62	57	61	56	53	58.0	0.64	15.5%	Procurement & O&M
Morocco (MAR)	MENA · Q3	62	58	61	57	54	58.6	0.65	15.6%	Subnational Sync
Colombia (COL)	LAC · Q3	63	60	62	55	54	59.1	0.65	15.7%	Environmental Permit
Indonesia (IDN)	EAP · Q3	64	60	63	57	55	60.1	0.66	15.9%	Land Cadastre & O&M
India (IND)	SAS · Q3	64	61	66	55	53	60.4	0.66	16.0%	Land ROW & Discom O&M
Brazil (BRA)	LAC · Q3	65	62	67	56	54	61.3	0.67	16.2%	Permitting & Subnat.
Vietnam (VNM)	EAP · Q3	65	61	66	60	57	62.0	0.68	16.3%	Defensive Audit Delay
Turkiye (TUR)	ECA · Q3	64	59	65	62	59	61.8	0.67	16.3%	PPP Contingent Liab.
Rwanda (RWA)	SSA · Q3	66	65	64	61	56	62.7	0.68	16.4%	Domestic Cap Market
Malaysia (MYS)	EAP · Q4	75	72	76	70	69	72.7	0.78	18.4%	State Land & Procurement
Chile (CHL)	LAC · Q4	78	76	77	71	72	75.1	0.80	18.8%	Permitting Clocks
United States (USA)	OECD · Q4	83	81	84	72	76	79.6	0.83	19.7%	NEPA/Permit Litigation
Singapore (SGP)	EAP · Q4	93	92	94	91	91	92.3	0.94	22.2%	Global Frontier Anchor

Notes & Sources: H Heuristics HH-ICIM v2.0 country diagnostic database (2025 baseline). Full 148-country spreadsheet available in digital supplement.

REPLICATION & DATA ARCHIVE **Open-Access Codebase, Imputation Validation & Subnational Extensions**

All econometric scripts, 5,000 Latin Hypercube Monte Carlo draws, country-pillar time series (2000–2025), and coupled HH-ICIM v2.0 × HH-PRM v1.0 scenario notebooks are archived in the H Heuristics Open Research Registry (`HH-2026-50-ICIM-v2`). Out-of-sample 10-fold cross-validation confirms that random-forest PIMA imputation achieves $R^2 = 0.87$ ($\mathrm{RMSE} = 4.1$ pts) with zero regional bias.

SECTION 15 | BIBLIOGRAPHY & ARCHIVAL SOURCES

80+ PRIMARY CITATIONS

References and Primary Institutional Data Sources

SECTION THESIS — This bibliography compiles the peer-reviewed econometrics, public administration scholarship, engineering-economic lifecycle studies, and primary multilateral datasets underpinning the HH-ICIM v2.0 model and this monograph.

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15.2 Primary Multilateral & Open-Access Archival Data Repositories

Table 17 registers the seven primary institutional, procurement, fiscal, and project-level data repositories harmonized in the **HH-ICIM v2.0** dataset (2000–2025 panel).

TABLE 17 **Primary Empirical Archival Sources and Harmonization Coverage in HH-ICIM v2.0** *— Institutional custodian, observation unit, temporal coverage, and primary HH-ICIM pillar mapping*

Dataset / Archival Repository	Custodian Institution	Sample Size	Time Span	Primary HH-ICIM v2.0 Pillar Mapping
1. Project Performance & ICR Ledger	World Bank IEG / ADB / AfDB	1,840 Projs	2005–2025	Realized ERR, cost overrun, delay & $\theta_{p,i,s}$
2. Public Inv. Mgmt. Assessment (PIMA)	International Monetary Fund	86 States	2015–2025	Pillars I, II, IV & V design vs. effectiveness
3. PEFA Fiscal & Accountability Assessments	PEFA Secretariat / WB	620 Reports	2005–2024	Pillar I (PI-1, PI-16, PI-21, PI-22 arrears)
4. Open Contracting & e-GP Portal Registry	Open Contracting Partnership	14,600 Lots	2010–2024	Pillar II (e-GP share, bidder count, amendments)
5. V-Dem v14 & QoG Expert Bureaucracy	Univ. of Gothenburg	148 States	2000–2024	Pillar III (Weberian meritocracy, rigor, autonomy)
6. RONET & IBNET Utility Lifecycle Panels	World Bank Transport / Water	410 Utilities	2005–2024	Pillar V (O&M spend, NRW, road roughness IRI)
7. Sovereign Spreads & Disaster Shock Panel	J.P. Morgan EMBI / EM-DAT	112 States	2000–2025	Section 10 IRF hysteresis & sovereign spread trap

Notes & Sources: H Heuristics HH-ICIM v2.0 data harmonization protocol. All public replication series available at https://www.hheuristics.com.

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